

Dakota State University

General Chemistry Laboratory Manual

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Introduction

Chemistry is a discipline based on observation (as are all sciences). In lecture, you will learn the principles and theories that, to date, best explain the observations that have accumulated. The problem is that, if all you have is lecture, then it is all too easy to forget that these theories apply to the “real world.” The laboratory experience is, by design, your opportunity to see these principles and theories in practice.

This laboratory manual has been written not only to enhance your understanding of chemistry, but also to utilize the computing device that you, as a student at Dakota State University. You will be required to bring it to lab so that you can use it to capture data and take observations. You are also strongly encouraged to bring it to class. In addition, you will be required to purchase a copy of Hyper Chem for students (available at <http://www.hyper.com> for download) which will be used both in lecture and lab.

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Special thanks to Amanda Miller for her help as my photographer.

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Laboratory Safety Guidelines
Dakota State University
Chemistry Laboratories

Legal Notice: These safety guidelines are just that; guidelines. To the best of the author's knowledge, this is as complete a document as can be reasonably expected, however, following these guidelines does not guarantee that an individual may not be harmed in a lab, and because situations can arise that are not expected and there may be guidelines that have been overlooked by simple mistake, the College of Arts and Sciences, Dakota State University, and the author claim no responsibility for any reason whatsoever by those who choose to use this document. This document is provided to the general public as a courtesy; any individual, institution, or organization who chooses to use this document, either in its original or in a modified form, do so at their own risk, assume all responsibilities and, by use of this document, implicitly agree that they shall not hold the aforementioned College of Arts and Sciences, Dakota State University or Dr. Richard E. Bleil liable for injuries or accidents that occur in any lab.

Introduction

Very rarely will an injury or accident occur in a well-supervised laboratory. When an injury or accident *does* occur, it is generally brought about by complacency. In this laboratory, you will hear a LOT about lab safety; you will be given safety instructions at the start of each lab, you will be told of the major hazards of each chemical you will be using, and you will be quizzed on safety. Sometimes, such an emphasis makes a student nervous about what may be a new learning environment for them. This is an unfortunate and unintentional side effect, but it is important to give such emphasis on safety to reduce the odds of injuries in the lab by being sure that students know what hazards exist, how to avoid them and how to respond if something does go wrong. Knowledge is the best defense against injury in a chemistry lab.

The best way to prevent accidents is for you to know the possible hazards of the laboratory. Any experiment, no matter how often it has been performed in the past, has the potential to fail with hazardous results. By knowing the hazards, you will develop a healthy respect for what is happening around you, and with this respect, heightened levels of observation are sure to follow. This implies that potential accidents can be spotted *before* they can occur. If there is ever anything that does not seem right to *you*, it is not only your right, but also your obligation to point them out to *me*, your instructor. I will do my part to keep you safe, but I will need your help.

The following sections present some general guidelines. These are *not* arbitrary rules set down to make your life less enjoyable. Each and every one of them has a specific purpose, which will hopefully be made clear to you. If not, ask! There will be a safety exam which you will be required to pass (90% or greater) before the fourth lab day. Even though this grade will not be a part of your final grade, you must pass this exam to continue in the lab, so take it seriously. On the other hand, it is not designed to trick you or to be particularly difficult. If you understand the following guidelines and the reasons behind each point, you will pass the exam.

Most of laboratory safety is common sense. Remember that this is a general guideline, and therefore may be incomplete. If you are ever unsure about safety, *please* ask.

Laboratory Apparel

- (1) **Safety goggles are required** in the laboratory AT ALL TIMES! Splash hazards are perhaps the most significant danger present in the lab, and eyes are extremely sensitive.
- (2) **Contact lenses are not permitted in the lab.** Your goggles will protect your eyes from spill hazards, but do nothing to protect you from fumes, which can dry your contacts out and may result in the necessity of an operation for their removal. Contact lenses can also absorb chemicals from the air (especially the new “breathable” lenses), concentrate and hold them against the eye, and prevent proper flushing of the eye should a chemical be splashed into the eyes.
- (3) **Laboratory aprons must be donned at all times.** In the event of a spill, these aprons are chemical and flame resistant, and could save you from scar tissue!
- (4) **Sandals, open-toed shoes and high heels are not permitted in the lab.** This is to protect your feet from splashes and spills. The restriction on high heels is for balance.
- (5) **Shorts or skirts cut above the knee are not permitted in the lab.** Again, should a spill occur, it will be your clothing that will be your protection from direct exposure of the skin to that chemical. The idea is to put as many layers of clothing as possible between you and a chemical spill. The more clothing, the more diffuse the chemical will be by the time it reaches the skin, and the greater the chance to remove the chemical *before* it reaches your skin.
- (6) **Careful consideration should be given before wearing any jewelry into the lab.** Some chemicals evaporate very quickly and therefore pose relatively little danger should they get onto your skin. However, if they get beneath a ring, watch or some other form of jewelry, they can be prevented from evaporating, held against the skin longer and greatly increase the risk of injury. Should you decide to wear jewelry to the lab (as I will be wearing my watch), be particularly mindful of itching, burning or any other irritation under or around your jewelry. (By the way, NEVER wear opals, pearls, or other “soft gems” in the lab. The harsh laboratory environment may dry them out or otherwise damage them, and neither your instructor nor DSU will replace or repair such items.)
- (7) **Never wear clothes that hang,** such as loose sleeves. Be sure ties and scarves are tucked well inside your laboratory apron. These pose fire hazards (if you are reaching or bending down near an open flame) as well as chemical hazards (if they accidentally get dragged through a chemical, they can transport that chemical directly to your skin). In fact, you may want to give very serious consideration to wearing only very old clothes. Some of my students have, in the

past, brought old clothes with them in a gym bag and changed right before and after lab. Be especially careful of sleeves around open flames..

(8) **Long hair is to be constrained.** Like hanging clothes, long hair is subject to fire and contact with chemicals. A rubber band *will* be used to constrain particularly long hair if necessary.

(9) **No radios, tape players, CD players or any other devices of this type will be permitted in the laboratory at any time.** Loud music is distracting, and headphones prevent you from hearing announcements or verbal warnings given in the lab.

Safety Equipment

(1) Take **the time to identify all of the laboratory safety equipment, and keep their location in your mind at all times.** You should be able to close your eyes any time during a lab and point to such safety equipment as the fire extinguisher, the emergency eyewash stations, the fire blankets, the safety shower, etc. If you were to splash a chemical in your eyes, you'd better be able to find that eyewash station without your eyes well before permanent damage can occur (which can be seconds depending on the nature of the chemical).

(2) **Check all safety equipment.** I'll keep as close an eye on it as possible, but I need your help as well. Is the fire extinguisher charged? Does it have the plastic "seal"? Is there enough sodium bicarbonate in case there is a chemical spill? If anything does not look right to you, report it to your lab instructor IMMEDIATELY!

(3) **Material Data Safety Sheets (MSDS's) are available to you on request only.** Basic safety information will be given during the safety lecture before each lab. You can also find links for MSDS's on my homepage at <http://www.homepages.dsu.edu/bleilr/> if you are interested.

General Behavior

(1) **ABSOLUTELY NO HORSEPLAY WILL BE TOLERATED IN THE LABORATORY!** Offenders of this one will be unceremoniously cast out with a zero resulting for that day's work. I realize that at times it is awfully tempting to grab that water bottle and squirt your friends, but many hazardous chemicals look like water. The humor will be lost if something other than water is in that bottle.

(2) Always **read the upcoming experiments carefully and thoroughly, being sure to understand all of the directions before entering the lab.** This will help you to be prepared to handle any hazards of the experiment, and will also help you to perform the experiment more quickly resulting in less "fumbling around" and reckless work as you rush to finish on time. To ensure that you have read the upcoming experiment, you are required to complete the pre-lab

assignment before entering the lab. **If you fail to complete the pre-lab assignment on time, you will not be allowed to perform the experiment.**

(3) Be **in the lab and ready promptly when the lab begins.** The safety lecture (specific to that day's experiment) will be the first item of business each day. **If you are not present to get this important information, you will *not* be allowed to do the experiment.**

(4) **Absolutely no food or beverages will be permitted inside the lab.** They can absorb chemicals from the air (and concentrate them), or can pick them up from the bench, causing ingestion of these chemicals. Everything possible will be done to be sure the laboratory air is safe for working in without the use of special respiratory equipment. Please don't complicate the issue by eating these chemicals as well!

(5) **WASH YOUR HANDS!** Wash your hands frequently during lab, and definitely wash your hands twice at the end of the lab, once in the lab itself, and again outside of lab (as in a public rest room), **ESPECIALLY** before eating. Once you get home, you should wash your face as well. You don't want to drag too many chemicals around with you on your skin.

(6) **Do NOT apply makeup (including Chapstick and other lip balms) in the lab.** In fact, you may want to seriously consider not wearing makeup to the lab at all. Makeup can also pick up and concentrate fumes from the air, and hold them against the skin causing irritation. Perfumes, colognes or other fragrances may also interfere with the olfactory senses when an experiment calls for "smelling" something.

(7) **Should an injury occur, regardless of how minor it is, report it IMMEDIATELY to the lab supervisor.** The smallest puncture wound allows for chemicals to enter the blood stream directly. By notifying your supervisor, even if no action is taken, the incident will be reported to the student health center. In the event that this wound should become infected later, having this information on file may prove to be of extreme importance for prompt treatment.

(8) **Never pick up broken glassware with your bare hands, regardless of the size of the pieces.** Typically, puncture wounds occur with the largest pieces in such a situation, because they look to be the most harmless. A brush and dust pan is provided for broken glassware. Please place all broken glassware in the appropriate broken glassware container, and never put caps, paper or other waste in this same container. Very small bits of broken glassware (as in the bottom of a drawer) can be picked up with a damp paper towel.

(9) **NEVER put broken glass in a regular garbage can.** A container is provided that is especially designed for broken glassware.

(10) **Always read the labels to reagents (chemicals used in an experiment) *twice!*** Many chemicals look identical on first glance, and may differ only slightly in their spelling or

concentration. Sodium sulfate may look similar to sodium sulfite, but they are most certainly different and confusing them in the lab may result in dire consequences. Therefore, read the label as you grab the bottle, and holding it in your hand, look carefully at the label a second time and verify that it is exactly what you want.

(11) **Never make unauthorized substitutions.** If you are wondering what would happen if you used this instead of that, ask me. If it's safe, I may let you try it. If not, I'll let you know what would have happened if you tried it.

(12) **Never use reagents from an unmarked bottle.** All reagent bottles will have proper labels, so if a reagent bottle is unlabeled, it is the incorrect reagent.

(13) **In any emergency, the fastest way to get the lab supervisors attention is to SCREAM!**

(14) **If you are not feeling well, report it to the laboratory supervisor immediately.** If your supervisor should lose consciousness during a lab period, it may be due to chemical fumes. Evacuate the lab immediately and seek another professor for help. Should anybody else lose consciousness in the lab, the lab supervisor will determine whether or not evacuation of the lab is warranted (it probably will not be).

(15) **Avoid bringing excess coats, books, backpacks or other personal items to the laboratory.** There is always the danger of spilling chemicals on them, and they create a fire hazard if left in the isles. In the general chemistry lab, you may use the small cabinets underneath each drawer to store personal items during an experiment

(16) **Close your lab drawer!** Once you have retrieved the equipment you need from your equipment drawer, be sure to close it again. Open drawers can pose tripping hazards (especially bottom drawers) and obstruction of walkways. Thump! OUCH!! The reason we do not have stools in the lab is to avoid similar obstruction.

(17) **Never smell a chemical straight out of a container.** Some chemicals are extremely caustic (fumes severely irritate delicate tissue) and the fumes should be avoided. To safely smell a chemical, hold it two to three feet from your nose, and with your other hand cupped, waft the fumes towards you. You may slowly move the chemical closer to your nose if you cannot smell it all the while taking only small sniffs.

Fire

(1) **In the event of a fire, *DON'T PANIC!*** This is probably good advice for a lot of sections of this outline.

- (2) If a small portion of your clothes catches fire, the fire may be extinguished by patting it out.
- (3) If a larger portion of your clothes should catch fire, there are three options for putting the flames out. (1) Drop to the ground and roll. (2) Use the safety shower. (3) Use the fire blanket.
- (4) NEVER use a fire extinguisher on a person. Carbon dioxide fire extinguishers (distinguishable by their flared out nozzles) are extremely cold and may cause shock to the person or frostbite of the eyes. Chemical fire extinguishers cause excessive scarring by mixing of the chemical in the extinguisher with the damaged skin. All fire extinguishers have the potential of causing asphyxiation.
- (5) If a fire should occur in a beaker or some other container, cover it with a glass dish or other flame-retardant item.
- (6) NEVER move ANY object that is burning. If you try to pick up a beaker that is on fire, should you drop it, the burning chemical will spill making the situation even worse.
- (7) Never use water to extinguish a chemical fire. Many flammable liquids float on water, meaning that the water will have no effect but to spread the fire. Other chemicals may even react explosively with water!
- (8) If a fire is large enough to warrant the use of a fire extinguisher, the proper use of the extinguisher is as follows; (1) Be sure there is an exit behind you in case you cannot get the fire under control; (2) pull out the restraining pin (which requires breaking the plastic seal); (3) point the extinguisher hose at the base of the fire; (4) holding the extinguisher UPRIGHT, squeeze the handle to release the extinguishing media; (5) sweep the spray back and forth at the front of the fire. There are two important things to remember when using a fire extinguisher. (1) You may only have about a 30 second blast of extinguishing media, so extinguishers are only for use on relatively small fires. (2) Some fires may be inappropriate for a fire extinguisher. Be sure you have the right rating of the extinguisher, and never try to extinguish a fire on a vertical surface!

Chemicals and Chemical Spills

- (1) Report all chemical spills IMMEDIATELY to your lab supervisor. The chemicals you will be handling are NOT "scaled down" chemicals-they are exactly the same chemicals any professional chemist would order and use. Keep a healthy respect for them, or they may bite you!
- (2) Should a chemical spill on your person, immediate remove all affected clothing (tops from the back forward to avoid dragging the chemical across your face) and wash the affected body area with copious amounts of water. Unfortunately, chemicals have no respect for modesty

and will cause permanent damage if not treated immediately. If a large portion of your clothing is affected, immediately get to the safety shower and remove the contaminated clothing while the water is running.

(3) **Small spills on the bench or floor must be cleaned up immediately.** Sodium Bicarbonate and vinegar are included as part of the safety equipment for neutralization of acids and alkaline (basic) solutions respectively. Neutralize all acid and alkaline spills before cleaning. If you are not sure how to clean a spill, let your lab supervisor know immediately.

(4) Be **especially careful of spills around the balances.** These electronic devices are EXTREMELY sensitive to corrosion. A brush is kept near the balances so you can brush the balances thoroughly after EACH use (even a single grain of a reagent can cause irreversible damage). Clean up ANY spill near the balance IMMEDIATELY, and report it to your laboratory supervisor.

(5) **Mercury, lead, and other heavy metals pose a particular health hazard in that the human body cannot get rid of heavy metals.** Any heavy metals you've ever been exposed to are still with you today (including mercury if you ever played with it, or lead if you've ever eaten lead paint, a favorite activities of children as it tends to have a sweet taste). As a result, although most heavy metal poisons are not *particularly* toxic, the effects of heavy metal poisoning are typically only seen long-term, and can include uncontrolled trembling, insanity and death. The only way to combat these effects is through minimization of exposure to heavy metal poisoning. Mercury poses a particular hazard as vapors from the liquid accumulate in a room and quickly are at dangerous concentrations in the air. As a result, report ANY spills of mercury, as, for example, from a broken thermometer, as quickly as possible so it can be cleaned up immediately.

Laboratory Equipment

(1) **Never heat a piece of glassware (beakers, flasks, etc.) that is chipped or cracked unless otherwise told to do so by your lab supervisor.** Heating defective glassware can cause that glassware to break (or explode!), resulting in a spill.

(2) **If you have chipped or cracked glassware, or glassware with sharp or jagged edges, inform your lab supervisor immediately.** The equipment will probably be replaced, or you may simply be given special instructions on using that bit of equipment.

General Guidelines

(1) **Epilepsy, pregnancy, dyslexia as well as other medical conditions can be hazardous in the laboratory.** Every effort will be made to keep you safe, but I will need some help. *IF YOU HAVE ANY MEDICAL CONDITION WHICH YOU THINK MAY ADVERSELY AFFECT YOUR ABILITY TO SAFELY PERFORM IN THE LABORATORY, OR THAT MAKES YOU*

PARTICULARLY AT RISK TO BE IN THE LABORATORY, PLEASE INFORM ME AS SOON AS POSSIBLE! Many such conditions may be deemed personal, but the chemicals themselves cannot tell the difference. Therefore, please feel free to stop in my office as soon as possible so you can tell me in private, and, of course, anything you do tell me will be kept in the strictest confidence.

- (2) **To turn on a Bunsen burner**, first turn the nozzle on the bottom of the burner all the way off, then turn it back on about 2 turns. With a LIT MATCH in one hand, slowly turn on the gas at the spigot. Hold the match near the edge of the burner as you do so the air being pushed out by the propane does not blow it out. Such a procedure will avoid "explosions" when lighting the burner.
- (3) **Before using a burner, be sure nobody else on the bench has any organic solvents.** Organic solvents are flammable, and heavier than air, meaning that as they evaporate, they creep down the edge of their container to the bench top, whereupon they spread out horizontally. Once these fumes reach an open flame, they can ignite causing "flashback", thereby causing the beaker of solvent to catch fire from four feet or more away!
- (4) **Before getting any organic solvent, be sure nobody on your entire lab bench has an open flame.**
- (5) **Never take more of a reagent than you need.** This means that if you need about 5 mL of a solvent, use your 10 mL beaker to get it, NOT your 600 mL beaker.
- (6) **NEVER return an unused portion of a reagent to its original container.** See if anybody else at your bench, or in the lab, needs it. If not, give it to your instructor, who will look at you in a forlorn and sullen manner but will appreciate that you did not put it back in the original container. Returning unused portions of reagent greatly increase the odds of cross contamination, that is, getting the reagent contaminated with an unwanted chemical.
- (7) **NEVER pour a waste chemical in the drain, or put it in the garbage, unless otherwise instructed to do so by your lab supervisor.** Waste bottles will be provided. Always pour waste into the appropriate and labeled waste bottle (reading the waste bottle label twice).
- (8) **If you have glass stirring rods or glass tubes with sharp or jagged edges, fire polish them.** This means holding the sharp end in a Bunsen burner flame and rotating the rod or tube until a bright orange flame begins to show on the end being heated. Continue to heat while rotating another minute or so, effectively melting that end a little bit. Be SURE to let it cool COMPLETELY before attempting to fire polish the other end.

(9) Many items (glass, metal, etc) look exactly the same HOT they do cool. Be VERY careful whenever working with flames that ALL of your equipment (beakers, flasks, ring stands, etc.) are cool before handling them.

(10) If you are inserting glass tubing into a rubber stopper, use the following technique to avoid jamming a jagged piece of glass through your hand; (1) use glycerol or water to lubricate either the end of the glass tubing being inserted, the hole in the stopper the tubing will be inserted into, or both; (2) protect your hands by using a paper towel to hold both the glass tubing as well as the rubber stopper; (3) hold the rubber stopper in such a way that the tubing cannot go through the hole and into your palm (your fingers should actually curve, holding the edge of the stopper, as if to make the letter "C"); (4) hold the glass tubing, also with your palm away from the end, near the end being inserted into the rubber stopper; (5) insert the glass tubing with a twisting motion; (6) clean up any excess glycerol; and (7) live your life free from scar tissue on your palms that everybody for the rest of your life will ask about by saying "how did that happen?", to which you will have to reply that you didn't listen to your dedicated and caring chemistry professor.

(11) Improper heating of a test tube can result in the chemicals within the test tube shooting out, possibly resulting in injury to anybody in the path. When heating a test tube, use the following procedure; (1) unless directed otherwise, always place a few (five or six) boiling chips in the test tube; (2) use a test tube clamp to hold the test tube; (3) hold the test tube at about a 45° angle; (4) be sure the opening of the test tube is pointing away from anybody else (preferably towards a wall in a low-traffic area of the lab); (5) NEVER heat the bottom of the test tube (unless otherwise directed); instead heat the middle of the test tube just at the level of the liquid in the test tube; (6) move the test tube horizontally back and forth across the flame to prevent the liquid from heating too quickly; (7) should the liquid begin to overheat (heat too rapidly), remove the test tube from the flame and allow the contents to cool for a minute or so.

(12) NEVER look down the opening of ANY container, including beakers, flasks, and test tubes (as well as any other piece of equipment). Should something happen to cause the chemicals to "blast out" of the container, they will go directly into your face if you are looking down the opening at the time.

(13) Do not use graduated cylinders for any purpose other than to measure a volume of a liquid. Graduated cylinders should not be used to get reagent for an experiment (use a beaker for this) or to run reactions (use a test tube for this).

(14) Never put a dropper into a reagent bottle. Instead, put the reagent in a beaker so you can bring it back to your desk and use a dropper there.

I hope you see that these guidelines are for YOUR benefit, and follow them faithfully; they will become habit more quickly than you can imagine. Most importantly, if you have ANY questions

or comments, *please* tell me as quickly as possible. I will be more than happy to clarify any questions you may have.

Chemistry Laboratory Name and Section Number: _____

Date: _____

Name: _____

I, the undersigned student, have received safety training, understood it and agree to abide by the safety guidelines. I understand the importance of proper eye protection in the laboratory at all times. I have been warned about the dangers of wearing contact lenses in the laboratory and understand that I should not wear contacts in the laboratory. I also understand that if I do wear contacts in the lab or fail to abide by the safety rules, I am doing so at my own risk and will not hold Dakota State University or Dr. Richard Bleil liable for any injuries that result.

Signature of Student: DO NOT SIGN-FOR YOUR RECORDS Date: _____

Using the Pasco System

Your first question ought to be “What is Pasco and why do I need it?” To answer that question, we need to discuss analog and digital devices (starting with the latter). Your computer is a digital device, which means it only can think in terms of “Ones” and “zeros”, or, if you prefer, “on” or “off”. For example, take your plain old-fashioned light switch: it only has two settings, it is on, or it is off.

Analog devices, on the other hand, can take any value we set. When I was in high school, my best friend was (and still is) Mitch. Now, Mitch’s parents had a cleaning woman stop by once a week, who had a child of her own. She would often bring her child with her as she came to clean their house. The child took great delight in going into Mitch’s room and turning the volume of his stereo all the way up. When Mitch would turn the stereo on, then, it blasted like you cannot believe. Now, if the stereo was digital, he would not have been able to turn it down; his only choice would be to turn it on or off. Fortunately, it was an analog stereo, so he could set the volume to any value he wanted between the stereo’s lower and upper limits.

“But wait,” some of you are surely thinking, “I have a stereo at home that is digital, and I can set the volume on that stereo as well.” Ours has become a digital society; digital signals are cleaner and more reliable than analog signals, so they are used for all kinds of things, like television, radio and even telephone signals. What makes these devices digital is that they “think” in terms of ones and zeros. The problem is that, while we might not care how the device works internally, we do care how devices present their output to us. If all we got out of our stereo was a stream of ones and zeros, it would not be of much use to us. We need an analog output to make sense to us (since, after all, we are analog creatures). To accomplish this, our digital devices have “digital to analog converters”, or “DAC’s”. These convert the streams of ones and zeros into an analog signal that sounds like music to us, and even allows us to choose the volume we want.

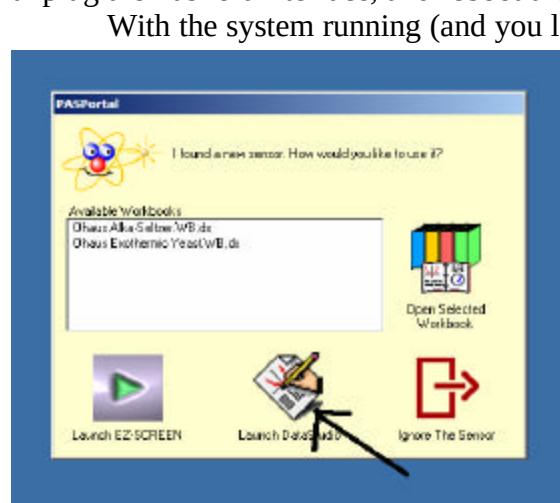
As you might well imagine, if we can convert a digital signal to an analog signal, then we should be able to turn it around and convert analog signals to digital. We can, and, not surprisingly, to do so we need an “analog to digital converter” or “ADC”. Your mobile phone has one of these (as well as a DAC) which it uses to convert your spoken (and analog) words into a digital stream of ones and zeros that it can transmit.

Essentially, this is what the Pasco system is: the black box (literally) is nothing more than an analog to digital converter, albeit somewhat larger than the one in your cell phone. The Pasco probes are really just devices that convert certain measurements into voltages; for example, the “temperature probes” give off higher voltages as the temperature increases. These voltages, just like temperature, are analog in nature. When you plug this probe into the Pasco box, the box converts this voltage into a digital signal, that your computer can interpret, store and manipulate.

Naturally, your computer has to know how to deal with this data, so, of course, you will need the appropriate Pasco software. This is to introduce you to the Pasco software and provide you with the basic process for using your Pasco system to collect data on your computer.

Starting Pasco

Of course, we will begin by assuming that you have already installed the Data Studio software on your computer. Make sure that you have installed both the Data Studio software as well as the PasPort hardware driver. If this is the first time you've used PasPort sensors, get the CD and install Data Studio. Keep the CD in the drive as you plug in the PasPort interface and Windows will automatically install the software. When the question is asked regarding "Windows Signing" answer "proceed anyways." Once Data Studio and the driver are installed, unplug the PasPort interface, and reboot the system.

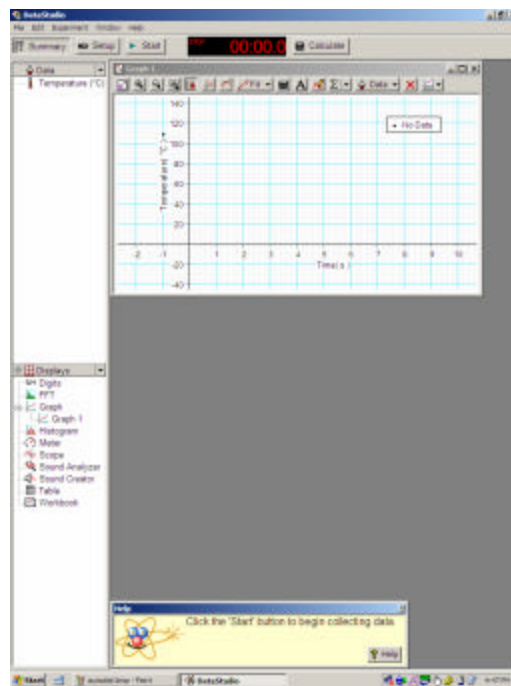


With the system running (and you logged in), plug the PasPort interface, WITHOUT a sensor in it, into a USB port. Make sure the green light is on on the Pasport sensor front. Next, plug the probe you want into the PasPort interface; make sure the writing on the probe and the interface are in the same direction, and it should plug in smoothly. Your system will recognize the sensor, and bring up a dialog screen asking you what you would like to do; choose "Launch DataStudio."

Once DataStudio is launched, it looks as if you should just be able to hit "start" and it will begin collecting data. Indeed, it would, but it might not be what you want it to do, so we must customize the software so it knows what we want it to do.

Calibration of the System

Many of the probes are pre-calibrated, however, there are times that we will want to calibrate them ourselves. The probes work by voltage; whatever they are measuring is converted into a voltage, which is read by the interface and fed to the computer. For example, the temperature probe has some given voltage that corresponds to a given temperature, and the factory calibration for this is relatively good. However, differences in manufacturing from one probe to the next means that there are small differences that might throw the sensor off a bit. For many experiments, these differences are not important and will usually cancel each other out; however, if we are doing a highly precise experiment where we need the absolute temperature (instead of the relative temperature), then we will want to calibrate the probe.

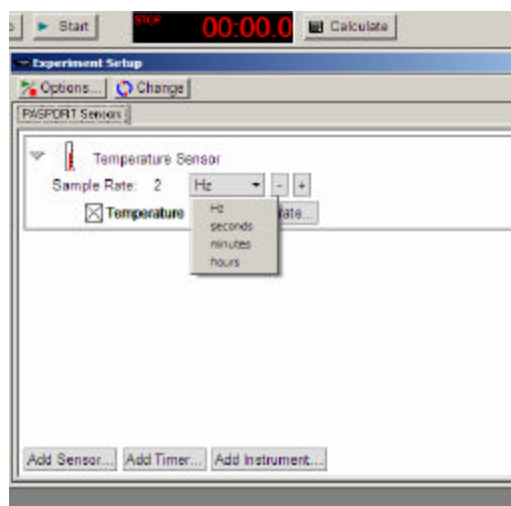
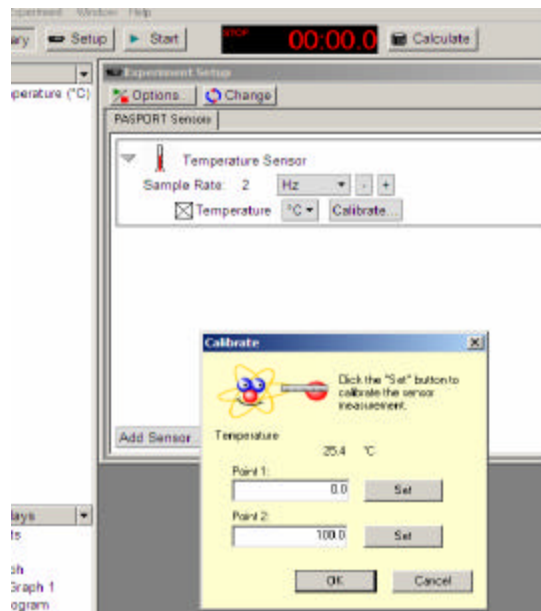


To calibrate the probe, choose the “set-up” icon near the top of the display. There you will see a variety of choices in the new dialog screen; to calibrate the probe, choose “calibrate.” To perform a proper calibration, Pasco will typically ask you for two set points, a high set point and a low one. They assume that the system will act linearly; as a chemist, this kind of bothers me. I’ve learned a long time ago that a minimum of three points is necessary to assure linearity, but we will discuss this, as well.

The set points can be done in one of two ways; either you can measure the value relative to a source that you trust more, or you can measure a fixed point. For example, for the temperature probe, we can use a high precision mercury thermometer in the lab to compare the values with Pasco, or we can use a well-known phenomenon like melting or boiling water. For the former, simply place the Pasco temperature probe and the thermometer into the same material (perhaps a beaker of water). Give both the probe and the thermometer a minute or so to equilibrate, and read the temperature as indicated by the thermometer. Type this value in for point 1 and click “set.” Notice that you will have to do this for two different temperatures; you can repeat the procedure for point 2 using a warmed or cooled beaker.

To use melting and boiling points, we take advantage of the fact that water freezes at 0°C, and boils at 100°C. For point 1, put the probe in an ice-water bath, and after it has a minute or so to equilibrate, type “0” in for the point and click “set.” Do the same for point 2 in the boiling water, only type in “100” before clicking “set.” This method is not as accurate as the former, because for these values to be true, the water must be absolutely pure, and the pressure must be exactly 760 torr; any deviation will result in slightly lower melting and slightly higher boiling points.

Once calibration is complete, be sure to press “OK” rather than “cancel.”



Automatic Data Collection

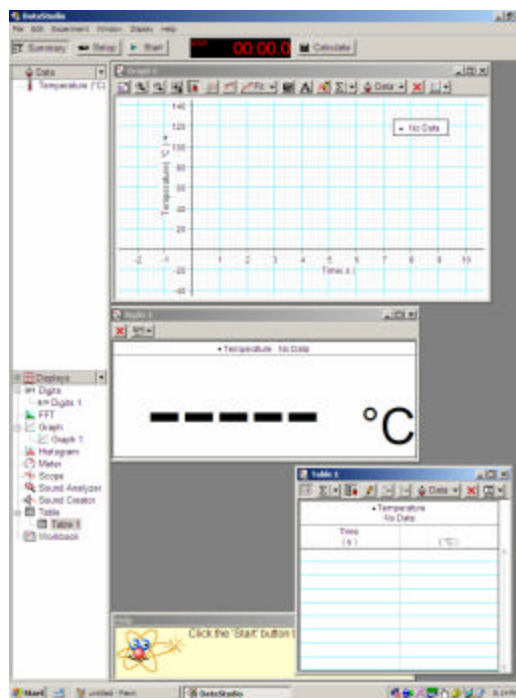
If you have not opened it yet (or you have closed it), open the “Setup” dialog box. For most probes, you will see “Sample Rate” followed by a number and a pull-down menu. This is for automated data collection. If “Hz” is in the pull-down menu, this means “per second.” For example, the temperature probe defaults to 2 Hz; this means that Pasco will collect 2 data points every second, or one data point

every 0.5 seconds. If it were at 10 Hz (which you can change by pressing the “+” and “-” icons after the pull-down menu), then there would be 10 data points per second, or one data point every $1/10^{\text{th}}$ of a second.

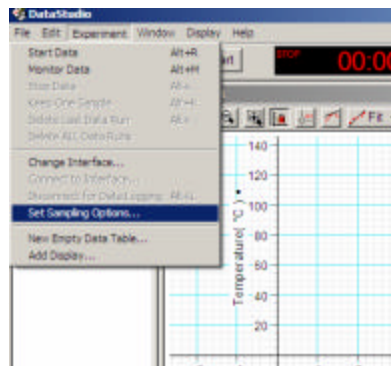
Sometimes you want this kind of rapid data collection, but usually it serves to do nothing but sop up valuable hard drive space and slow down your computer. Think about what it is you are measuring, and decide on how rapidly you would like the data points to be taken. For example, if I wanted to measure the temperature under my armpit, I don’t need a temperature update every 0.5 seconds; instead, maybe I’ll choose 10 seconds instead, that is, one new data point every 10 seconds. So, I will go to the pull-down menu, and choose “seconds” rather than “Hz.” Then I will click “+” until I get to 10. Once I close the window, Pasco will remember my choices.

Choosing displays

Now, there are a variety of ways we can view the data as we are collecting it. The default is usually to bring up a graph, which I usually like to keep. Other options include Digits (my other usual choice), FFT (for “Fast Fourier Transform”; we usually will not use this), Histogram, Meter, Scope, Sound Analyzer, Sound Creator, Table (another common favorite) and Workbook. You will see these to the left of the screen near the bottom (if not, click on the “Displays” tab on the left near the bottom). For our armpit experiment, I want to see the digits, and keep a table of the data, so I will click and drag the digits icon and the table icon onto the view screen. For each of these displays, I recommend playing with the options so you can see what they can do. At this point, if you press “Start” you will see that Pasco begins taking data at the rate of 1 point every 10 seconds.

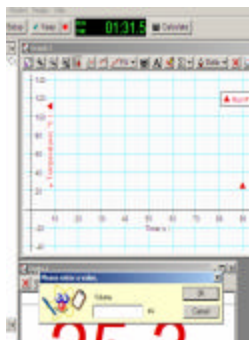


Manual Data Collection

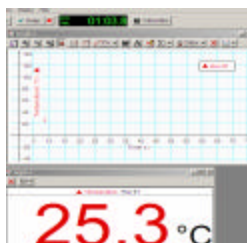


The default data collection is time-based, but there are experiments in which you do not want the data to be collected automatically. For example, in a titration, you might want to measure pH as a function of the volume of base that you have added. In this case, at each data point, you want to tell the system exactly what the volume is according to the buret, and have it record pH as a function of that specific volume.

To do this, go to “Experiment” and “Set Sampling Options.” This will open up a new dialog box for you in which you can be what you want it on “Keep data value.” These are the first tells



keyboard (in our example, volume) and to prompt for this value. In “Name” put down the name that makes sense to you (such as “Volume of Base Added”), and do the same for Units (for this example, probably “mL”). Click “OK”. Now, when you click “record,” instead of

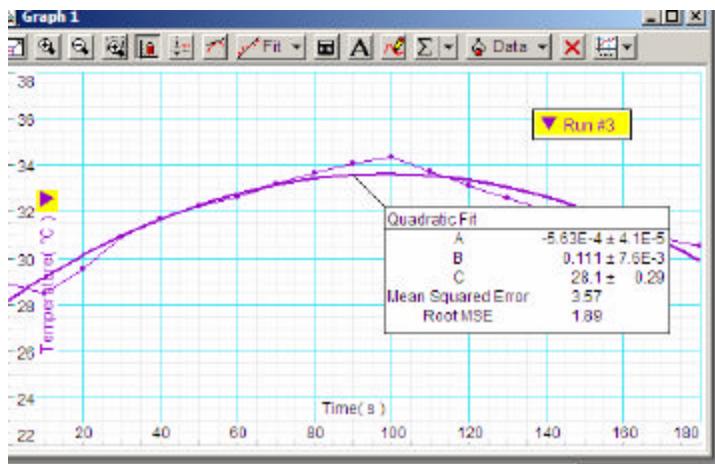


more precise in telling Pasco to do. In this dialog box, click values only when commanded.” automatically choose “Enter a value...” and “prompt for a typically what you would want; Pasco to take the values from the

automatically starting to collect data points, the system will begin collecting data, but not recording it. When you click on “Keep”, a new dialog box will open asking you what value to associate that reading with. Type in the value you want associated with this reading, and click “OK.”

Manipulating Data

Data Studio does have some ability to manipulate data. You will notice, on the right side of the screen, that each data set has been automatically stored. To delete one of these sets, just click on the data set once to highlight the specific run you want deleted, and press “del.” Notice that each screen (in this case, the graph and the table) allows you to manipulate the data. For example, look at the graph. Suppose we want to expand the scale so we can see it better; to do this, simply double click on one of the scales in the graph, and choose the values you want. You can also go to “Data” and choose which data sets to display, or not. There is even a curve fitting tool that can be used. The manipulation of data depends on what you are taking and what you need to do with the data.



Exporting data

Finally, suppose you want to manipulate the data using Excel, so you can pull the graphs directly into a lab report. To do so, highlight the table so it is the active screen, and go to “File” and “Export”. Choose the run you want exported, and click “OK”. Save it as a “txt” file in an

easy location to find. In Excel, go to “Data” and “Import external data”, and import the file you just saved.

Using Hyper Chem

See? It's an old game now. Yow are already expecting me to start with "So what is Hyper Chem?" Well, I can't let you down. Hyper Chem is a molecular visualization and calculation package. Do you know how you wished that you could see things like molecules and orbital so they were not so abstract? Well, this is what Hype them does. At its simplest level, Hyper Chem allows you to visualize molecules in three dimensions, including zooming in or out, or even rotating the molecule so you can see it from different perspectives. Hyper Chem allows you to import molecular structures or build your own, and change the way it looks, but it can do much more.

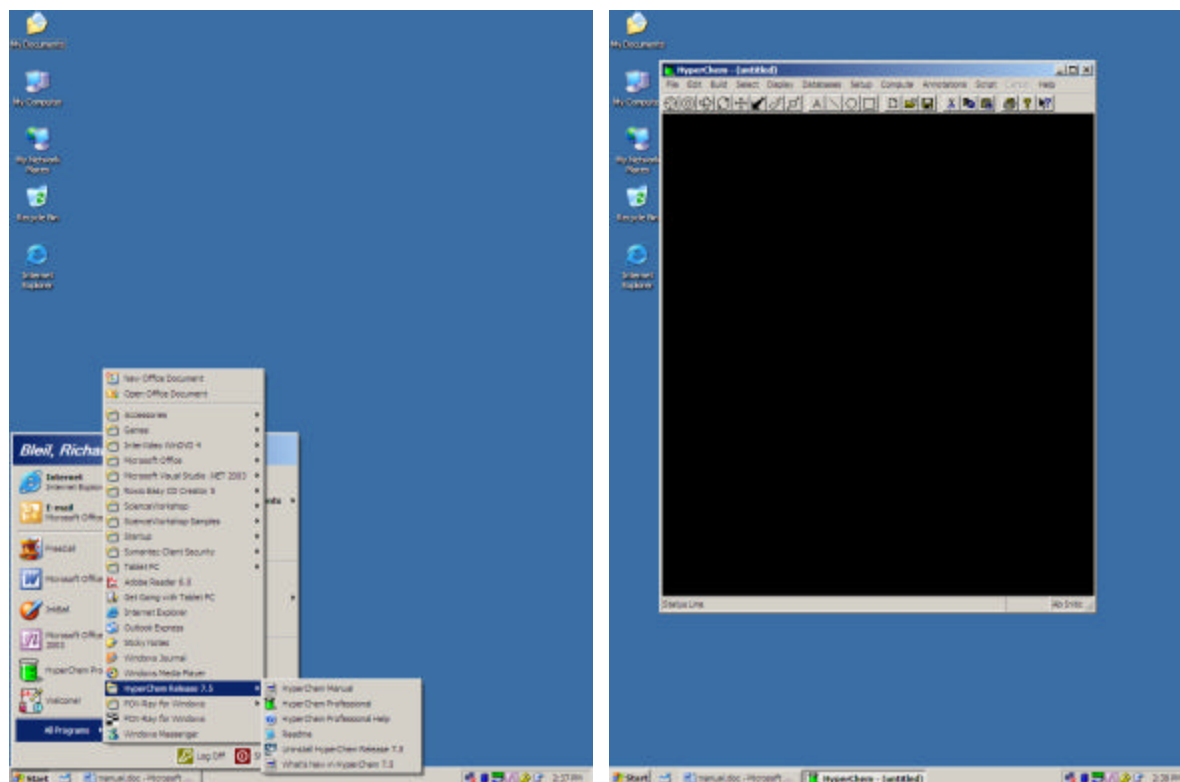
In addition to simple visualization, Hype Chem is a powerful tool for calculating the properties of molecules. At this point, we will be using these tools blindly; that is, I will not be giving you a lot of details on how it is working, but we can still get a lot out of it. Some of these are more or less automatic. For example, when you build your own molecule, Hyper Chem automatically calculates the most likely bond angles and lengths. In addition, Hyper Chem can do things like simulate molecular motion, or calculate and display electron orbitals. Using these tools help make chemistry feel less abstract.

Starting Hyper Chem

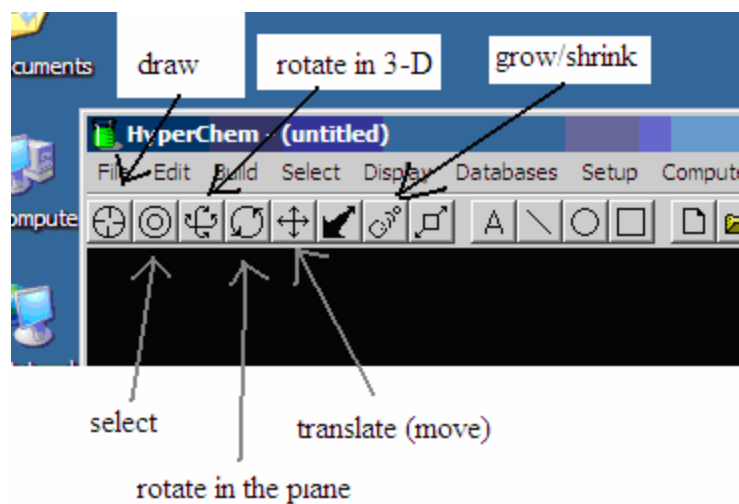
I will assume you hove already bought your copy of Hyper Chem If not, it is available for download at <http://www.hyper.com> (go to the student version) and can be downloaded and used for free for 30 days. However, yow will want to purchase the program for this course. In this introduction, I will show you the basics of Hyper Chem, but more specific instructions will be included with those experiments that utilize it.

Basic Hyper Chem Building Tools

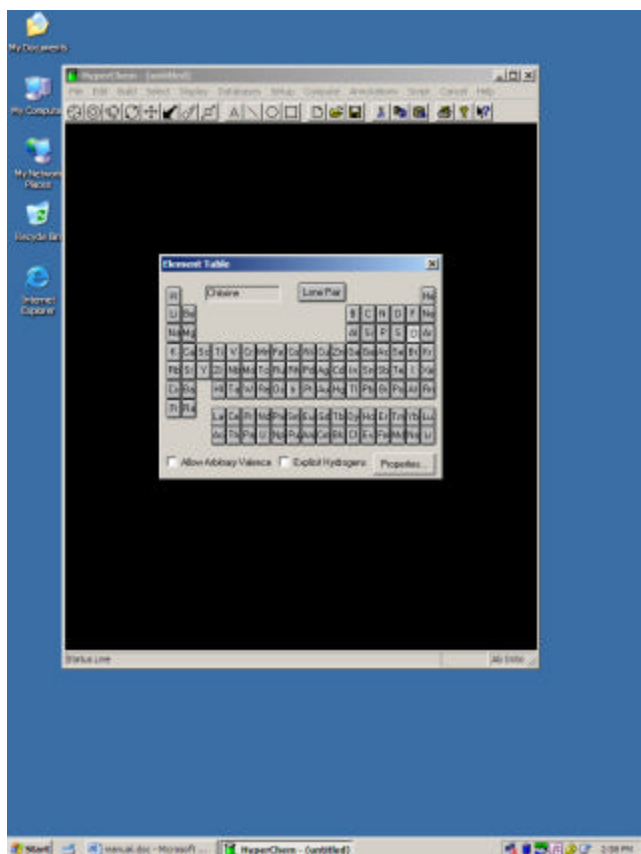
If you have purchased and installed Hyper Chem, it should be in your program files. Just start it up normally. By default, it starts with a black screen, which can be intimidating, but don't let it be.



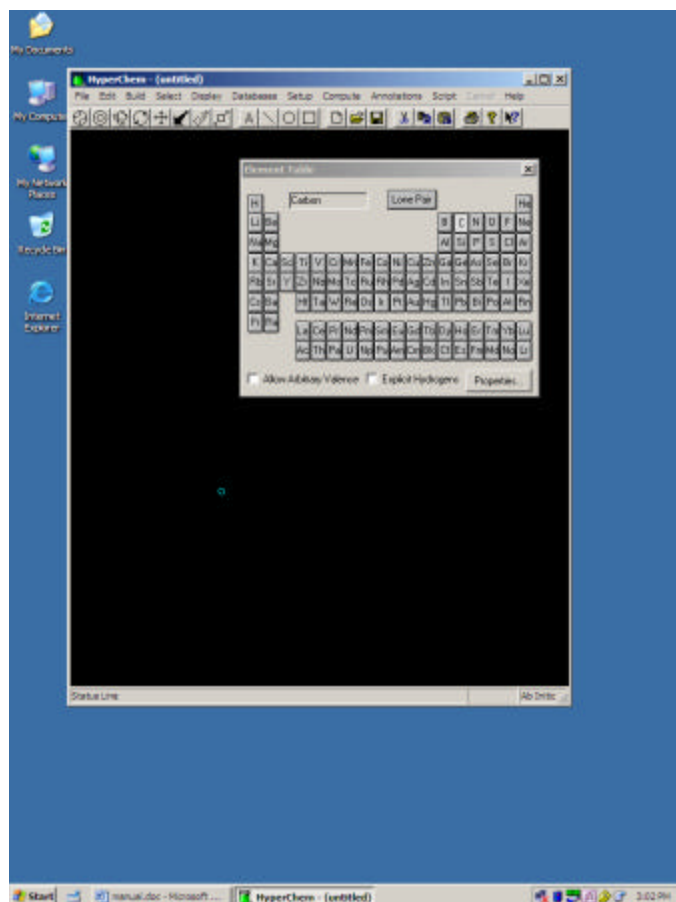
The most important keys are on the bar; although hovering your mouse above them will bring up an explanation, you will find that you quickly learn what they are and their function.



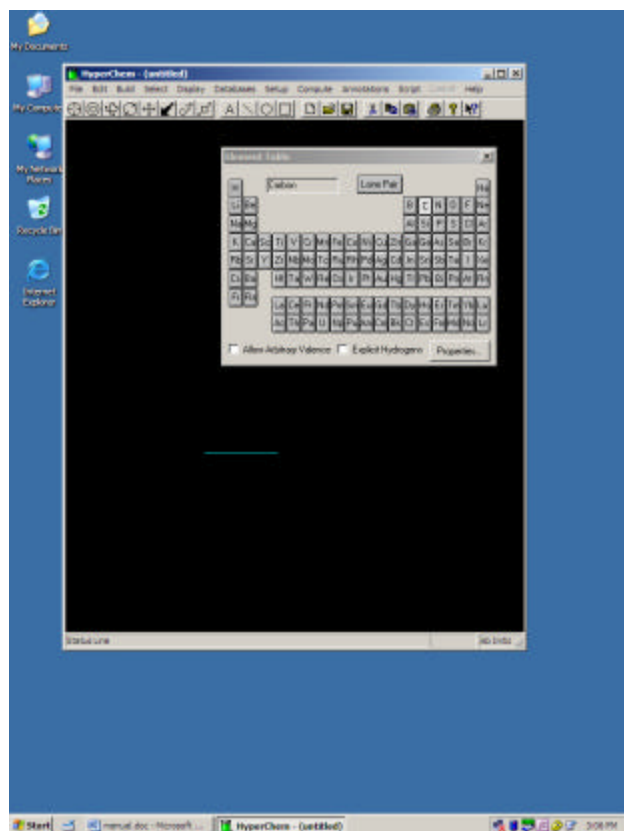
To build a molecule from scratch (say, for example, isopropanol), begin by double clicking on the draw key. This will open up a periodic chart for you (to the best of my knowledge, this is the ONLY way to get the periodic chart to open).



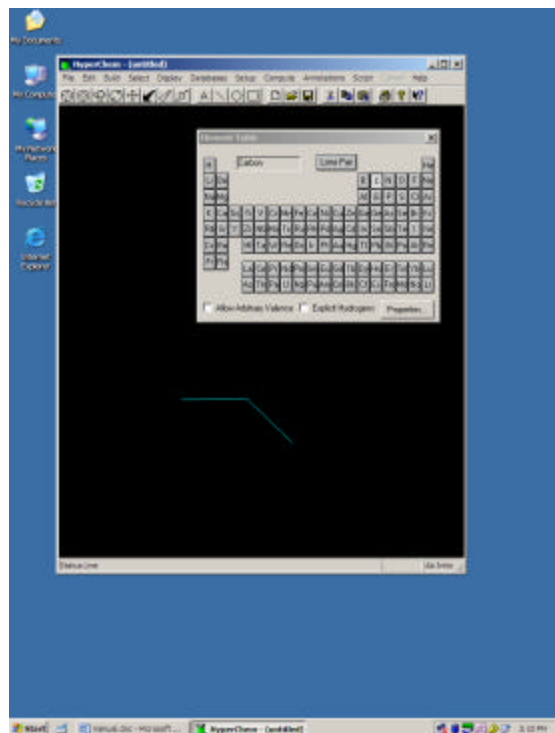
Notice that every element is here (as is on ANY periodic chart). To build our molecule, select carbon (C), and “place” a carbon somewhere in the black area. Notice that you might have to click twice to get the carbon to show (it will appear as a small blue circle); this is because when you click on the periodic chart, that becomes the active window. The first time you click on the black background, it makes the Hyper Chem window the active screen, so the second click is required to place the carbon.



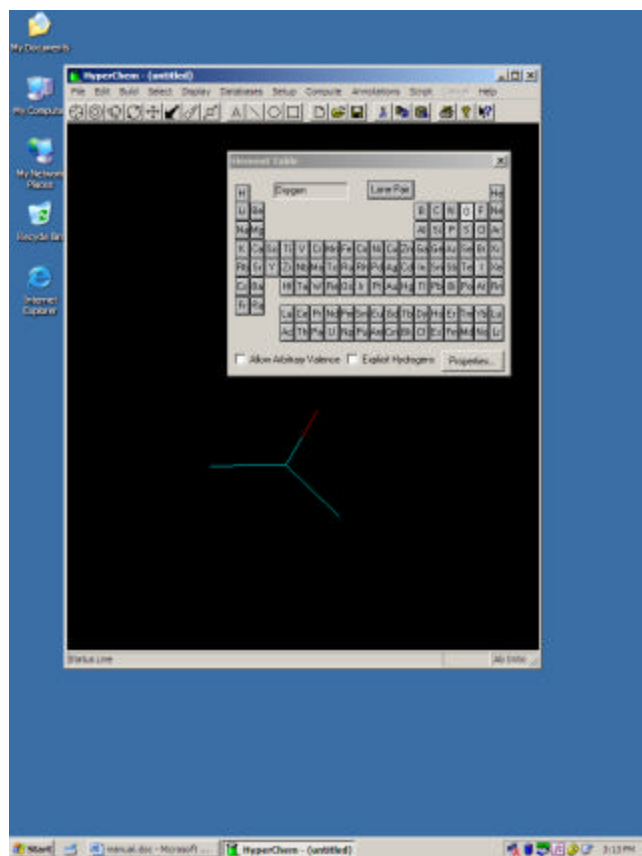
Now, we want to attach a second carbon to our first, so begin by clicking on the carbon already there, and drag a line a little ways to represent the bond. You will find that the circle disappears, and all that remains is the line. This is OK; the default rendering (that is, how the program displays molecules) is a short-hand “stick” form. This line represents the single line between the two carbons, with a carbon on each end. If the line does not draw the first time, try it again (again, if the Hyper Chem window is not the active display, the first time you click on it all it does is activate the window).



Now, we need to add another carbon to our chain of two, so, simply click on one end of the line, and draw another line. Here's a hint, though; make the line at a slight angle, so you can easily see each atom. By doing this, there is a carbon at each end, and one carbon at each bend (in this case, only one bend).

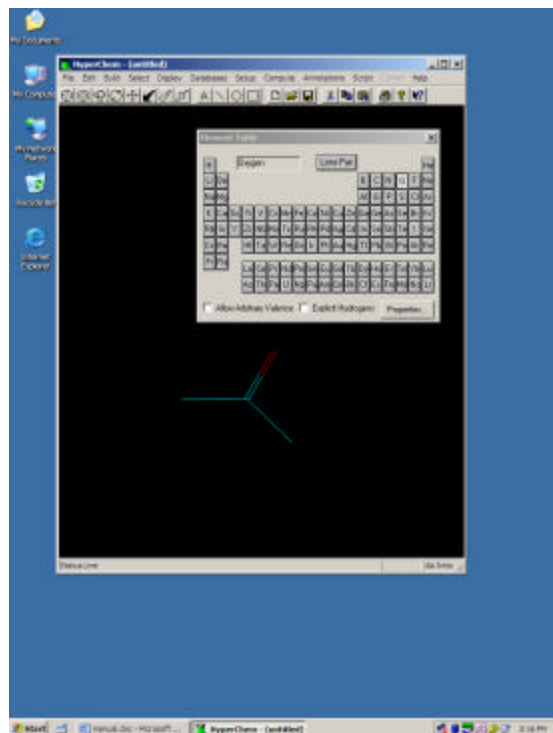


Now, isopropanol (which is present in trace amounts in isopropyl alcohol) has a double bonded oxygen to the central carbon (here, the one in the bend). So, select oxygen in the periodic chart (O), click the bend in our line (the central carbon), and draw a line up for the oxygen. Notice again that we will only get a line, but THIS time, the line is blue on one side (where the carbon is) and red on the other (to represent oxygen on the other side).

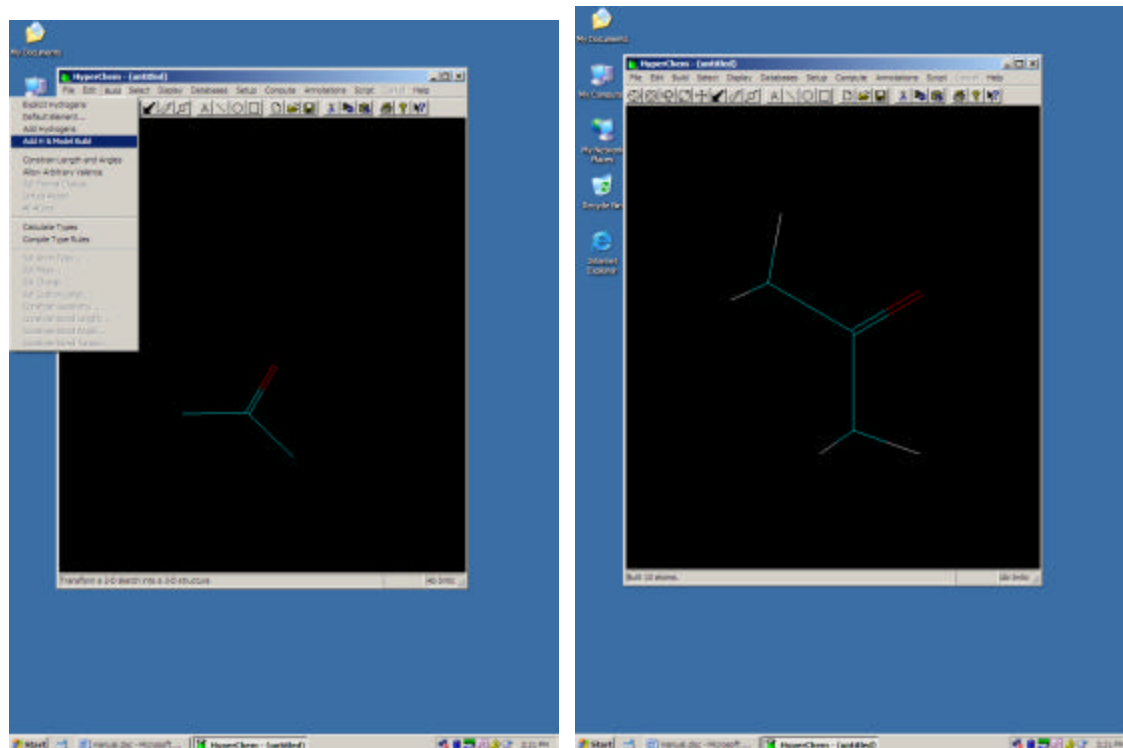


If you accidentally clicked on the middle carbon twice, you will notice that it changed from blue to red. Hyper Chem assumed you wanted to change the identity of that atom (which you did not). No problem, just choose Carbon in the periodic chart and double click the red bend to turn it back to blue; then click on Oxygen, click and drag the line from the middle carbon to put the oxygen on just as before.

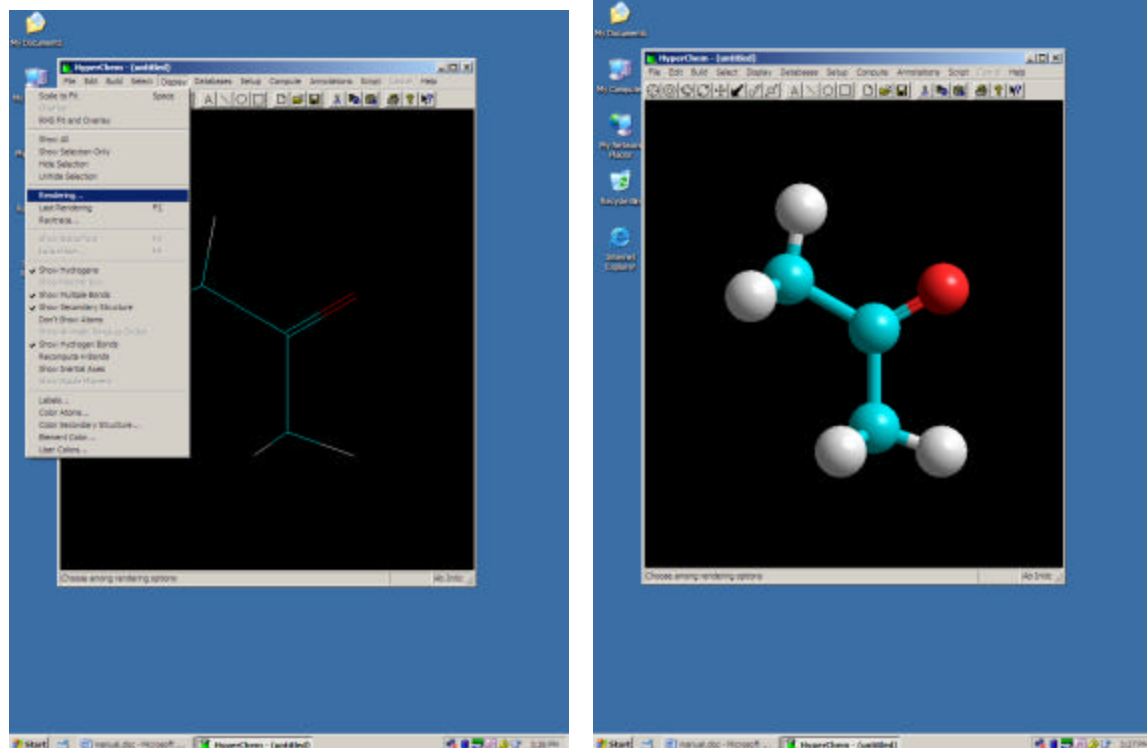
Now, we have a problem. We don't want a single bond between the carbon and the oxygen, but rather, a double bond. Well, this really is not a problem; with oxygen still selected in the periodic chart, just click once in the middle of the line between the carbon and the oxygen. You will see it change from a single line to two lines, to represent a double bond.



Now, isopropanol actually has six hydrogens, but we are not going to add them manually. HyperChem actually has a very nice feature to do this automatically. Close the periodic chart, and under “Build” choose “Add H & Model Build”. This feature automatically completes your molecule by adding hydrogen to any “open valences” (that is, location where it is expecting another bond but there is not one) and selecting the best bond lengths and angles for all of the atoms in the molecule.



Now, I am very comfortable with this shorthand notation since I have been through organic chemistry, but maybe you are not. If you want to change the way the program displays our molecule, simply go to “display” and choose “rendering...”. I recommend “balls and cylinders” which will make the molecule look like it would if you built it out of small plastic balls (like when I was taking chemistry).



If you are following along on these instructions, I recommend now that you choose some of the other tools (begin with rotate in 3D, then rotate in the plane and translate) to see what these do. For example, choose “rotate in 3D”, click and hold the molecule while moving the stylus and you will see it rotate as you do in any direction you like.

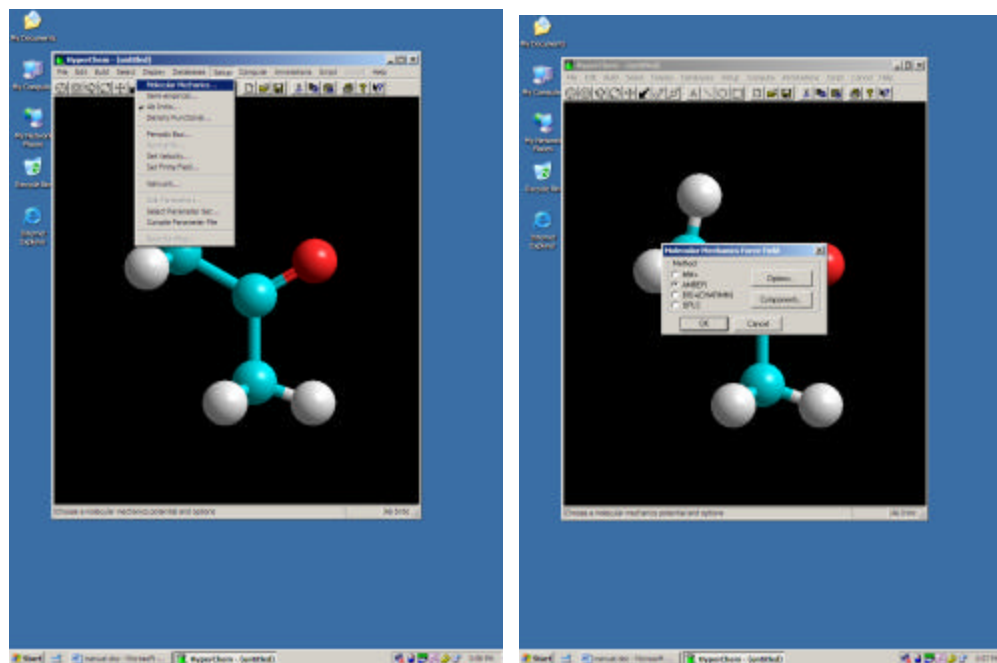
Now, obviously, this technique is suitable only for very simple molecules. If you have a much more complicated molecule, such as DNA, RNA, proteins or crystals, Hyper Chem comes with some very powerful tools built into the program to help you with these. They are found under “Databases”. We will not cover these here because you probably will not need them for this course, however, I invite you to play with them. You will find they are fairly self-explanatory; it is a very simple thing to very quickly build a double-stranded DNA of any sequence that you choose.

Introduction to Hyper Chem calculations

Hyper Chem calculations are basically broken up into three basic steps; (1) define the calculation, (2) run the calculation, and (3) display the results. We will not go into this in great detail here as there are a LOT of possible calculations (an exceptionally impressive amount) that come with this package. However, we’ll run through one simple one so you can see the theory; molecular dynamics. Molecular dynamics (or MD) simulates the motions of molecules by taking into account attraction and repulsion of every atom in the system with every other atom. While it must be remembered that these calculations are only as good as the programmers who developed

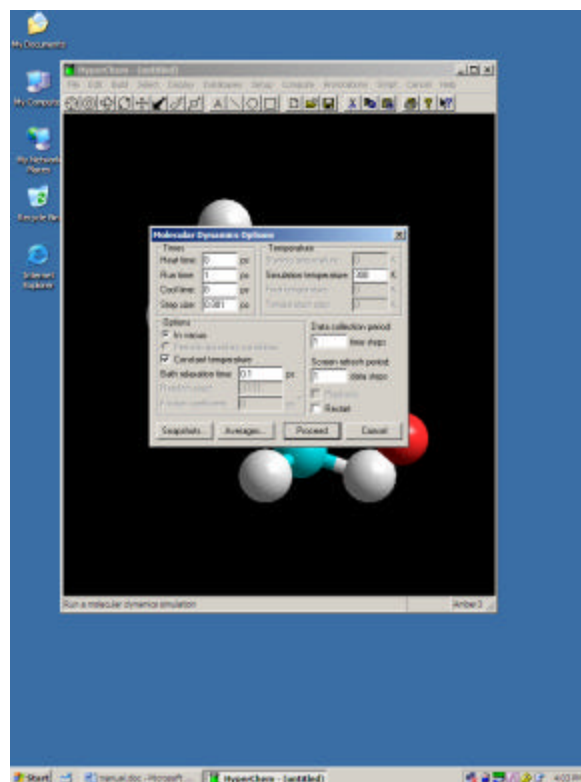
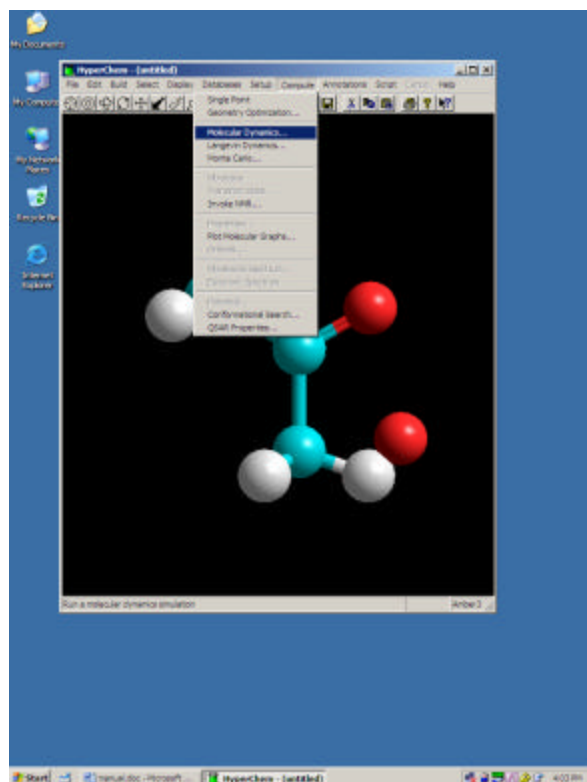
them, through the years they have become so accurate at reproducing experimental outcomes that some scientists today consider MD calculations to be legitimate experimental result in and of themselves (I disagree with this philosophy as I recognize that there will always be factors that we, as human beings, will simply not recognize, regardless of how accurate we try to get the program).

To run MD simulations on isopropanol, begin by going to “set-up”, and choosing “Molecular Mechanics...”. Once in there, select one of the options (I usually choose AMBER, since I am familiar with this potential function).



There are two basic ways to run MD calculations, “In Vacuo” (as I’m sure you guessed, Latin for “In a Vacuum”), or in water. If you want to add water, under “Set Up” choose “periodic box”; I will run this one in vacuo, however, since it often becomes difficult to see the molecule with a lot of water molecules around it.

Step one is complete; the calculation has been set up. Now, step 2; run the calculation. Go to “Compute” and “Molecular Dynamics”; Hyper Chem will remember your choices from the previous step. For example, my MD will run using the AMBER force field. You will notice there are more choices here, such as step size and time of simulation. Typically, the smaller the step size and the longer the run, the more accurate the results, but the slower it will be. For our purposes, I recommend simply choosing the results. You will see the isopropanol begin to dance on your screen; this is a simulation of how the molecule can be expected to actually move, if we could see it so clearly, in reality.



Now, aside from the really cool ways the molecule moves, there is not much more that we need here today. However, while this simulation was running, there were certain properties that were calculated as it ran. Although we don't need them now, if you want to display these properties (the third and final step of the sequence), go to "compute" and you will see several options that were not available before (such as "properties" and "plot molecular graphs"). Feel free to take a look at what is in these if you are curious.

Basic Laboratory Procedures

Laboratory Equipment:

Most basic laboratory equipment is made of glass so one can easily see what is happening inside. When I think of the most common equipment, the beaker and the flask come to mind.



Beakers are probably most commonly used; their wide mouth and spouts make it very easy to transfer solutions from one beaker to another. The flask (or, to be more precise, the “Erlenmeyer flask”) is an excellent choice to run reactions if you do not plan to transfer solutions frequently. The tapered neck of the flask makes it very easy to grab, hold, swirl and manipulate.

Other pieces of common equipment includes the scoopula (for manipulating moderate sized amounts of solids), the spatula (for smaller amounts of solid), The stirring rod and the rubber policeman (on the end of a stirring rod, for scraping crystals out of beakers). For this lab, we will replace the thermometer with a Pasco temperature probe. Rather than droppers, we will use disposable pipettes.



Balance:

The balance is used to determine the mass of an object. Like so many other things today, modern balances just continue to get easier to use. At DSU, we use digital balances. The basic operation of these balances is trivial; place your object on the pan, and the mass appears on the display. However, also like so many modern devices, there are advanced features that may not be so obvious. In this section, I will not only

describe these features, but also the basics of maintenance that must be observed by all users and, finally, how to correctly read the display to avoid errors.

Let's begin with maintenance. Careful use of the balance is critical because of three factors that cross to make the balance, perhaps, the single most critical piece of equipment in the chemistry laboratory today. First, you will learn that there is a close relationship between the mass of a substance, and the number of molecules present. Chemists think in terms of molecules (or, more precisely, moles of molecules), but there is no instrument capable of counting the exact number of atoms or molecules. The next best option is to measure mass, which can easily be converted to and from the number of molecules. This in and of itself makes a good balance, to a good extent, the life blood of a chemist. The second factor is one of simple economics. A good quality balance easily will cost several thousand dollars, while high end ("analytical") balances will cost tens of thousands of dollars. Finally, balances are extremely precise instruments. This means that balances are very easily damaged, susceptible to both mechanical and chemical damage.

Some forms of damage are obvious, such as mechanical damage. If you drop or hit the balance, you can quickly and easily damage the mechanical components that do the work in a balance, especially the "knife edge". Modern electronics balances are also designed to work on a level, draft free surface. These high tech devices still rely on the good old-fashioned low-tech "bubble" leveling device. It is good practice to check the level bubble to ensure that the balance is level before you begin.

Balances are also very susceptible to corrosion. For the reason, you should never weigh any reagent directly on the weighing pan, even if it is a solid. Always use a piece of weighing paper, or a piece of laboratory glassware such as a beaker, flask or a watch glass. If you should inadvertently spill something on the balance, clean it up as soon as possible. Liquids must be prevented from getting inside the balance (paper towels will be near the balances) and solids must be removed as well (a brush works well for this, and will be located near the balance).

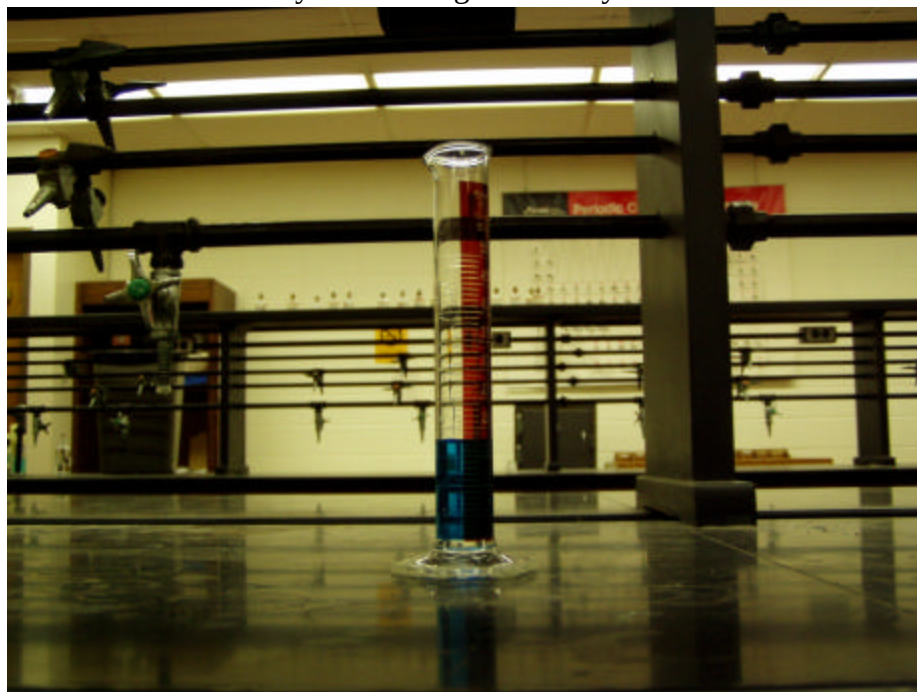
To use the balance, first you must decide what the plan is. This may sound odd, but there are a couple of ways that the balance can be used. Always check the bubble to be sure the balance is level. If you have both the reagent and a container ready that you want to measure the reagent in, then put the container on the balance and press the "tare" button. This will set the balance to "zero", even with the container on the weighing pan. Now, pour your reagent into the container; the mass shown on the scale is the mass of the reagent alone. Remember to add the reagent slowly, so you don't have to remove excess reagent (remember, if you do have to take some of the reagent out, do not put it back into the original container).

Sometimes, you will need the mass of the container so you can take it back to your bench, put something in it, and weigh it again. You can then get the mass of the contents by taking the mass of the beaker and contents and subtracting the mass of the beaker. This is known as "mass by difference." In this case, start by checking to see that the balance is level, and press the "tare" button to set the balance to zero. Place your container on the balance pan, and record the mass of the container. Once the container has the material in it, repeat the procedure (check to see that the balance is level, press the "tare" button to set the balance to zero, place your container on the balance pan, and record the mass of the container and material.)

Finally, just a couple of hints to improve your results when using a balance. First always use the same balance. If the balance is off slightly, these errors will usually cancel themselves out if you are always using the same balance. Secondly, avoid drafts, vibrations or anything that might give you an erroneous reading. The best way to do this is to avoid motion near the balance when it is in use, avoid weighing objects when they are hot, and do not lean on the counter when the balance is in use.

Graduated Cylinders:

Graduated cylinders are used to measure volume. They are the most commonly used devices for volume measurement in the lab because of their accuracy, speed and ease of use. NEVER use the graduations on a beaker or flask for volume measurement; their accuracy is not sufficient for laboratory use. Most graduated cylinders are accurate to three significant figures



(as opposed to flasks and beakers that are accurate to only two significant figures, and burettes and pipettes that are accurate to four significant figures).

Before we begin, it is important to note that graduated cylinders are to be used for measuring volume only. NEVER use the graduated cylinder to mix reagents or to heat a substance!

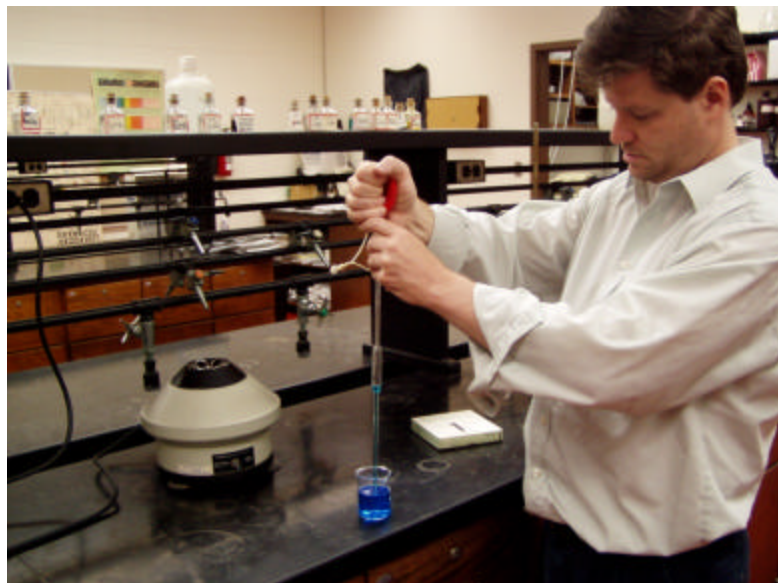
With a liquid in the graduated cylinder, always read the bottom

(or top) of the meniscus. A meniscus is a curvature to the liquid caused by intermolecular forces between the liquid and the glass. If you have attractive forces between the glass and the liquid, such as water, the liquid will “creep up” the sides of the glass slightly to cause the normal downward curvature. If these forces are repulsive, then the liquid will not move up along the walls as far as the liquid, creating an inverted meniscus. Always look past the wall of the glass, and read the volume at the center of the liquid.

Remember to estimate the last significant figure when reading the volume. This means that you simply guess how far in between the two closest graduation lines the top of the meniscus is. If it looks to you like the top of the meniscus is right on one of the graduation lines, then record an extra “0” at the end of the recorded value so the reader knows that this is the case.

Pipette:

Like the graduated cylinder, the pipette is used to measure volume. Unlike the graduated cylinder however the pipette is designed to measure one, and only one, volume, as indicated on the pipette. Never use a pipette that has a chipped or cracked tip, as these are no longer properly calibrated.



Begin by cleaning the pipette according to standard methods. Be especially careful to avoid bumping the tip against the sink or other surfaces. If the pipette is clean, liquid should flow out of it smoothly without leaving spots.

To use a pipette, begin by verifying that it is the correct type of pipette. Read the volume on the side (and record it in your notebook), and verify that it says “TD”, not “TC”. “TD” stands for “To Deliver”, which means that the volume that comes OUT of the pipette is exactly the amount that

the pipette is calibrated for (NOT the amount the pipette will hold). Although “TC” (or “To Contain”; that is, it is calibrated so the amount of liquid actually IN the pipette is the recorded volume) pipettes are rare, they do exist. We will ignore the “TC” procedure, and focus only on the “TD” procedure, as these are the commonly used pipettes today.

The bulbs we use in the chemistry lab have a hard plastic base attached to a rubber bulb. These give the user more finesse and better reproducibility than a mechanically designed device or a “three-port” bulb. Notice that the bulbs we use are NOT designed to fit onto the top of the pipette; they are designed to be placed there and quickly and easily removed.

Begin by holding the pipette vertically. Do not hold the pipette by the “fat” part of the glass; warmth from your finger will cause it to expand, and the pipette will lose its calibration. Instead, hold the pipette near the top (above the calibration mark) so it is easy to get your finger over the top. Squeeze the air out of the bulb (NOT on the buret) and place the bulb on the buret top. Place the buret into the liquid to be drawn up and slowly release the pressure on the bulb, allowing the vacuum created to suck up the liquid. If your bulb completely expands before the liquid is above the calibration mark at the top of the pipette, quickly remove the bulb and cap the top of your pipette with your INDEX finger (not thumb). Squeeze the air out of the bulb, put it back on the pipette, and continue to draw up the liquid.

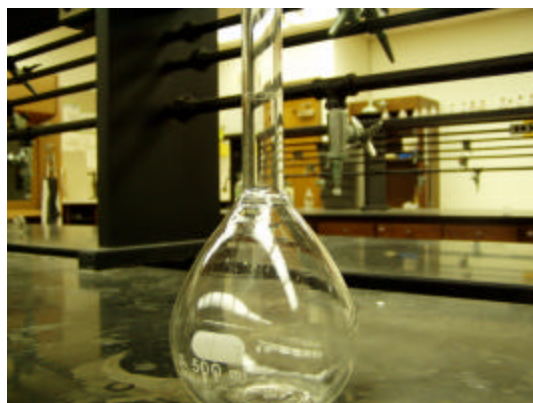
Once the liquid is above the calibration mark, take the pipette bulb off and cap the pipette with your index finger (again, not your thumb; you will have better control and get better results with your index finger). Put the bulb down, and slowly allow the liquid to flow out of the pipette until the bottom of the meniscus is right at the calibration mark. (If you are having trouble controlling the flow, try these few tricks; if you cannot hold the liquid in the pipette, moisten your finger slightly to get a better seal; instead of trying to lift your finger up to get the fluid to flow out, try rolling it slightly to one side instead; if the fluid still is too rapid, CAREFULLY put the tip of the pipette direction onto the bottom of the container with the fluid.)

Once the fluid is at the correct level, lift the pipette out of the fluid, and touch the tip to the side of the container to get any excess drops off. Put the pipette over the container you want the liquid in, and take your finger off of the top of the pipette. Holding the pipette vertically, allow the fluid to flow out on its own WITHOUT trying to force the liquid out. Once the flow stops, touch the tip of the pipette to the side of the container to get any last drops off and remove the pipette. You will notice that a small amount of liquid remains in the pipette; do NOT try to “blow” this last drop out. The pipette is calibrated to keep this amount of liquid in the pipette, so if you blow this last bit out, you have ruined the calibration and do not know precisely how much liquid you have.

If necessary, wash the reagent down the side of the container with a distilled water bottle.

Volumetric Flask:

The volumetric flask is a piece of volumetric glassware (calibrated to four significant figures) designed to contain the volume on the flask. Notice on the neck that there is a single calibration mark; when the flask is filled to this mark (with the bottom of the meniscus), it contains the volume indicated.



There are a couple of tricks that are necessary to make it easier to work with a volumetric flask. First, make sure you have the correct size lid. Plastic tops are probably better than ground glass for general purpose since ground glass tops can easily dry out and get stuck or damaged. The T/S number on the flask should be identical to the one on the stopper, so if the flask reads T/S 19, get a T/S 19 stopper.

If you are dissolving materials in the flask, once you have placed in your reagent, do not fill the flask to the graduation mark initially. Instead, fill the bulb about half full; this will allow you to swirl more vigorously to get the solid to dissolve. If you want to shake the flask, put the stopper on it first. If the solid does not dissolve immediately, add a little more water and continue. Once the solid has dissolved completely, fill the flask to the graduation mark.

Notice that if you stop here, the solution in the neck is not mixed thoroughly with the solution in the bulb. The next step is to put the stopper on it, and hold the flask such that your

palm is holding the stopper relatively firmly on the flask. Invert the flask, and swirl it relatively vigorously. Return the flask to its upright position, allow the fluid to flow out of the neck of the flask, and repeat the procedure. Typically you will want to invert and swirl a minimum of three times.

Finally, avoid handling the volumetric flask by the bulb at the bottom. It is best to handle it with the neck above the graduation mark for the same reason that you want to handle a pipette in an analogous fashion; if you hold the bulb, the glass will warm and expand, thereby throwing off the actual volume.

NEVER heat anything in a volumetric flask, and NEVER run reactions in a volumetric flask. Only store solutions in a volumetric flask on rare occasion and for short periods of time. If you overshoot the calibration mark, do NOT try to backtrack by removing some of the fluid from the neck. While this fluid may not be well mixed yet, it does have some solute in it, and this maneuver will decrease the accuracy of the concentration. Instead, discard the solution, and start over again.

Gravity Filtration:

In chemistry, even filtration is more sophisticated than it might seem. Take gravity filtration; all you do is stick a piece of filter paper in a funnel and let it go, right? Wrong. The reason it is called “gravity filtration” is because we employ gravity to help us out, if we are careful enough.



Begin with a clean long-stem funnel. Place it in an iron ring. Take an appropriate piece of filter paper (on the back of Whatmann boxes, you will find a table of types of filter paper; the slower the paper, the finer the porosity, so the longer it will take, and the smaller the particles it will catch). Fold the filter paper in half, and fold it in half again, but not perfectly; there should be a little angle, about 5° , made from the corner of the second fold when you compare the back of the folded paper with the front.

Tear a small corner off of the front fold; this will help the filter paper to lie more smoothly next to the glass of the funnel so there are no bubbles between the funnel and the paper.

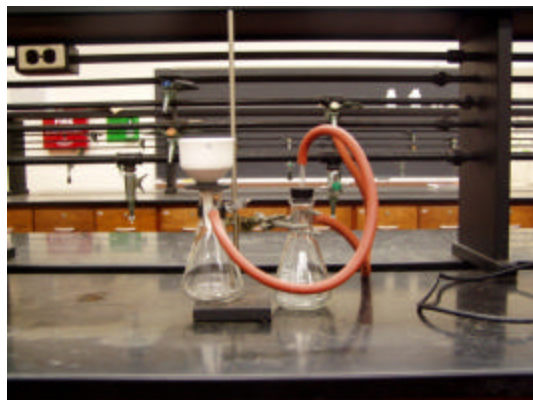
Moisten the filter paper completely with a bit of distilled water and carefully press the filter paper against the funnel. Be very careful to avoid tearing the paper, but you want to be sure there are no bubbles between the filter paper and the funnel. Place a clean receiving vessel underneath the funnel. Add your solution, and allow the solution to filter through the funnel naturally.



If you have set the filtration up correctly, you should see a “plug” of liquid forming in the stem of the funnel. If there are no bubbles between the funnel and the filter paper, this will create a little vacuum that will help the filtration proceed more rapidly.

Vacuum Filtration:

Choose a clean, dry side-armed flask, and secure it to a ring stand so it will not tip over. Attach a piece of vacuum hosing from the side arm, using water for lubrication if necessary, to a water trap. Attach a second piece of vacuum tubing from the water trap to an aspirator or vacuum line. The water trap prevents both liquid drawn from the faucet in an aspirator into the filtrate, and keeps filtrate from accidentally being pulled into a vacuum line. This is an important step even if you do not plan on using the filtrate because, if you need to re-filter, your filtrate will not be contaminated.



Place a collar on the top of the side-armed flask, and a Buchner funnel onto the collar. This collar is not intended to fit snugly; it will be quite loose, but the vacuum, once applied, will draw it in tightly. Be sure that the vacuum works by turning it on and testing to see that a vacuum is created. Turn off the vacuum, and place a piece of filter paper into the Buchner funnel; the paper must not be larger than the funnel, but it must be large enough to cover all of

the holes in the bottom of the Buchner funnel. Moisten the filter paper completely with a little distilled water.

Add your solution to be filtered, and turn on the vacuum. If your crystals are to be washed, first, turn off and break the vacuum by lifting the Buchner funnel slightly. Add the wash liquid to the original container, and pour it onto the crystals in the Buchner funnel. Using a rubber policeman, VERY carefully stir the crystals to break them up and wash them thoroughly, but do NOT tear the filter paper. Re-apply the vacuum. This step is usually repeated three times.

Decanting:



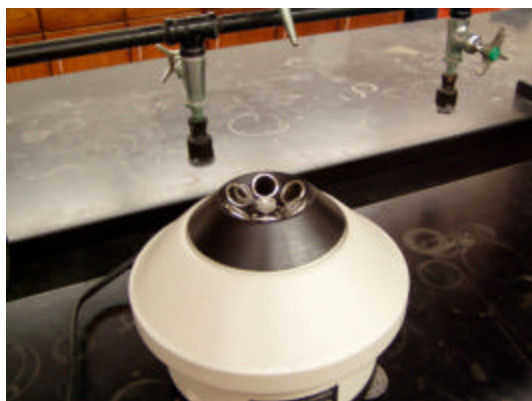
There are times that you would like to separate a mixture, but it is not necessary to do so with extreme care. Decanting is a method in which one can separate liquids from solids in a mixture rapidly, but relatively sloppily. Typically, one begins by centrifuging the mixture; this forces all of the solid to the bottom of the test tube (although if you are decanting from a larger container, say, a beaker, this step is obviously impossible with standard laboratory equipment). Place a stirring rod across the top of the container, not to stop the solid from flowing out, but

rather to help the liquid flow out more easily because it breaks the surface tension which can form without it. Slowly and carefully pour the liquid from the mixture into another container; stop when the solid is about ready to pour out as well.

It must be kept in mind that this is not a good separation technique; it is designed to be fast and crude when this is all that is required. The solid will still have a considerable amount of liquid left on it, and the liquid will have some of the solid in it as well. Of course, do not use this as an excuse to be sloppy. It is really the analysts call as to when the separation is complete; if you try to get too precise with it, you lose the speed (which is the only real advantage), but if you are not precise enough, you may as well not be decanting at all. The only real hint I can give you is to pour slowly and try to avoid agitating the solution.

Centrifuging:

Centrifugation is a process which uses centrifugal force to separate mixtures by density; the more dense material will be on the bottom (typically solids) while the less dense will be on top (typically liquid, although a similar technique is used to separate proteins by biochemists). The most important thing about centrifuging is to balance the centrifuge; put a test tube of the same size and design opposite the test tube to be centrifuged. The test tube can be filled with tap



water if necessary (do NOT dilute another solution with tap water; instead, place it in its own slot and balance the centrifuge). If the centrifuge is not properly balanced, it is very easy to severely damage the centrifuge. If you understand the concept of vectors, you can balance the centrifuge with three test tubes as well; ask your instructor for more information.

Once the centrifuge is balanced, turn it on. Allow it to run for one or two minutes. If the centrifuge begins making a lot of noise, turn it off and check the balance. You must stay with the centrifuge

the entire time it is running, since minor vibrations can cause a centrifuge to “walk” off of the

bench. When you turn it off, allow it to come to a stop itself; NEVER put your hand (or anything else) above the centrifuge.

Ice baths:

Although there are times that one would like to simply pack ice or ice chips around the outside of a container to keep it cool, it is far more common to use an ice bath. Ice baths are easier to work with and tend to be more efficient, since they will make more thorough contact with the container. Start with a bath that is ice, and half or less of water; if too much ice melts (if the ice does not touch the bottom of the ice bath), pour some of the water out and add more ice, since this can create isotherms that are warmer than you would like. If you need the ice bath colder, add salt (usually low grade since it will not be used for a chemical reaction).

Litmus Paper:

Litmus paper will be blue in a basic solution and pink in an acidic solution. Because it is easier to see color changes, use blue litmus paper to test for acidic solutions (it will turn from blue to pink) and red litmus paper to test for alkaline (or basic) solutions (it will turn from pink to blue). Do NOT place the litmus paper into the solution to be tested; instead, put a piece of litmus paper on the desk, and use a clean glass stirring rod to transfer a drop from the container to be tested onto the paper. In this fashion, the same piece of litmus paper can be used many times. Read the litmus paper BEFORE it dries, and never try to use the same spot of a piece of litmus paper twice.

Glowing Splint:

The oldest trick in the book for testing gas is the “glowing splint” test. Depending on your reactants, there are a variety of gases that can possibly be given off; this simple little test helps to distinguish four of these gases: oxygen, hydrogen, carbon dioxide and nitrogen. You perform it by taking a glowing splint, lighting it on fire, extinguishing the fire such that the splint continues to glow, and using the splint to test the gases coming off from the reaction. One of three things can happen.

<i>If the splint...</i>	<i>The gas is likely to be...</i>
Goes out completely	Carbon dioxide or nitrogen
Re-ignites or glows brighter sometimes accompanied by a “pop”	oxygen
Explodes (in the form of a loud “pop”) but does not glow brighter	hydrogen

To perform this test, there are only a few things to keep in mind. **Be sure the test tube or container is set in a holder; not held by you.** If you are holding it and it “pops,” you could be

startled into dropping it. Simply take a glowing splint (looks like a tongue depressor), and start it on fire in a Bunsen burner. Allow it to burn for a few moments, so when you gently blow or shake it out, the wood is hot enough to continue to glow. Put the glowing part of the splint into the top of the test tube or container, but do NOT drop it or allow it to touch the liquid (remember you are testing the gas, not the solution). Note the IMMEDIATE reaction of the splint; if you wait too long, there will not be enough oxygen to support the glow, and the splint will give a false-positive for carbon dioxide or nitrogen.

Remember that the hydrogen test is an explosion; we must keep it contained. **Never use any container other than a test tube unless otherwise instructed to do so.**

Buret:

Like the pipette, the buret is a precision instrument for measuring volumes of liquid. However, the buret is different in two major differences; first, the volume it measures is variable. Secondly, even though the buret is often read to four significant figures as well, it is not quite as accurate as the pipette. Because of human error, there tends to be larger variance in the last significant figure.



Careful inspection of the buret reveals that the volume measurements appear to be “backwards”, with 0 at the top, and the maximum volume (we will use mainly 50 mL burets) at the bottom. This is because the buret is designed to show how much volume has been delivered, rather than how much it contains (like a graduated cylinder, for example).

There is a special clamp and ring stand for use with burets. Because burets are primarily used for titrations, and titrations usually require the ability to see a color change (indicating the endpoint), when you use a buret, you want to take a ring stand with a white ceramic base. Avoid using these ring stands for anything else, because the more stained they become, the harder they are to use. The clamp has two positions for burets, and is designed to hold the buret(s), and to be very easy to take the burets out of the clamp. This is for safety reasons; you must never fill a buret while it is in the clamp.

Notice the valve at the bottom. The valve is open when it is parallel with the buret (vertical), and closed when it is perpendicular (horizontal). Begin by cleaning the buret by standard methods. To fill the buret, take the buret out of the clamp. Check the valve to be sure it is closed. Pour the reagent from a beaker into the top of the buret. A funnel may be used if necessary, but is discouraged



because of mechanical difficulties this tends to produce. As you are filling the buret, at some point, pause and look at the tip to be sure the liquid is not pouring out of the bottom. If it is, take necessary action immediately to contain the reagent and clean up the spill. Fill the buret slightly above the “zero” mark and return it to the buret clamp.

The buret is still not quite ready to use, because the tip of the buret is probably still filled with air. If you try to measure the volume of liquid now, you will think you’ve added more reagent than you actually have because this volume of air will show in the reading. Put either the reagent beaker, or, better still, a waste beaker, under the buret tip and open the buret tip fully to expel the air and fill the tip with reagent. Sometimes it helps to “flick” the buret tip to dislodge any stuck bubbles. When the tip is full of reagent (no air remaining), close the stopcock, and check to be sure that the volume in the buret is now BELOW the zero mark. Do not waste time trying to get the volume EXACTLY to zero; it does not matter what volume you actually have, as long as you can read it (which is why it must be below zero).

Record the starting value; remember to estimate the volume to the nearest 0.01 mL (one more significant digit than the graduations on the buret). Follow the titration procedure. If you are right handed, the correct way to use the buret is as follows: use an Erlenmeyer flask (rather than a beaker) for the titration. Once you have added the titrant (the chemical to be titrated) to the flask, you can add additional distilled water as needed since this will not change the amount of the chemical already in the flask. This is convenient for washing down droplets as they splash onto the sides of the flask.

There is a proper, and an improper, way to add reagent from a buret. The proper way feels a little bit cumbersome at first, but will give you better results. Begin by noting if you are right or left handed; I will refer to them as your “dominant” and “secondary” hand. Position the buret such that the stopcock control is on the same side as your dominant hand, and the scale is facing you. The burets we use all have stopcocks that can be twisted around to accommodate your dominant hand; be careful to do this before filling the buret, though, so the stopcock does not fall out causing a chemical spill.

Now comes the part that throws most students. Even though the buret now looks like it is set up to be controlled by your dominant hand, you will actually use your secondary hand to handle the stopcock; your primary hand will be used to swirl the flask. Reach your secondary hand around the barrel of the buret and the stopcock to control the flow from the buret, and use your primary hand to swirl the flask. We do this so the tendency is to pull the stopcock in tighter, rather than looser, so we don’t have to contend with a leak halfway through a titration. This is not a two-person operation; the same individual who is controlling the flask also controls the buret, and reads the volumes on the buret.

Once the endpoint is reached, read the final volume in the buret. The volume added to the flask is this volume minus the initial volume. You need not refill the buret after each run; if you still have enough liquid in the buret for a second run (which should have the same volume as the first run), you can just use the liquid that is already in the buret. Only refill the buret if necessary (and remember; do NOT refill the buret in the buret clamp!). If you underestimate the volume in the buret, do not try to “save” the run by letting the liquid run below 50 mL in the buret, or by adding liquid mid-titration. These are designed to be exceptionally high precision

experiments; either of these techniques will introduce undue error into your calculations. Just refill the buret, and do another run.

Cleaning laboratory Equipment:

Finally, we come to the general process of cleaning laboratory equipment. This is critical; contaminants can have a profound influence on an experiment, and since equipment is shared, you will want to be sure the glassware is clean from the other groups. Heck, you don't know what they did! So how does one go about cleaning laboratory glassware? Easy; you make your lab partner do it.

But what if YOU are the lab partner? OK, well, it's really not that bad. The first thing you will want to recognize, however, is the simple fact that glassware is easiest to clean before the gunk has the opportunity to dry into it. However, the same basic rules apply even if it is old. Begin by rinsing very well with tap water (surprised?). You will find an assortment of brushes near each sink; rinse out as much of the bulk of the contaminants as possible. Then, still using tap water, add Alconox. Alconox is a white cleaning agent at each sink; it is a detergent made BY chemists FOR chemists. This is good stuff; it's like Lava soap with an attitude. Use the brush to clean the glassware thoroughly; if you are cleaning something that cannot be cleaned with a brush (like a pipette); dissolve a little Alconox in a beaker of water and run it through as best you can.

After cleaning with Alconox, rinse the glassware VERY THOROUGHLY with TAP water to get rid of all of the excess soap. If you are not happy with how clean it is, repeat the step with Alconox. Finally, rinse it THREE TIMES with SMALL amounts of distilled water (in the large carboys near each sink). Do NOT allow the distilled water to run the entire time during the rinsing process; open the spicket just long enough to allow a small volume (maybe 5 mL) out, and close it again. Rinse thoroughly and repeat two more times. It does take time to make distilled water, so if we run out during a lab, there is nothing I can do about it. If you are working with something like a pipette, run distilled water through it using a wash bottle.

If the glassware is clean, the water should form a smooth sheath as it runs off of the sides. If there are any spots where a "bubble" forms, you might want to re-clean the glassware. At this point, decide if you need to dry the glassware or not. If you are washing it for next week, it will dry without spots after the distilled water rinse. If you are going to put water based solutions into it, you might not have to dry it. If it is a qualitative lab and the exact concentration of the reagent is not so important, you can pour the reagent directly into the wet container. If, on the other hand, it is a quantitative lab, you might want to repeat the rinse cycle with even smaller volumes (say about 1 mL portions) of the reagent you will use. For example, draw a little of the reagent up into the pipette, wet the entire inside of the pipette, and repeat three times, discarding each wash. On the other hand, if you are working with something like an organic solvent, you will want to dry the container with paper towels before proceeding.

One of the easiest way to dry a test tube is to begin with a paper towel twisted small enough to fit inside. This in and of itself may not sound terribly insightful; however, once you put the paper towel inside the test tube, twist the paper towel in the opposite direction of the twist

you used on the paper towel itself. This will force the paper towel to open up, and, therefore expand, to dry the test tube.

Experiment 1: Density of a Liquid and a Solid**Purpose**

- (1) To practice the procedures commonly used in a laboratory
- (2) To learn how to use some of the common laboratory devices
- (3) To distinguish between chemical and physical properties

Pearly

Having characteristics of pearls

Background:

See “Basic Laboratory Procedures”: pipette, graduated cylinder, balance

Introduction

In just about any lab manual you look at, you will notice that the first experiment is something like the determination of the density of various objects. The reason for this is quite simple to understand: the author wants you to learn how to use some of the basic pieces of the lab while performing an experiment that is relatively safe. In time, you will be performing experiments that do, out of necessity, have inherent dangers, but before you do, you want to be comfortable with your own laboratory skills through practice.

That is really what this experiment is all about. You will be performing a series of relatively simple procedures, but as you do, keep in mind that these are skills and tools you will need in future experiments, so be sure to get any questions that arise answered, and be sure to take many notes and observations for yourself for future reference, especially potential problems and thing to watch out for when there techniques show up again. Remember to refer back to laboratory procedures for any techniques you do not know.

Procedure:**Part I: Density of a Liquid**

You will measure the density of the same liquid three times to demonstrate the difference in various techniques for measuring the volume. In a clean, dry beaker of an appropriate size, get approximately 30 ml of the unknown liquid and bring it to your desk. Get the dry weight of a second clean and dry container. Using a pipette, put 20.00 ml of the unknown liquid into the second beaker. Determine the mass of the second beaker with the liquid in it.

Return the liquid to the first beaker, and dry the second beaker. Using a clean and dry graduated cylinder, measure out 20.0 ml of the unknown liquid and put it into the second beaker. Again, get the mass of the second beaker with the liquid in it.

Once again, return the unknown liquid to the original beaker. Measure out 20. ml of the unknown liquid using an Erlenmeyer flask. Determine the mass of the beaker with the liquid in it.

Dispose of the unknown liquid as instructed.

Part II: Density of an Unknown Solid

From time to time, a chemist has to be clever enough to find an indirect method to measure some quantity. For example, how would one go about measuring the volume of an unusually shaped solid. Archimedes faced this problem when he had to determine the density of a crown for the king in order to determine whether or not the blacksmith stole some of gold and substituted copper for it. To do so, he used water displacement to determine the volume of the crown, as you will do for this part of the experiment.

Get a solid object from your instructor. Determine its mass on an electronic balance. Choose a graduated cylinder of an appropriate size. Fill it approximately half full with water. It is not important to fill it to exactly half, but it is important to determine exactly what the initial volume is. Once you have recorded the volume, carefully lower the solid into the graduated cylinder so as to avoid splashing the water or breaking the graduated cylinder. Once you have recorded the final volume, dry the solid and return it.

Part III: The volume of a grain of sand

Now let's try our hands at critical thinking and experimental design. I have a very simple question for you; what is the average volume of one grain of sand? Talk with your lab partners and come up with a few approaches to answering this problem. Then, discuss each procedure in turn, critically analyzing it for weaknesses and difficulties. Decide upon a best procedure and give it a try. You will be graded on accuracy.

Calculations:

Part I:

For each of the three trials, determine the mass of the liquid by subtracting the mass of the container from the mass of the container and liquid. Divide the mass of the liquid by the volume (20 mL) to determine the density.

Part II:

To determine the volume of the object, subtract the volume of the liquid in the graduated cylinder from the volume of the liquid in the graduated cylinder with the object. To get density, divide the mass of the object by its volume.

Data and Observations

Part I:

	Pipette	Graduated cylinder	Flask
Mass of empty beaker			
Mass of beaker and liquid			
Mass of Liquid			
Density of liquid			

Observations:

Part II:

Mass of object	
Volume with object	
Volume without object	
Volume of object	
Density of Object	

Observations:

Pre-Lab Questions

1. Why are we measuring the density of the same liquid using three different techniques?
2. How do we determine the volume of an oddly shaped solid?
3. What volume of liquid are we using to determine the density?
4. Who developed the method of volume by water displacement?

Post-Lab Questions

1. What is the quickest method for determining the volume of a liquid? Which is the most accurate?
2. How do your densities compare with the three methods of volume determination from part I for the liquid?
3. How would you measure the volume of a sample of sand?
4. For each of the following, what technique would you choose for measuring the volume?
 - (a) You want to take 50 mL of a reagent from the area that it is stored to your desk
 - (b) A titration requires 10.00 mL of a reagent measured as accurately as possible
 - (c) A synthesis requires 35 mL of acid

Experiment 2: Compound types

Purpose: To examine the difference between ionic and covalent compounds and understand how their properties give rise to this categorization

Percolate: To bubble, usually as a result of applied heat

Background:

See “Using the Pasco System”

Introduction:

It is amazing what early chemists accomplished even without an understanding of atomic make-up. I love reading old chemistry textbooks to see how they justified some of their conclusions, which, with a few exceptions, were right on target. I even have one book that discusses the octet rule in terms of “valencies”, and have it exactly right. What is interesting about this is that at the back of the book, they talk about this new sub-atomic particle that they are tentatively calling the “electron”.

Another thing they had correct was the categorization of compounds into “covalent” and “ionic”. In class, we discuss these compounds in terms of electrons, wherein ionic compounds transfer electrons and covalent compounds share electrons. How did the early chemists classify compounds, though, when they did not know what electrons were?

They used properties, such as solubilities, melting points, and conduction. Solubility helps us to classify compounds as polar or non-polar, because, as a general rule, polar solutes dissolve in polar solvents (like water), while non-polar solutes dissolve in non-polar solvents (like oils). Conductivity means whether or not a compound will conduct electricity when it is dissolved in water. We call these “electrolytes”, which are just like regular electrolytes, but with 1/3 fewer calories. An electrolyte will conduct electricity when dissolved in water, while a non-electrolyte will not. Finally, ionic compounds tend to have higher melting points than covalent compounds. These are summarized as follows:

	Non-polar Covalent	Polar Covalent	Ionic
Solubility	non-polar solvents	polar solvents	polar solvents (or generally not soluble)
Conductivities	Non-electrolytes	Non-electrolytes	Electrolytes (even if apparently not soluble)
Melting points	very low	low	high

Procedure:

You will find a series of solid compounds. Run the following tests to decide if each compound is ionic, polar covalent or non-polar covalent based on the above table. Begin by taking very careful observations of each compound, and run each of the following tests on each compound.

Solubility:

You will need two clean test tubes (one of which is dry) for each of the unknown solids. Make sure that these test tubes are cleaned very well, and rinsed very thoroughly with distilled water. Any contamination from tap water or other sources will seriously affect your conductivity experiment.

Put about 1 mL of water (a polar solvent) into one series of test tubes, and 1 mL of the non-polar solvent (probably Hexane) in the other series of test tubes. Place just enough of each solid into one water and one non-polar solvent test tube. Agitate each test tube by flicking it several times while holding it such that it does not fly out of your hands. Note whether or not the solid dissolved completely, dissolved partially, or did not appear to dissolve at all. Record your observations.

“Partially dissolved” means that it is apparent that there is not as much of the solid in the test tube remaining as you put in initially, but there is still some solid left. If the amount of solid did not seem to decrease, it is “insoluble”. To be truly dissolved, the solution must be clear (not necessarily colorless, but clear). If it appears cloudy, then there is still undissolved solid in the test tube deflecting the light (called the “Tyndall effect”).

You may discard the solutions with the non-polar solvent according to the instructions provided in lab. Keep the water solutions for the next step.

Conductivity:

Whether the solid appears to have dissolved or not, perform this conductivity test on each water solution. If a Pasco probe for conductivity is available, connect the probe to your computer and set it up according to the standard Pasco procedure. You will want the “digits” to be displayed, and probably a table. Set up the Pasco software for manual data collection, so you can type in the sample number for each test. If the Pasco probe is not available, you will find a multimeter set up with conductivity available.

Rinse the probe tips off very well with distilled water (you need not dry them). Rinse of a clean watch glass with distilled water as well. Pour a little bit of the liquid from the test tube onto the watch glass; if solid remains in the test tube, you need not include the solid in the quantity you pour out. Place the probes into the water solution on the watch glass, and give the probe a few seconds to equilibrate. Once equilibration seems to have been reached, record the conductivity for that solution. Discard the solutions according to the instructions given in lab, and remember to clean the probes and watch glass and rinse them all off with distilled water very well. Repeat for each water solution.

Melting Points:

Here we are not interested in absolute melting point temperatures, but rather, relative melting points. Place a VERY SMALL AMOUNT of each solid (just enough to see it) onto a scoopula, close enough that they can be viewed and heated more or less at the same time, but far enough apart that you can easily remember which is which. Light a Bunsen burner, and, being very careful not to burn yourself, pass the scoopula through the flame several times. Note the order of melting (which melts first, second, etc) and whether or not any appear to burn rather than melt. Record your observations. Remember that the compounds that melt first have the lower melting points. You might not be able to get all of the compounds to melt; these are all very high melting point compounds.

Calculations:

Based on the results from the procedure, categorize each compound as non-polar covalent, polar covalent, or ionic.

Data and Observations:General Observations:

Compound Number	General Observations

Solubility:

In the following table, record whether each substance is soluble, partially soluble or insoluble in each of the solvents.

Compound Number	Polar Solvent	Non-Polar Solvent

Observations:

Conductivity:

Compound Number	Conductivity

Observations:

Melting Point:

In the following table, record which melted first, second, third, etc. in “order of melting”

Compound Number	Order of Melting	Observations

Observations:

Classification:

In the following, record for each compound if you believe it to be non-polar covalent, polar-covalent or ionic.

Compound Number	Compound Type

Pre-Lab Questions:

1. Describe briefly how we are determining the order of melting.
2. What test are we using to determine if a compound is an electrolyte or a non-electrolyte?
3. If you have a compound that is not soluble in hexane or water, and is an electrolyte, what kind of compound do you expect you will have?

Post-Lab Questions:

1. Is it possible to have an ionic compound that is not water soluble?
2. Is it possible for an electrolyte to be covalent?
3. Is it possible to have an insoluble electrolyte? What does this imply about the term “insoluble”?

Experiment 3: Chemical Reactions

Purpose: To gain experience with the various types of possible chemical reactions.

Perturb: To agitate or disturb

Background:

See “Basic Laboratory Procedures”: pipette, graduated cylinder, balance, glowing splint and Bunsen burner

Introduction:

Chemistry doesn't get really interesting until we begin talking about how matter interacts with each other and changes; that is, until chemical reactivity. As with anything else, chemists classify reactions into several different categories, as summarized below.

<i>Reaction Type</i>	Number of Reactants	Number of Products	Symbolic reaction	<i>Example</i>
Addition	2 or more	1	$A+B \rightarrow C$	$4 \text{ Fe (s)} + 3 \text{ O}_2 \text{ (g)} \rightarrow 2 \text{ Fe}_2\text{O}_3 \text{ (s)}$
Decomposition	1	2 or more	$A \rightarrow B+C$	$\text{C}_6\text{H}_{12}\text{O}_6 \text{ (s)} \rightarrow 6 \text{ C (graphite)} + 6 \text{ H}_2\text{O (g)}$
Single Replacement	2	2	$AB+C \rightarrow CB+A$	$\text{Cu (s)} + 2 \text{ AgNO}_3 \text{ (aq)} \rightarrow 2 \text{ Ag (s)} + \text{Cu(NO}_3)_2 \text{ (aq)}$
Double Replacement	2	2	$AB+CD \rightarrow AD+CB$	$\text{NaCl (aq)} + \text{AgNO}_3 \text{ (aq)} \rightarrow \text{NaNO}_3 \text{ (aq)} + \text{AgCl (s)}$

There are several means by which one can tell if a chemical reaction has occurred. Essentially, we are closely looking for observations to denote that something has happened. They can be obvious, like a color change, evolution of a gas (by “effervescence”, or fizzing), evolution of a smell, or the formation of a precipitate (a solid that forms when two clear solutions are mixed together). However, some of these observations can also be very small, such as evolution or consumption of heat (the container gets hot or cold), or changes in acidity (pH).

The oldest trick in the book for testing gas is the “glowing splint” test. Depending on your reactants, there are a variety of gases that can possibly be given off; this simple little test helps to distinguish four of these gases: oxygen, hydrogen, carbon dioxide and nitrogen. You perform it by taking a glowing splint, lighting it on fire, extinguishing the fire such that the splint continues to glow, and using the splint to test the gases coming off from the reaction. One of three things can happen.

<i>If the splint...</i>	<i>The gas is likely to be...</i>
Goes out completely	Carbon dioxide or nitrogen
Re-ignites or glows brighter	oxygen
Explodes (in the form of a loud “pop”)	hydrogen

In writing a chemical reaction, it is important to remember the law of conservation of matter; one cannot create or destroy matter, but it can change form. From Dalton's Law, we know that we cannot change the subscripts of the chemicals, because that would change what the chemical is. What we can change is the numbers in front of the chemicals in a chemical reaction (the “stoichiometric coefficients”); these can be used to balance a chemical reaction (the same number of each type of atom on each side of the reaction).

An unbalanced chemical equation:

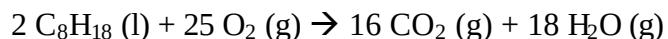


C: 8 on reactant side, 1 on product side

H: 18 on reactant side; 2 on product side

O: 2 on reactant side; 3 on product side (notice it is in both compounds on the product side)

The same chemical equation balanced:



C: 16 on reactant side, 16 on product side: balanced

H: 36 on reactant side; 36 on product side: balanced

O: 50 on reactant side; 50 on product side: balanced

(Notice that in a balanced chemical equation, the stoichiometric coefficients are the smallest possible whole numbers that balance the equation; remember to take out any common denominators from your final answer and leave no fractions.)

Sometimes a chemical is added to speed up a reaction. If these chemicals are not consumed in the reaction (that is to say, there is exactly as much of the chemical after the reaction as we had before the reaction), then the chemical is called a “catalyst”. For example, your car has a catalytic converter to speed up the reaction of carbon monoxide (a poisonous gas) and oxygen to form carbon dioxide (a less toxic gas). The catalyst is platinum; if the platinum were consumed in the reaction, we would periodically have to pay to have new platinum put into our cars; how much would you enjoy that? However, because the platinum is a catalyst, exactly the same amount of platinum is in your car today as when you first bought it. These chemicals either do not show up in the chemical equation at all, or they are placed above the arrow and are not balanced with the rest of the reaction.

Experimental Procedure:

WARNING: Wear your safety goggles, safety aprons, perform all tests carefully and follow any additional warnings that follow!!!!!!!!!!!!!!

REMEMBER! Be SURE to use CLEAN EQUIPMENT that has been THOROUGHLY rinsed with distilled water at least three times.

Preliminary steps:

Although we will not get to the single replacement reactions for a bit, we'll start a couple early so they have time to react as these tend to be slow reactions.

1. Copper (Cu) Metal in Silver Nitrate (AgNO_3)

After taking careful observations of both the copper metal and the 0.4 M silver nitrate solution, place the copper metal in about 1.0 mL of the solution in a SMALL test tube. Set aside and allow it to react undisturbed until called for later in this experiment.

2. Zinc (Zn) or Aluminum (Al) and Copper (II) Sulfate ($\text{Cu}(\text{SO}_4) \cdot 6\text{H}_2\text{O}$)

After taking careful observations of both the zinc or aluminum metal and the 0.4 M copper (II) sulfate solution, place the metal in about 1.0 mL of the solution in a SMALL test tube. Set aside and allow it to react undisturbed until called for later in this experiment.

Addition Reactions:**1. Cyclohexene (C_6H_{10}) and Bromine (Br_2) **DEMONSTRATION ONLY****

In the fume hood, you will see Cyclohexene and Bromine. Carefully take observations regarding these two reagents. Decide what color the mixture of the two should be if no reaction were to occur. (Your professor might be kind enough to show a mixture of a hydrocarbon that will not react with bromine as an example)

Your brave and illustrious professor will add some of the bromine to the cyclohexene in the fume hood. Take observations. Did a chemical reaction occur?

2. Magnesium (Mg) and Oxygen (O₂)

Get crucible tongs (NOT coated with rubber) and a watch glass ready. Light a Bunsen burner. Take observations on a SMALL piece of magnesium. Holding the magnesium with the tongs, ignite the magnesium in the flame. **WARNING! Do NOT look directly at the burning magnesium!** Once the reaction is complete, put the burnt magnesium into the watch glass and allow it to cool a few minutes. Take careful observations on the product.

3. Anhydrous Copper (II) Sulfate (CuSO₄) and Water

Put a SMALL AMOUNT (about the size of a pea) of anhydrous copper (II) sulfate into a small test tube. Take careful observations on it. Add one drop of water to the copper (II) sulfate (just enough to wet it; not enough to cover it completely). Take observations on the final product.

Decomposition Reactions:1. Sucrose **DEMONSTRATION ONLY**

In the fume hood, you will see table sugar (C₆H₁₂O₆, also known as “sucrose”) and concentrated sulfuric acid (H₂SO₄). Carefully take observations regarding these two reagents.

Your poor underpaid and greatly misunderstood professor will pour sulfuric acid onto standard table sugar. Take careful observations as the reaction occurs. What is the “smoke” that is given off? (HINTS! First of all, sugars are often referred to as “carbohydrates”, because they all have the formula C_n(H₂O)_n and sulfuric acid is a dehydrating agent.) What is the product left behind? Balance the equation.

2. Hydrogen Peroxide

Take a SMALL AMOUNT of hydrogen peroxide (about half a milliliter) and put it into a SMALL test tube. Take careful observations. Be sure you have a glowing splint handy and a lit Bunsen burner nearby (to light the glowing splint). Add a SMALL AMOUNT of Manganese Dioxide (a catalyst). Quickly ignite and extinguish your glowing splint and test the gas given off. What is the gas? Balance the equation.

3. Sodium Bicarbonate (baking soda)

Take a small amount of sodium bicarbonate and put it into two test tubes distributed approximately equally. Vigorously heat one of the two test tubes for several minutes and note any observations. During the heating, test for gases released using the glowing splint test. Allow the test tube to cool.

When the heated test tube is cool, add about half a milliliter of water to both the heated and the unheated test tubes. Test the pH of both test tubes using pHHydron paper. There were two gases released in this reaction; what do you think they were? Balance the equation.

4. Potassium Perchlorate

In a test tube, place about one scoopula full of potassium perchlorate (KClO_3) and a little bit (about a pea size) of manganese dioxide (a catalyst). Shake the test tube to mix the contents. Using proper procedures, heat the test tube and its contents over a Bunsen burner and note the reaction. Test the gases in the test tube with a glowing splint.

Single Replacement Reactions:1. Copper (Cu) Metal in Silver Nitrate (AgNO_3) (conclusion)

Take careful observations of the test tube you prepared earlier with the copper metal and the silver nitrate solution. If copper has a +2 charge, determine the formulas of the products. Balance the equation.

2. Zinc (Zn) or Aluminum (Al) and Copper (II) Sulfate ($\text{Cu}(\text{SO}_4) \cdot 6\text{H}_2\text{O}$) (conclusion)

Take careful observations of the test tube you prepared earlier with the zinc or aluminum metal and the copper (II) sulfate solution. If zinc has a +2 charge (you should already know the charge of aluminum if this is the reagent you used), determine the formulas of the products. Balance the equation.

Double Replacement Reactions:

1. Hydrochloric acid (HCl) and sodium hydroxide (NaOH)

The reaction between an acid and a base is called a “neutralization reaction”, and typically results in the formation of some type of salt and water. Add about half a milliliter of 2.0 M HCl to about half a milliliter of 2.0 M NaOH. Note any temperature changes that occur. Balance the equation.

2. Reactions of Salt Solutions

You'll find a table for this section of the experiment listing several salts (use 0.4 M salt solutions). Fill in the table, mixing 2-3 drops of each salt solution with every other solution in the table in separate SMALL test tubes (that is, one test tube for each salt solution pair). If there is no noticeable change after careful observation, write N.R. ("No Reaction") in the table. Balance all chemical equations; if no reaction seemed to occur, simply write "N.R." as the product.

Data Page*Copper and Silver Nitrate Solution:*

<i>Reactant</i>	<i>Observations</i>
Copper	
Silver Nitrate	

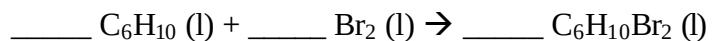
Zinc or Aluminum in Copper (II) Sulfate Solution:

<i>Reactant</i>	<i>Observations</i>
Zinc or Aluminum	
Copper II Sulfate	

Cyclohexene and Bromine:

<i>Reactant</i>	<i>Observations</i>
Cyclohexene	
Bromine	

Observations of the product:



Magnesium and Oxygen:

<i>Reactant</i>	<i>Observations</i>
Magnesium	
Oxygen	

Observations of the product:

_____ Mg (s) + _____ O₂ (g) → _____ Mg_ O_ (s) (HINT! Using charges, figure out the chemical formula for the magnesium oxide first.)

Anhydrous Copper (II) Sulfate and Water:

<i>Reactant</i>	<i>Observations</i>
Anhydrous Copper (II) Sulfate	
Water	

Observations of the product:

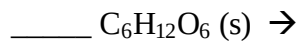
_____ CuSO₄ (s) + _____ H₂O (l) → _____ CuSO₄·6H₂O (s)

Sugar:

<i>Reactant</i>	<i>Observations</i>
Sugar	
Sulfuric acid (the catalyst)	

Observations of the products:

<i>Product</i>	<i>Observations</i>
“smoke”	
“cylinder”	

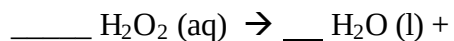


Hydrogen peroxide:

<i>Reactant</i>	<i>Observations</i>
Hydrogen peroxide	
Manganese Dioxide (the catalyst)	

Observations of the products:

<i>Product</i>	<i>Observations</i>
remaining liquid	
gas	
Result of the glowing splint test	



Sodium Bicarbonate:

<i>Reactant</i>	<i>Observations</i>
Sodium Bicarbonate	

Observations of the products:

<i>Product</i>	<i>Observations</i>
solid	
gases	
Result of the glowing splint test	

Result of the pHydron test

<i>Unheated test tube</i>	<i>Heated test tube</i>



Potassium Perchlorate:

<i>Reactant</i>	<i>Observations</i>
Potassium Perchlorate	
Manganese Dioxide	

Observations of the products:

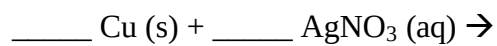
<i>Product</i>	<i>Observations</i>
solid	
gases	
Result of the	

glowing splint test	
---------------------	--



Copper and Silver Nitrate Solution (conclusion):

<i>Product</i>	<i>Observations</i>
Solid	
Liquid	



Zinc or Aluminum in Copper (II) Sulfate Solution (conclusion):

<i>Product</i>	<i>Observations</i>
Solid	
Liquid	



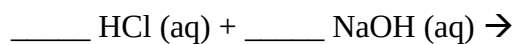
or



Hydrochloric acid and sodium hydroxide:

<i>Species</i>	<i>Observations</i>
Hydrochloric acid	
Sodium hydroxide	
Product	

Observations on mixing:

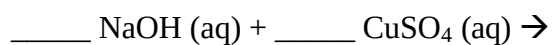
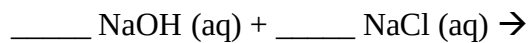
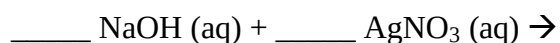
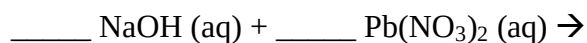
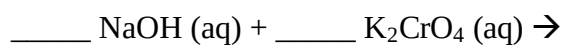
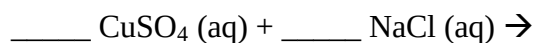
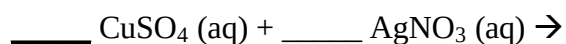
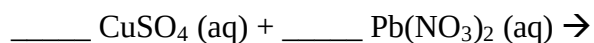
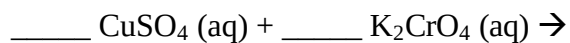
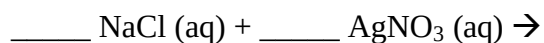
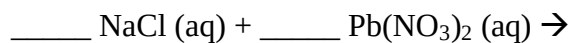
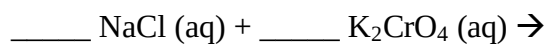
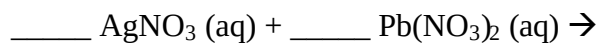
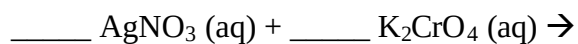
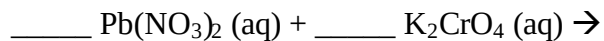


Reactions of Salt Solutions:

<i>Salt solution</i>	Observations (before reaction)
0.1 M Potassium Chromate (K_2CrO_4)	
0.1 M Lead (II) Nitrate ($Pb(NO_3)_2$)	
0.1 M Silver Nitrate ($AgNO_3$)	
0.1 M Sodium Chloride ($NaCl$)	
0.1 M Copper (II) Sulfate ($CuSO_4$)	
0.1 M Sodium Hydroxide ($NaOH$)	

Observations:

	K₂CrO₄	Pb(NO₃)₂	AgNO₃	NaCl	CuSO₄
Pb(NO₃)₂					
AgNO₃					
NaCl					
CuSO₄					
NaOH					



Pre-Lab Questions:

1. What is a catalyst?
2. What is the only type of reaction with only a single product?
3. Why is it important to balance chemical equations?

Post-Lab Question:

1. Balance all chemical equations
2. Chemists discuss the concept of “spectator ions”; look at the reaction of salts and try to determine what is meant by this term.
3. Silver nitrate is a very sensitive test for any solution which has chloride ions; where might a test like this be useful to the average consumer?

Experiment 4: Synthesis of a Compound

Purpose: To synthesize alum from recycled aluminum.

Persona: The traits that makes one's personality.

Background:

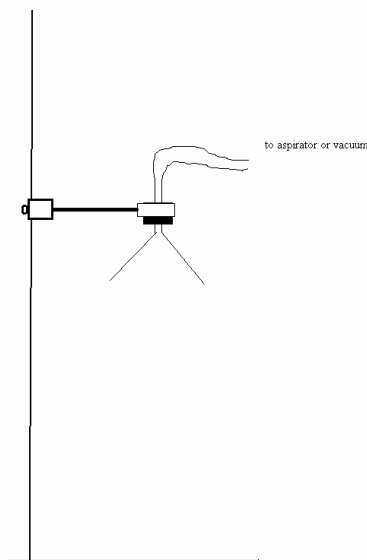
See "Basic Laboratory Procedures": balance, Bunsen burner

Introduction:

As I am writing this experiment, I am drinking a nutritious delicious diet cola. Well, maybe not nutritious. Who am I trying to kid; not delicious either. But it is a diet cola (I'm not gonna tell you which; after the way I opened this introduction, the Coca-Cola Bottling Company would sue my butt off if I did!). AND, it is an aluminium can. I know, because it has a "recycle" symbol on it. Now, let me ask you a question; if you knew that the can that YOU were drinking from at some point touched MY lips, would you STILL be drinking from it? I didn't think so. Good thing, too, because you're not supposed to be drinking in lab anyway.

So what CAN we do with...well...cans? One possibility is to turn them into something else, like, oh, for instance, Alum. Now, alum is not just a great joke in Bugs Bunny every time they need to make somebody pucker up, oh no no no NO. It is also an astringent, used in dyeing and printing fabrics, tanning, baking powders, gelatin and several other things as well.

In this experiment, we are going to do our first synthesis. In a previous experiment, you had the opportunity to see some chemical reactions occurring, but we weren't being very careful about how to control the reaction to try to maximize our yield. In today's experiment, we will be much more careful with our measurements and try to maximize the yield.

**Specialized equipment:**

Fumes can be a particular problem in this procedure, and because of limited fume hood space, we will need to construct a specialized means to remove them. Using proper procedure (see safety instructions), put the stem of a long-stem funnel through a rubber stopper, and attach a piece of vacuum hosing to the stem as well. Invert the funnel and secure it to a ring stand with a clamp. Attach the other end of the vacuum tube to an aspirator (preferably) or vacuum line; be sure that a

vacuum is drawn. You have just built a personal fume hood. Check with your instructor to be sure it has been constructed correctly, and remember, it will only work if it is *directly over* (in physical contact) with the container.

Procedure:

WARNING: This is a hazardous experiment; be sure to follow ALL safety instructions!

Measure out about 0.5 gram of aluminium pieces. It is not important to have exactly one gram, but it is important to know exactly how much you do have. It is better to have slightly more than one gram than slightly less because of significant figures. Place these pieces into a 50 mL beaker; add about 5 mL of water, and CAREFULLY add 4.0 mL of concentrated sulfuric acid. Place the beaker on an iron ring with a wire mesh and lower the inverted funnel over it. Heat gently with a Bunsen burner, bringing the solution to a gentle boil. Do not allow it to boil too vigorously, and add water if the volume starts getting too low. Continue heating until you are convinced that there is no more aluminum, but no more than 30 minutes.

Allow the solution to cool, adding water if necessary. Use gravity filtration to remove undissolved paint. Put the filtrate in a clean 50 mL beaker, and place the beaker into an ice bath. To the filtrate, slowly add 4.0 mL of 10. M KOH (aq) with stirring; be very careful to avoid splashing or overheating during this step. Check to make sure the solution is still acidic to litmus paper; if it is not acidic, add concentrated sulfuric acid dropwise until it is acidic.

Return the beaker to the iron ring, and give the solution a few minutes to warm up to room temperature. Slowly apply heat until the solid dissolves. If the solution is boiling gently, and there is still some solid in the beaker, add a little more water, as little as needed, until all of the solid dissolves.

When the solid dissolves completely, remove the heat. Put the beaker (carefully) back into the ice bath. If crystals do not begin forming in a few minutes, try scratching the bottom of the beaker with a glass stirring rod. Allow the solution to continue to cool, while setting up a vacuum filtration apparatus.

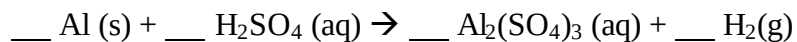
When the solution is cold, filter it using vacuum filtration. Wash the precipitate three times with cold methanol. Allow the vacuum to continue to run for a few minutes to dry off as much of the methanol as possible.

Weigh a watch glass and record the mass. When the crystals are cool, scrape them onto the watch glass, and weigh it again. Dispose of the crystals as instructed to do so in class; your professor may or may not want to collect them.

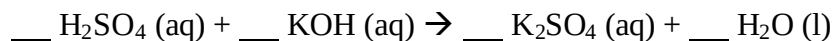
Calculations:

Balance the reactions of each step of the reaction:

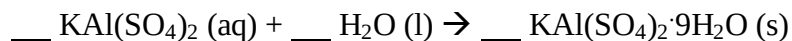
(1) Dissolving the aluminum:



(2) Neutralizing the excess acid:



Precipitate the product:



And balance the overall reaction:



From the mass of the aluminum, calculate the theoretical yield of the Alum using standard stoichiometric calculations. From the theoretical yield and the actual yield, calculate the percent yield.

Observations:

Mass of Aluminum:

Observations:

Mass of Watch Glass + Product:

Mass of Watch Glass:

Mass of product:

Actual yield:

Theoretical Yield:

Percent Yield:

Pre-Lab Questions:

1. How are we dealing with excess fumes in this experiment?
2. What is necessary for the *ad hoc* fume hoods to work? Here, I mean what must you do, not what it is hooked up to.
3. What is the concentration of potassium hydroxide?

Post-Lab Questions:

1. Excluding human or calculation errors, how is it possible to have a percent yield greater than 100%?
2. What are the steps where your product could be lost and why?

Experiment 5: Empirical Formula

Purpose: To determine the empirical formula of a hydrate

Purr: Phrase uttered by happy cats everywhere.

Background:

See “Basic Laboratory Procedures”: Bunsen burner, balance

Introduction:

In the previous experiment, we made a hydrated compound. These compounds are excellent for demonstrating the principles of how mass can be used to determine empirical formula, because, provided the salt is stable enough, all you have to do is heat them up to drive off the water. The weight difference, then, between the hydrate and the anhydrous salt is the mass of the water. In essence, this is all we are really doing in today's experiment.

Procedure:

VERY carefully, clean a crucible with a little bit of concentrated nitric acid. If all of the stains do not come out, don't worry about it; after a nitric acid cleaning, nothing will come out anyway. Follow that by washing and drying the crucible.

Place the crucible on an iron ring attached to a ring stand using a clay triangle. Place the lid on it slightly ajar. Heat the crucible to remove the water from it, gently at first, then with a strong heat until the bottom of the crucible glows red hot. Remove the heat, using crucible tongs, put the lid fully on the crucible, and allow it to cool. Avoid touching the crucible because moisture from your fingers will affect your results. By putting your hand near, but not ON the crucible, you should tell if it is still hot or not.

When the crucible seems to no longer be hot, use crucible tongs to bring the crucible and lid over to the balance. Obtain the mass of the empty crucible and lid. Return the crucible to the clay triangle, and repeat the drying step. If the difference in the mass of the crucible and lid is greater than 0.02 g between the two dryings, repeat the drying step one more time; otherwise, proceed to the next step.

Still using tongs, take the crucible dried to a constant mass to the solid hydrate provided for you. Put enough of the hydrate into the crucible to fill it perhaps 1/3 to 1/4 full. Still using the tongs, get the mass of the crucible, lid and hydrated salt.

To dry the hydrate, place the crucible with the hydrate on the clay triangle with the lid slightly ajar. Heat it slowly; if it seems to be heating too rapidly, that is, if there is splattering or too much smoke from the crucible, remove the heat, allow the crucible to cool for a minute, and resume the slow heating. Once it seems as though heating no longer causes splattering, smoke or

even steam, heat the crucible vigorously until the bottom is, again, red hot. Once hot, remove the heat, put the lid fully on the crucible with crucible tongs, and allow the crucible to cool.

Once cool, using crucible tongs, determine the mass of the crucible, lid and the anhydrous salt. To be sure all of the water has been driven off, repeat the step for drying the salt, and again determine the mass of the crucible, lid and anhydrous salt. If the mass of the crucible, lid and anhydrous salt is greater than 0.02 g between the two drying steps for the salt, repeat this step one more time.

Dispose of the anhydrous salt as instructed to do so in lab.

Observations:

Mass of crucible and lid after heating 1:

Mass of crucible and lid after heating 2:

Mass of crucible and lid after heating 3 (if necessary):

Mass of crucible, lid and hydrated salt:

Mass of crucible and lid:

Mass of hydrated salt:

Observations:

Mass of crucible, lid, and anhydrous salt after heating 1:

Mass of crucible, lid, and anhydrous salt after heating 2:

Mass of crucible, lid, and anhydrous salt after heating 3 (if necessary):

Mass of crucible, lid and anhydrous salt:

Mass of crucible and lid:

Mass of anhydrous salt:

Mass of water:

Moles of anhydrous salt:

Moles of water:

Molar ratio of water to anhydrous salt:

Empirical formula of compound:

Calculations:

From the mass of the crucible, lid and hydrated salt, and the mass of the crucible and lid after the FINAL drying step, determine the mass of the hydrated salt. From the mass of the crucible, lid and anhydrous salt and the mass of the crucible and lid, both after the FINAL drying steps, determine the mass of the anhydrous salt. From the mass of the hydrated salt and the anhydrous salt, determine the mass of the water that was driven off.

From the empirical formula of the anhydrous salt provided by your lab instructor, determine the molecular mass of the anhydrous salt. From the molecular mass of this salt and the mass of the anhydrous salt, determine the moles of anhydrous salt. Similarly, determine the moles of the water. Divide the moles of water by the moles of anhydrous salt; from this, determine the empirical formula.

Pre-Lab Questions:

1. What do you expect the primary danger in this lab will be?
2. What do we need to convert from mass to moles?
3. How can we determine if the crucible is cool?

Post-Lab Questions:

1. If moisture was still on the crucible before adding the hydrated salt, which was then subsequently driven off in the drying step for the salt, would your calculated molar ratio of salt to water be too high or too low?
2. If you did not drive all of the moisture out of the hydrate, would your calculated molar ratio of salt to water be too high or too low?

Experiment 6: Probability

Purpose: To generate a fair quantity of numbers for students to practice certain statistical tests and methods.

Porpoise: A friendly aquatic animal.

Background:

None

Introduction: There is an old joke, regarding a famous concert hall, Carnegie Hall, in which only the most talented musicians in the world are invited to play. It goes like this; “How do I get to Carnegie Hall?” “Practice, practice, practice.”

And then there would be huge laughter.

That’s the idea behind today’s experiment. Sure, you can practice on the homework problems, but then I would be forced to think up a *real* experiment. So, we’re doing this instead. You will be generating your own data for statistical analysis. The concept is to have one data set that should be completely random, one data set that should not be random but should, none the less, contain a good amount of human error, and one data set that should provide a linear relationship, but with fairly good random error as well. I would like you to perform the statistical analysis using Excel, both by putting in your own formulae (from our textbook), and from using Excel’s built in functions. I also highly encourage you to repeat these calculations using your hand-held calculator as practice for the exam, since you probably won’t have Excel available to you then.

Good luck!

Procedure:

CAUTION! Wear safety goggles and aprons throughout the entire experiment! Unless you don’t want to.

A dicey problem: You will find dice in the lab. Throw this dice 100 times, keeping track of the sum total of each throw.

Return to Middle School: If a paper football does not exist, make one. On the floor, use a piece of tape to denote the “touchdown”. From the other side of the lab, slide the paper football 100 times; record the distance from “scoring” for each throw. Remember to keep track of the

signs of the distances as well (“+” if each throw was too far, “-” if each throw was not far enough).

Buns of Mossy Zinc: Using a balance, obtain the mass of one piece of mossy zinc. Remove the zinc, tare the balance, and obtain the mass of two pieces of mossy zinc. Continue in this fashion, increasing the pieces of mossy zinc by one at each iteration, until you reach 100 pieces of mossy zinc. Record the mass at each iteration.

Calculations:

The purpose of these calculations is to give you experience with them. Use Excel, calculating the values below using formulas that you build in the appropriate cells. If there is a built-in Excel function, calculate the same thing using their built-in function in a different cell. Account for any discrepancy between these two values. Repeat the calculations on your calculator as practice for using your calculator; be sure the values all agree.

A dicey problem: Calculate the mean, median and mode of your data, the variance, and the standard deviation. Using Student’s t-test, calculate the range of expectation values with 95% certainty. Repeat these calculations both assuming it is a sample, then a population. Plot the distribution of values (relative frequencies).

Return to Middle School: Calculate the mean, median and mode of your data, the variance, and the standard deviation. Using Student’s t-test, calculate the range of expectation values with 95% certainty. Repeat these calculations both assuming it is a sample, then a population. Plot the distribution of values (relative frequencies). Check the two extreme values (most overthrown and most under thrown) to determine if these data points can be discarded with 95% certainty.

Bloody students! You are a professor, and had a graduate student quit graduate school in a huff. He left behind two sets of data (represented by the above two sets). This was very expensive data to generate, taking a lot of time and (government) money, so you’d rather not have to repeat it, but you do not know if these two sets of data represent different trials of the same thing. Can you reject the “null hypothesis” with 95% certainty?

Buns of Mossy Zinc: Using formulas that you input, use the Linear Regression method to calculate the slope and intercept of this data. Also, calculate the R^2 value. Use Excel to plot this data. Click on one data point in the graph, and left-click the data. Choose “linear” and have Excel calculate the slope, intercept and R^2 value. Are the results the same as yours?

Experiment 7: quantum mechanics

Purpose: To understand the relationship between electronic structure and the periodic chart.

Purport: To claim, declare, profess or assert.

Background:

Using HyperChem

Introduction:

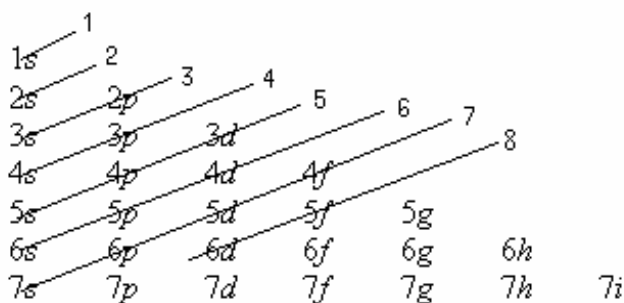
As atoms collide, it is the electrons that interact owing to the simple fact that they are the only subatomic particles that are not confined to the nucleus of the atom. Thus, the structure of the electron cloud around the nucleus is essential to the chemical and physical properties of an element, and the structure of this cloud of electrons is dependent upon the total number of electrons. From this argument, it might seem reasonable to assume that we could identify an element based on the number of electrons it has, but unfortunately, the number of electrons for an atom is subject to change.

Fortunately, for an element only (this is no longer true when the element becomes part of a compound), the total number of electrons is exactly equal to the total number of protons. What's more, the number of protons is not subject to change. Thus, we identify an element according to the number of protons it has; we call this its "atomic number."

Electrons are not bound by the laws of classical physics; they follow their own laws as delineated by Quantum Mechanics. Although we will never be able to fully understand these quantum laws (since we are classical creatures), we can learn from the lessons they provide. (1) Electrons will fill shells in order from the first shell, to the second, and onwards until all electrons are accounted for. (2) Each shell has a set of subshells equal to the number of subshells from the previous shell plus one; that is, the first shell has only the "1s" subshell, the second shell has the "2s" and "2p" subshells, the third shell has the "3s", "3p" and "3d" subshells, the fourth shell has the "4s", "4p", "4d" and "4f" subshells, and so on. (3) Each type of subshell has a set of orbitals equal to the number of orbitals of the previous subshell plus two more. That is, there is always only 1 "s" orbital, but there are 3 "p" orbitals, 5 "d" orbitals, 7 "f" orbitals, etc. (4) Each orbital can hold exactly two electrons, provided they are in opposite "spin." Thus, in any given shell, there are a maximum of 2 "s" electrons, 6 "p" electrons, 10 "d" electrons and 14 "f" electrons.

Experimental Procedure:*Part I: Electronic Configuration of the Elements*

The Aufbau process tells us what subshells (s, p, d, or f) are present in any given shell. Begin by drawing out a diagram of each shell and all associated sub shells with that shell, being *very* careful to keep consistent spacing both horizontally and vertically. Then draw angled lines through them beginning from the 1s orbital through the remainder of the diagram. Then simply read off the order of filling beginning from the first line and going through diagonally down each line from the upper right to lower left until you've gone as far as needed.



From this diagram we see that the order of filling is 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d 6p 7s 5f 6d 7p, which is as far as the periodic chart goes. Take a careful look at the periodic chart with this filling order and follow it along to see how it corresponds.

Once we know the order of filling, we can simply fill in the electrons, remembering of course the number of electrons each element can hold, until we've accounted for all of the electrons in the element, even if we are looking for an element that is beyond those that have been discovered to date!

Remembering the number of electrons each subshell can hold, and the subshells present for each shell, fill out the following table for the entire periodic chart. (One example has been included.)

Atomic Number	Element Symbol	Electronic Configuration
1	H	
2	He	
3	Li	
4	Be	
5	B	
6	C	
7	N	
8	O	
9	F	
10	Ne	
11	Na	
12	Mg	

13	Al	
14	Si	
15	P	
16	S	
17	Cl	
18	Ar	
19	K	
20	Ca	
21	Sc	
22	Ti	
23	V	
24	Cr	
25	Mn	
26	Fe	
27	Co	
28	Ni	
29	Cu	
30	Zn	
31	Ga	
32	Ge	
33	As	
34	Se	
35	Br	
36	Kr	
37	Rb	
38	Sr	
39	Y	

40	Zr	
41	Nb	
42	Mo	
43	Tc	
44	Ru	
45	Rh	
46	Pd	
47	Ag	
48	Cd	
49	In	
50	Sn	
51	Sb	
52	Te	
53	I	
54	Xe	
55	Cs	
56	Ba	
57	La	
58	Ce	
59	Pr	
60	Nd	
61	Pm	
62	Sm	
63	Eu	
64	Gd	
65	Tb	
66	Dy	

67	Ho	
68	Er	
69	Tm	
70	Yb	
71	Lu	
72	Hf	
73	Ta	
74	W	
75	Te	
76	Os	
77	Ir	
78	Pt	
79	Au	
80	Hg	
81	Tl	
82	Pb	
83	Bi	
84	Po	
85	At	
86	Rn	
87	Fr	
88	Ra	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^6 6s^2 4f^{14} 5d^{10} 6p^6 7s^2$
89	Ac	
90	Th	
91	Pa	
92	U	
93	Np	

94	Pu	
95	Am	
96	Cm	
97	Bk	
98	Cf	
99	Es	
100	Fm	
101	Md	
102	No	
103	Lw	
104	Rf	
105	Db	
106	Sg	
107	Bh	
108	Hs	
109	Mt	

Part II: Electronic Configuration and the Periodic Chart

I've provided for you a filled, and an empty, periodic chart. Be very careful to note the difference between the "classic" periodic chart (the filled periodic chart) and the empty periodic chart that more accurately reflects the filling of the shells in the sixth and seventh period. Fill in just the very last portion of the electronic configuration (from above). Answer the following questions:

- (1) Why are the first two columns offset?
- (2) Why are the last six columns offset?
- (3) Why are there ten columns of transition metals?
- (4) Why do I contend that it is more appropriate to put La and Ac in the bottom two periods, and put Lu and Lr in the transition metals?
- (5) Why don't we see the use of any subshells higher than the "f" subshell?

(6) If enough elements are created to necessitate the use of the “g” subshell (the one immediately following the “f” subshell), how many columns would this block need in the periodic chart?

Part III: HyperChem

Often, one of the biggest problems with quantum mechanics is the abstract nature of it. The concept of shells, subshells, orbitals and spin all arise as mathematical solutions to Schroedinger’s equation, and our only verification that the results are reasonable at all is by comparison with experiment. We will show in later experiments how these calculations can be used to calculate experimental results, but for the time being, let’s at least get used to what these mathematical tells us by looking at the shapes of orbitals.

Start HyperChem on your system. Open up the build tool/periodic chart by double clicking on the “draw” tool. Click on Krypton (Kr); this is the first noble gas that uses s, p, d and f subshells. Put a single Krypton atom on the screen (it should appear as a simple circle).

Choose “SetUp” and click on “Ab Initio.” This will solve Schroedinger’s equation for Krypton, built into the program with approximations to make it possible (called the Hartree-Fock approximation). In the dialog box that pops up, click the “small” radial button (3-21G). Click the “options” button, and be sure “RHF” is clicked, and charge is set to 0. The other default values should be fine (convergence limit 1e-008, iteration limit 50, accelerate convergence chosen, gradient and MP2 not chosen). Click on “polarizabilities” and be sure “do not calculate” is clicked. Click on “Configuration Interaction” and be sure “none” is highlighted. Click “OK.”

Click on “Advanced Options.” In the dialog box, be sure “five d orbitals” is chosen. For “MO initial guess” choose “core Hamiltonian.” Choose “regular” for integral format, and select “direct SCF calculation.” “Cutoff” should be 1e-010 and buffer size 32000. Click “OK.” Click “additional basis sets” and be sure “none” is selected. Click “OK” and then “OK” again to close the setup dialog box.

Go to “Compute” and click on “Single Point.” The system will begin calculating the electronic configuration for Krypton. You know the calculation is proceeding if “Cancel” is the only menu option available; it will be the only option that is NOT available once the calculations are complete. Let this calculation run to completion.

Once the computation is complete, go to “Compute” and choose “Orbitals.” This will open a dialog box. The shape of the orbitals has already been computed; this is just how you can bring out the properties of these orbitals. Click and drag the dialog box so you can see the Krypton atom in the main HyperChem window.

You will notice two radial buttons called “LUMO+” and “HOMO-.” LUMO stands for “lowest unoccupied molecular orbital;” these are orbitals that have no electrons in them (lowest means lowest energy), while HOMO stands for “Highest Occupied Molecular Orbital.” Be sure “3D isosurface” is chosen and “orbital squared” is also chosen.

For the occupied orbitals, put in the appropriate value to complete the table for part III in the results section. Be sure “HOMO-” is selected and put the appropriate number in the box below. Then click on “plot.” You need to close the dialog box by clicking “OK” if you want to

rotate the orbital; just re-open it as previously if you choose to do so. Repeat this for the unoccupied orbitals, being sure “LUMO+” is chosen. Notice that these orbitals do exist, even though they are not utilized.

Finally, arrange the shell/subshells from lowest to highest energy from both the occupied and unoccupied tables. How does this order compare with the order of filling using the Aufbau process?

Results

Occupied orbitals:

HOMO-	Shell/subshell	Energy	Shape*
17	1s		
16	2s		
15, 14, 13	2p		
12	3s		
11, 10, 9	3p		
8, 7, 6, 5, 4	3d		
3	4s		
2, 1, 0	4p		

*Notice that inner shells may not appear correct because of approximations and scale.

Unoccupied orbitals:

LUMO+	Shell/subshell	Energy	Shape*
0	5s		
1, 2, 3	5p		

*Notice that inner shells may not appear correct because of approximations and scale.

Experiment 8: periodicity

Purpose: To observe what is meant by “periodicity” and relate periodicity to electronic configuration

Perv: I think we ALL know a few people who fit into this category

Background:

None

Introduction:

In 1869, Russian Scientist Dmitri Mendeleev was busy considering chemical and physical properties of the known elements, and trying to find an underlying order to them. At this time, very little was known of atomic structure, and the concept of atomic number had not yet been proposed. However, a table of atomic masses were published by Stanislao Canizzaro, which Dmitri used to arrange the elements. Then, he noticed an odd thing; if he arranged the lists of elements in a table, based on atomic masses, the properties of the elements in any given column were similar to one another. Such became his “Law of Periodicity”; if you arrange elements according to atomic mass, the chemical and physical properties of these elements will repeat in a periodic fashion.

Now we understand his error; the elements are *not* arranged according to atomic mass, but rather according to atomic number. Careful examination of the periodic chart will show that there are only a handful of pairs of elements that are in their incorrect spots if you arrange the table according to atomic mass, however, so Mendeleev’s creation was adopted and exists even today as the basic form of the periodic chart.

Our purposes today are two-fold: (1) we wish to elucidate several of the trends of the periodic chart, and (2) we wish to learn of some of the on-line tools available to the chemist.

Experimental Methods:*Part I: On-Line Tools*

You will be given a presentation of current on-line resources available to chemists. **In your lab manual**, take note of these URL addresses, and the specific role of each site (a.k.a. the type of information each web site provides).

Part II A: Periodicity

For this part of the project, you may split this work with your lab partner and no more than one additional group (five people maximum), but you *must* report the data that you personally were responsible to obtain. Write down the information *you* obtained in your own lab notebook, and don't forget to give the URL address or reference the source from which it was obtained. This is an on-line project, so do not use any print resources (unless they are fully available on-line as well).

With the other people you are working with, obtain the following information:

For the first **18** elements (H through Ar), find the atomic radii, melting points, boiling points, electronegativities, and first ionization energies. Graph this data in your lab notebooks in accordance with the calculations section.

Part II B:

ON YOUR OWN, find the following information for the Oxide of ONE OF THE 18 elements for which the oxide exists (save, of course, oxygen): Reactivity with water, LD₅₀, and first aid treatment.

Calculations:

On separate pages in your lab notebook, graph the atomic radii, melting points, boiling points, electronegativities, and first ionization energies. Make sure that each graph fills the entire page by choosing appropriate scales for the x and y- coordinates (see me if you do not know how to do this). Connect your points, but also, using different color pens, connect the points for just the individual groups (that is, Group IA, Group IIA, etc.). Use these graphs to answer the post lab questions.

Periodicity Post-Lab Questions

- 1) For the atomic radii, melting points, boiling points, electronegativities, and first ionization energies, what is the general trend as you go across the periodic chart (left to right), neglecting "glitches"?
- 2) For the atomic radii, melting points, boiling points, electronegativities, and first ionization energies, what is the general trend as you go down the periodic chart (from top to bottom)?
- 3) For the atomic radii, melting points, boiling points, electronegativities, and first ionization energies, where are the major "jumps"? (That is, the greatest discontinuities in the graph occurs between which groups?)

Experiment 9: VSEPR

Purpose: To use the Valence Shell Electron Pair Repulsion Theory to predict molecular shapes

Pink: Not just a color, but also a singer

Background:

See "Using HyperChem"

Introduction:

Sure, Quantum Theory is powerful, but sometimes, simple is better. Simple implies more insightful, more intuitive, easier to utilize, and sometimes just as good as more difficult theory. One such simple, insightful theory is the Valence Shell Electron Pair Repulsion Theory (VSEPR, often pronounced as "vesper").

VSEPR is a simple model that was developed to predict the shape of covalently bonded compounds. It turned out to be exactly "on the mark". Nowadays, we can use quantum theory to predict the same thing, by forming "bonding orbitals", which are solutions to Schroedinger's equation created by the overlap of orbitals from each atom involved in the bond. The odd thing, though, is that while quantum mechanics has verified the validity of VSEPR, it gave us no real improvement to the model. Therefore, chemists today are more likely to turn to the older, simpler VSEPR model than they are the more difficult and cumbersome quantum mechanics.

VSEPR: The Premise

Let's begin by looking at the premise to VSEPR. It's amazing to me how simple and intuitive it is. Basically, we begin by recognizing that we are only interested in the valence shell electrons (the outermost electrons, which is where the "VS" comes from in VSEPR, the "valence shell"). Inner shell electrons do not interact with other atoms, and therefore play no part in the shape of the molecule. Secondly, we notice that electrons exist in pairs, either as bonding electrons or non-bonding electrons (hence the "EP" part of VSEPR, the "electron pair"). Therefore, instead of looking at single electrons, we deal with each electron pair (cuts our work in half right off the bat!).

Finally, we notice that, with the correct spin configuration, we can fit two electrons into a given orbital, but that, because the region of space defined by the orbital is occupied by negatively charged particles (electrons), that region of space has a negatively charged characteristic. Sure, the net charge for the electrons is counterbalanced by the positive protons in the nucleus, but the region of space occupied by the electrons is negative by virtue of the particles in that space, just as the region of space occupied by the nucleus is positive by the same token.

Well, seeing that the region of space around the nucleus of the atom occupied by electrons is negative in charge, we notice that these electron pairs will repel one another (the last part of VSEPR, "R" is the repulsion of the valence shell electron pairs). Since these electron pairs repel one another, we arrive at the *premise of VSEPR*: **electron pairs will arrange themselves in space about an atom such that their total repulsive energy is minimized**. Well, it sounds more difficult than it really is.

Simply speaking, the electron pairs will get as far apart from one another as possible. By doing so, their repulsive energy is minimized. Suppose you had to take a particularly foul tasting medicine, and you were told that you would have to take this medicine for at least 30 days. This is rather open ended, because we never said "at most" how long you should have to take it, so you have your choice of 30 days to the rest of your days. What would you choose? Assuming it's not a life-threatening condition, you would choose to minimize the repulsion as much as possible. Electrons minimize their repulsion by getting as far apart from one another as possible.

VSEPR mechanics I: The Lewis-Dot Structure

We begin by determining the Lewis-Dot structure of the molecule. We cannot proceed without the Lewis-Dot structure, because it is from the Lewis-Dot structure that we determine the number of lone pairs of electrons and how the elements are bonded to one another. **It is important to note that although the Lewis-Dot structure does provide us with bonding information, it does *not* speak to the true three-dimensional structure of the molecule.** Although there *are* some molecules restricted to a plane (two-dimensional), typically they are three dimensional figures, as we would expect. Since we draw Lewis-Dot structures on paper, we are confined to two dimensions. Therefore, although we gain a wealth of information from Lewis-Dot structures, actual molecular shape is not one of the pieces of information Lewis-Dot structures afford.

VSEPR mechanics II: The Central Element

Now, we must define a central atom. We have no way of describing the shape of a large, complex molecule in its entirety all in one step since the number of possible shapes is overwhelming. What we *can* do is describe the position of atoms around one single, or central, atom.

Usually the central atom is just that, the central atom to which most of the other atoms are bonded. For instance, we might choose the carbon in methane, CH_4 , or the nitrogen in ammonia, NH_3 . For more complex molecules, the central atom will need to be specified. Consider acetic acid, $\text{CH}_3\text{CO}_2\text{H}$. Here we have two carbons, four hydrogens and two oxygens. Even if we choose an element, we have to decide which one. For instance, if we choose carbon to be the central element, then we will have to further specify it as the carbon on the left, or the carbon on the right (or the first or second carbon).

VSEPR mechanics III: Sets

Next, we look for the number of "sets" about the central atom. A set is basically a pair of electrons, kind of. Hmmm, how to put this. Let's put it in equation form:

$$\# \text{ sets} = \# \text{ L.P.} + \# \text{ Bonded atoms}$$

where # L.P. is the number of lone pair, or non-bonding pairs, of electrons assigned to the central element, and #Bonded atoms is the total number of additional atoms to which the central atom is attached. The number of lone pairs, or the number of additional atoms bonded to any atom other than the central atom are not included; the number of sets, or set number, applies only to the central atom. There are two things that must be made note of.

Note 1: The number of bonded atoms refers to the number of atoms bonded directly to the central element (only) *regardless* of the order of the bond (single, double, triple). Even if it is a multiple bond, all electron pairs must occupy the region of space between the nuclei (to some extent), so it is irrelevant what type of bond it is.

Note 2: In the rare occasion that we are dealing with the shape of a radical (a molecule with at least one unpaired electron), each unpaired electron we will count as a lone pair anyway. This is because even unpaired electrons must exist within an orbital, so an unpaired electron would occupy the same region of space as a lone pair of electrons.

VSEPR mechanics IV: Molecular Shape

Now we know how many "sets" are around our central element. How do we know its shape? We look it up! The figures represented in the table below show the molecular geometry based on the set number and on the number of non-bonding pairs of electrons (lone pairs). They have been determined to be the shape that gets each electron pair as far apart from each other as possible.

Set Number	Number of Nonbonding Pairs	Electron-Pair Geometry	Molecular Shape
1	0	Linear	Linear
2	0	Linear	Linear*
2	1	Linear	Linear
3	0	Trigonal Planar	Trigonal Planar*
3	1	Trigonal Planar	Bent
3	2	Trigonal Planar	Linear
4	0	Tetrahedral	Tetrahedral*
4	1	Tetrahedral	Trigonal Pyramidal
4	2	Tetrahedral	Bent
4	3	Tetrahedral	Linear

5	0	Trigonal Bipyramidal	Trigonal Bipyramidal [^]
5	1	Trigonal Bipyramidal	See-Saw
5	2	Trigonal Bipyramidal	T-Shaped
5	3	Trigonal Bipyramidal	Linear*
5	4	Trigonal Bipyramidal	Linear
6	0	Octahedral	Octahedral+
6	1	Octahedral	Square Pyramidal
6	2	Octahedral	Square Planar+
6	3	Octahedral	T-Shaped
6	4	Octahedral	Linear*
6	5	Octahedral	Linear

* These shapes are highly symmetrical *if* all atoms bonded to the central element are identical.

+ These shapes are highly symmetrical *if* diametrically opposed elements are identical.

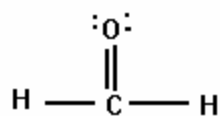
[^] Trigonal Bipyramidal is a highly symmetrical shape *if* both elements on the axis are identical *and* all three elements in the plane are identical.

VSEPR: Some Examples and Shapes

Let's take a look at some of these shapes. Doing this will also prove to be useful in showing how to use VESPR. The three-dimensional figures below were generated using Chem3D. Here at Dakota State University, we use HyperChem, which will be the last portion of this exercise.

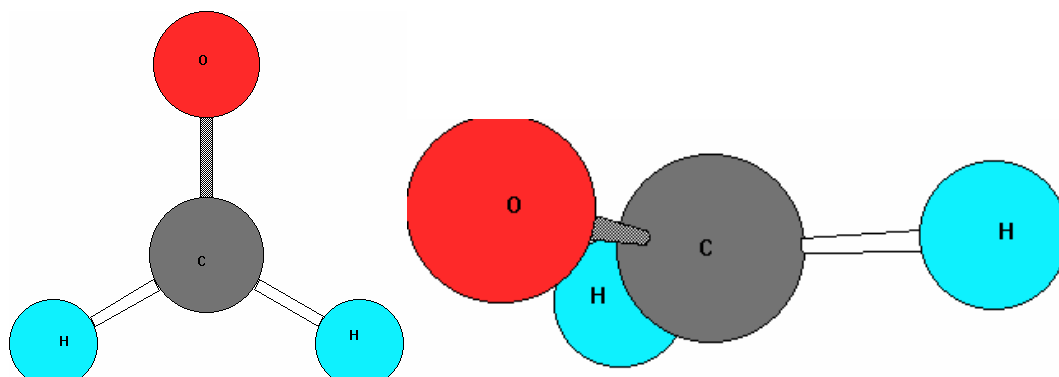
Example 1: COH₂

First, we determine the Lewis-Dot structure. Hopefully, by now you are relatively skilled at this. The Lewis-Dot structure would look as follows:



Now we determine the central element, which in this case is fairly obviously carbon. To determine the set number, we notice that carbon has no lone pair electrons (we ignore the lone pairs on the oxygen because they are not associated with the central element). We further notice that there are three atoms attached directly to carbon (even though there is a double bond to oxygen, it is only one element). Therefore, the set number is 3.

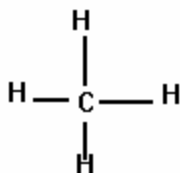
Looking at our table, we see that the parent structure for this molecule is trigonal planar. Looking further, with no lone pair electrons, the molecular shape is also trigonal planar. Here is how it looks from a couple of different viewpoints:



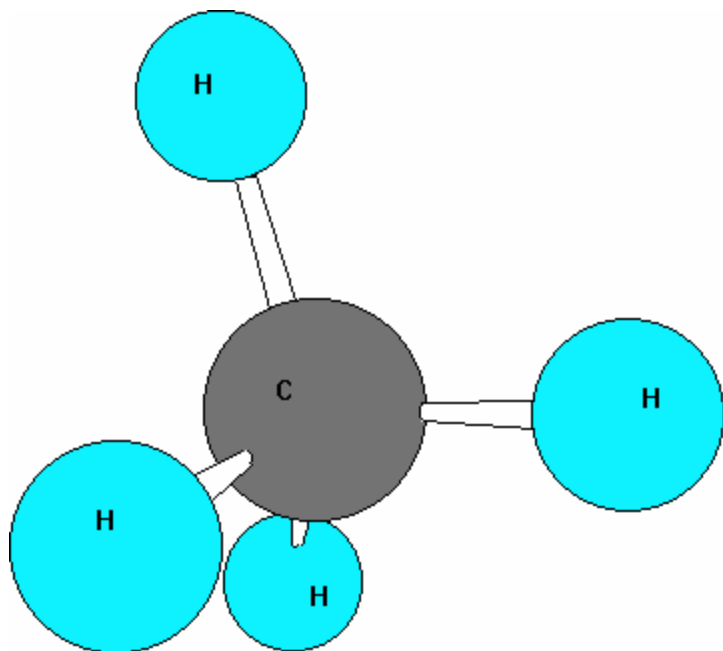
We probably could have drawn the Lewis-Dot structure in this case the way it would really look, since the molecule, as can be seen, is confined to a plane.

Example 2: CH₄

Here we have a classic example of VSEPR. Again, we start with the Lewis-Dot structure:



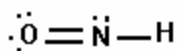
Again, it is obvious that the central element is carbon. To get the set number, there are (again) no lone pairs, and four atoms attached to the central atom, giving us a set number of 4. Looking at the table, set number four is tetrahedral, and with no lone pairs, the molecular shape is tetrahedral. Here is how it looks:



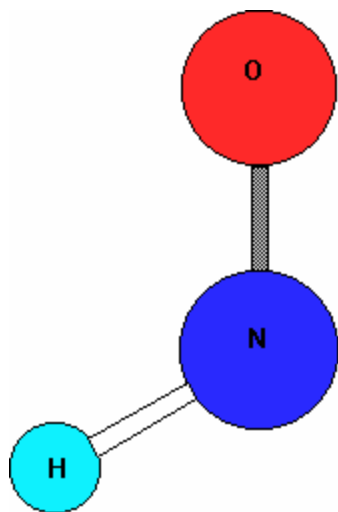
This looks quite different from our Lewis-Dot rendering. Each hydrogen is evenly distributed about the carbon in three dimensional space, with angles of about 110° . In our Lewis-Dot structure, it looked as if the molecule would be square planar. This point cannot be stressed too strongly: *Lewis-Dot structures give bonding information, but do not accurately represent the true shape of the molecule.*

Example 3: NOH

This is a hypothetical molecule, that would be called "Nitrosyl Hydride", except that I cannot find reference to it anywhere so I doubt that it actually exists (it more than likely immediately breaks down to more stable compounds). We can make a Lewis-Dot structure for it, though, whether it exists or not. The fact that it does not implies that it would be highly unstable. None the less, let's imagine what it should look like, if, indeed, we could make it. The Lewis-Dot structure would look like this:



So, let's go through it. Central element is nitrogen; there is one lone pair and two atoms giving a set number three. This gives us the trigonal planar parent structure, but the bent molecular shape. It would look as follows:



Polarity:

By now you should know how to determine whether or not a bond is polar based on differences in Pauling's electronegativity scale (a difference in electronegativity of less than 0.5 is non-polar, between 0.5 and 1.7 is polar, and greater than 1.7 is ionic). For a diatomic molecule, the molecule is polar if the bond is polar. This is a simple enough concept, and quite intuitive, but what do we do with polyatomic molecules? Such a question is important since polarity of molecules determines the intermolecular forces between such molecules, and therefore affect many of the physical properties.

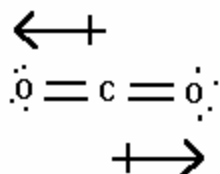
Well, for polyatomic molecules, we have a few additional steps:

1. Determine the Lewis-Dot structure.
2. Determine if *any* of the bonds are polar or not. If there are no polar bonds, you are finished; it is a non-polar molecule. If there is even one bond, you must proceed, because it may or may not be polar.
3. Determine the molecular shape using VSEPR.
4. The molecule is non-polar if it is highly symmetrical; otherwise, it is polar.

In a highly symmetrical molecule, the polar bonds will cancel exactly, but how can we tell if we have a highly symmetrical molecule? Once you see it, it becomes obvious. Unfortunately, it is often difficult to see, so let's go through these rules. A molecule is highly symmetrical *if*;

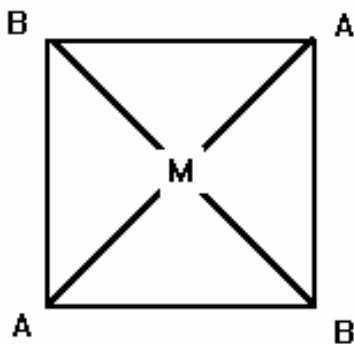
1. All elements bonded to the central atom are identical *and* we have no lone pairs, *OR*
2. All elements bonded to the central atom are identical *and* we have a linear molecule with either a trigonal planar or octahedral parent shape, *OR*
3. All elements on diametrically opposed sides of the central atom (exact opposite sides of the atom, so they could not "see" one another around the central element) are identical *and* the chemical structure is square planar.

Let's see why these molecules are non-polar. The classic example is carbon dioxide, CO_2 , so let's take that one as the example. The Lewis dot structure shows that each oxygen is double bonded to the carbon, the central element, and there are no hydrogens on the carbon. This gives us set number two, and the molecule's parent structure and shape are both linear. We also see that the carbon-oxygen is polar (quite polar) with the oxygen being more electronegative. But, since the molecule is symmetrical, the two dipoles on each side of the molecule are exactly identical, and therefore cancel one another out:



One way to think of this is if two children of exactly identical strength were pulling a wagon. If they were pulling in exact opposite directions, the wagon is not moving anywhere; it's stuck. However, if they were not equally strong, or if they were pulling in kind of similar direction, then the wagon would move, maybe not towards one of the children exactly, but it would move. As a result, carbon dioxide is non-polar because it is a highly symmetrical molecule.

To see why something that is, say, square planar might be non-polar, we have to consider what it would look like. An example of such a shape might be something like $[\text{Cu}(\text{NH}_4)_4]^{+2}$, which is square planar. Anyway, consider the following diagram:



As you can see, if the molecules on diametrically opposed corners are identical, any dipole present would cancel out exactly.

This has been kind of vague, but all the information you need is contained within this document. Any questions, comments or suggestions should be sent to me by email, phone, or by stopping by in a friendly little visit WITH NO VIOLINS!

EXPERIMENTAL PROCEDURE:

It seems to me that the best way to learn Lewis Dot Structures is to practice with them. Fortunately we have an entire class of compounds that are particularly well suited for such practice, a class so large, in fact, that it is often given a year of study in most Universities. It's called organic chemistry.

Yep, organic chemistry. I know what you are thinking, "mmmm...chocolate". But you are ready for this challenge. Below you will find a series of empirical formulas. For each formula, find an allowed Lewis Dot structure, noting special instructions where given. Some of these formulas exist in "isomeric" forms, that is, there is more than one possible Lewis Dot Structure. In these cases, try to find as many possible structures as indicated. For each valid Lewis Dot Structure, determine the molecular shape using VSEPR, and determine if it is polar or not. **Do this part of the experiment before the computer part!**

Let's get started! *Take notes in your lab notebook, as if this is a normal experiment.*

Formula 1: CH₄

Formula 2: CH₂O

Formula 3: CH₄O

Formula 4: C_2H_6

Formula 5: C_2H_4

Formula 6: C_2H_2

Formula 7: $\text{C}_2\text{H}_6\text{O}$ (2 isomers; hint: for one isomer, try placing the oxygen between the carbons for ONE of the isomers)

Formula 8: $\text{C}_2\text{H}_4\text{O}$ (2 isomers)

Formula 9: $\text{C}_2\text{H}_7\text{N}$

Formula 10: $\text{C}_2\text{H}_4\text{O}_2$ (at least 3 isomers)

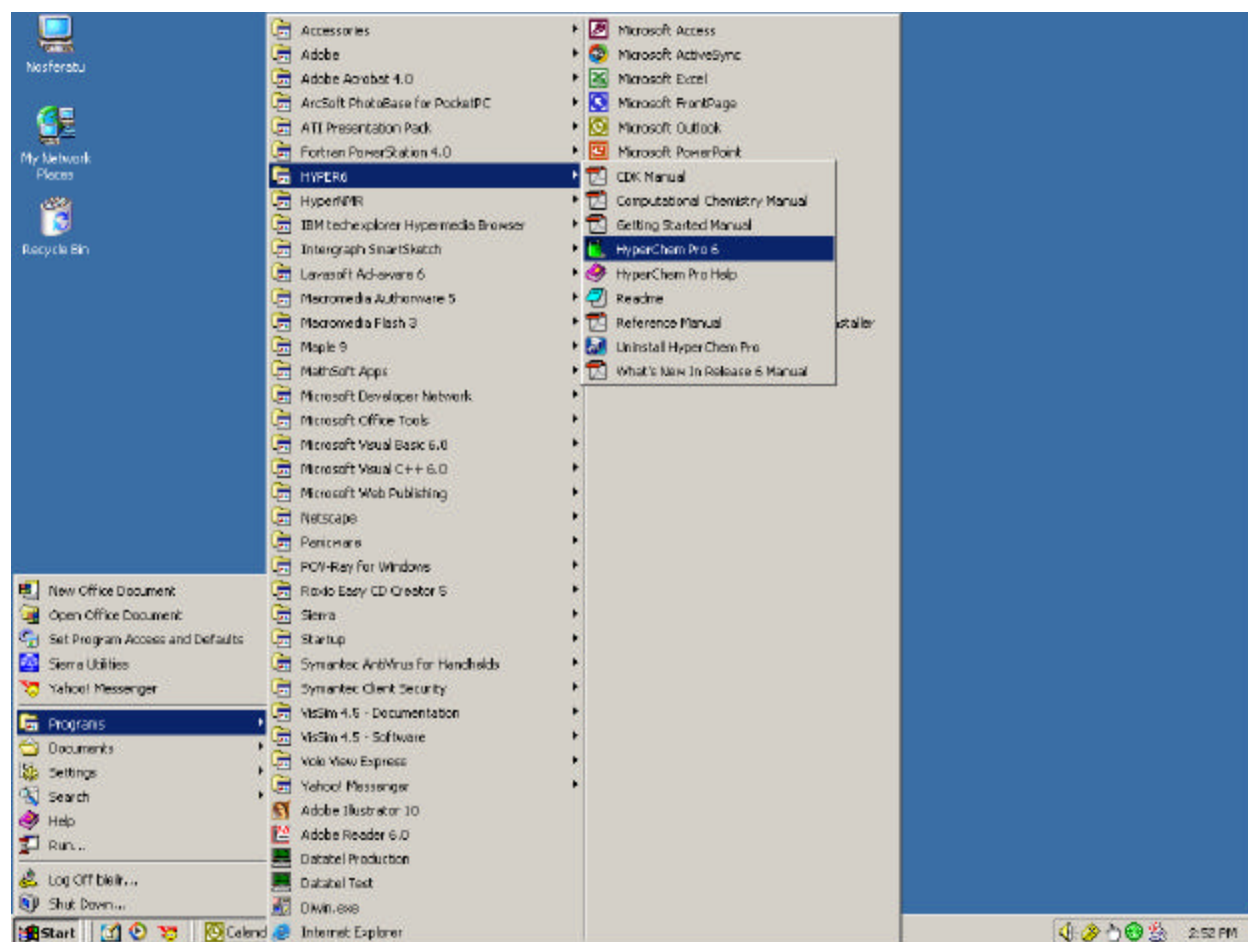
Formula 11: C_4H_{10} (2 isomers)

Formula 12: See how many formulas you can get from C_5H_{12}

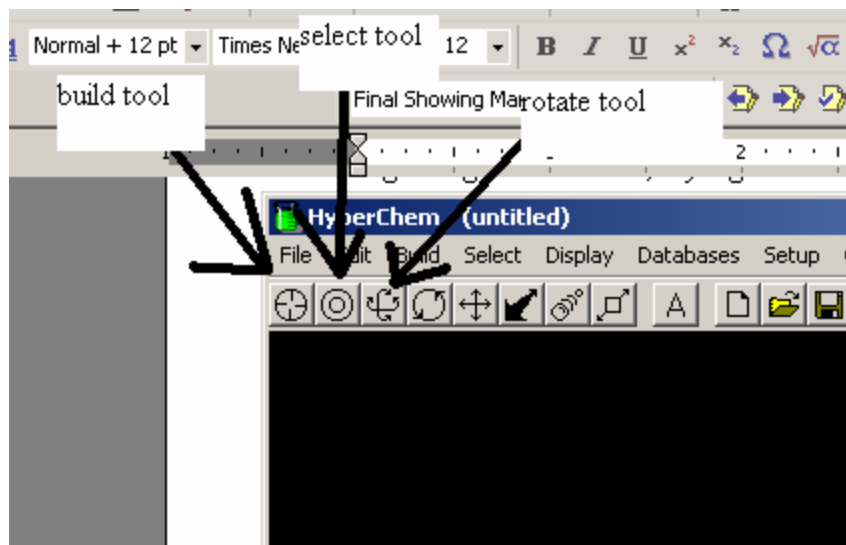
Computer visualization of molecules:**IMPORTANT NOTES:**

- (1) HyperChem is available ONLY on the computers in the computer lab in the science center. It is not available anywhere else.
- (2) We have only FIVE licenses for HyperChem; thus, if you log on and get an error message regarding the licenses, try again later. This implies that you must NOT wait too long before starting this assignment.
- (3) Be sure that the first part of the lab is completed BEFORE attempting to do this portion of the lab. The idea is to compare your results from the computer with your results from above.

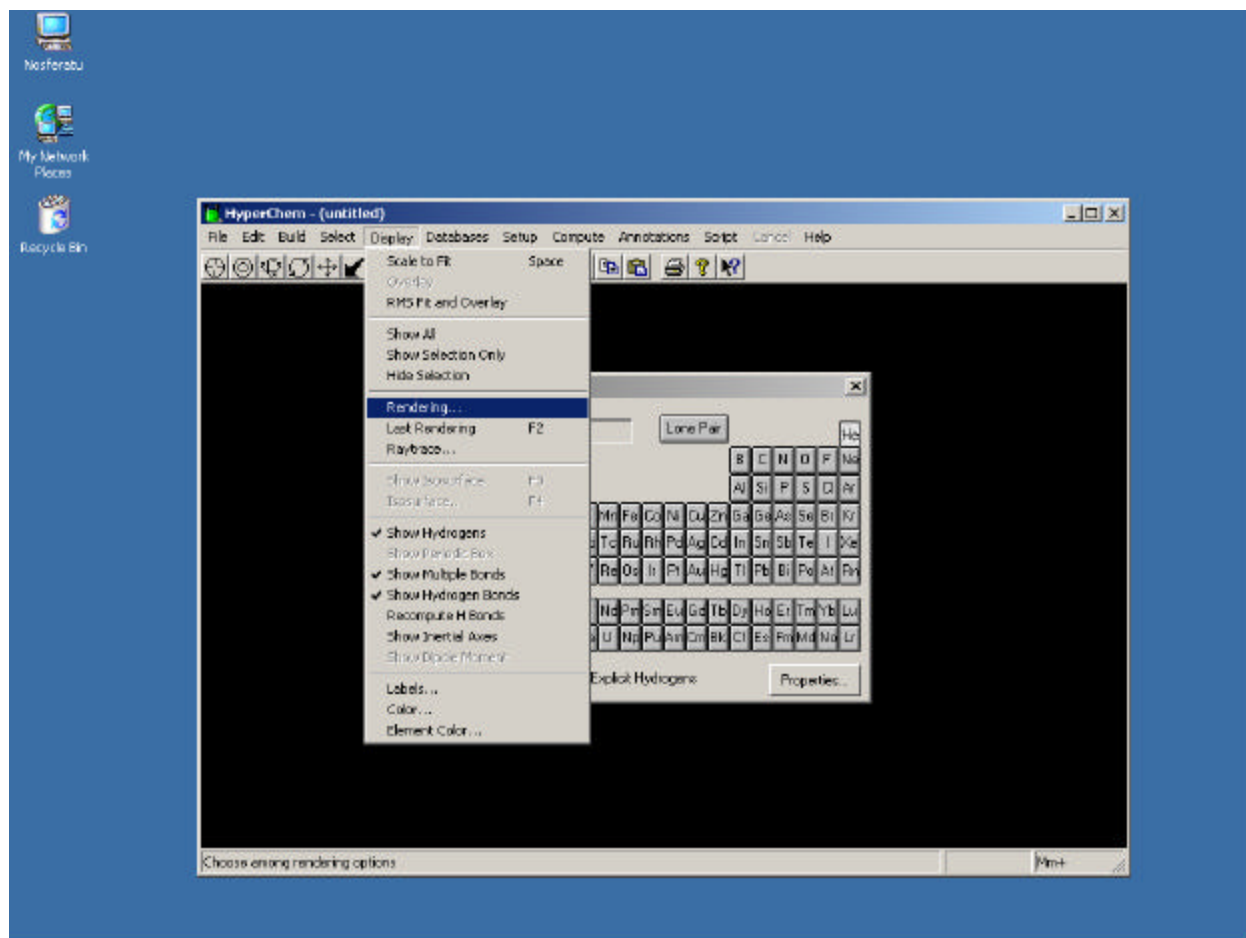
Log on to a computer in the science center computer lab, and start HyperChem Pro.



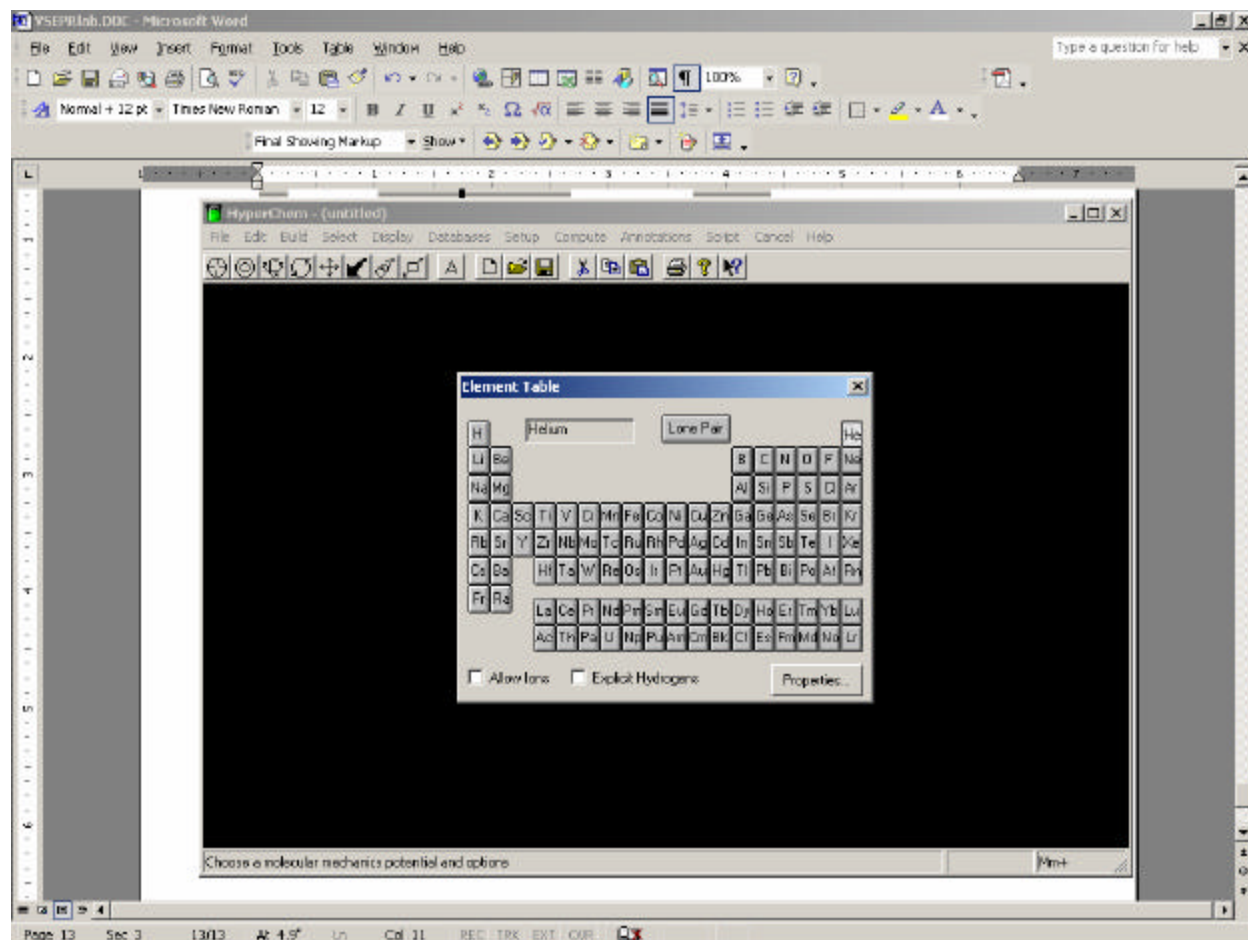
You will need to know a few key features of the hyperchem menu:



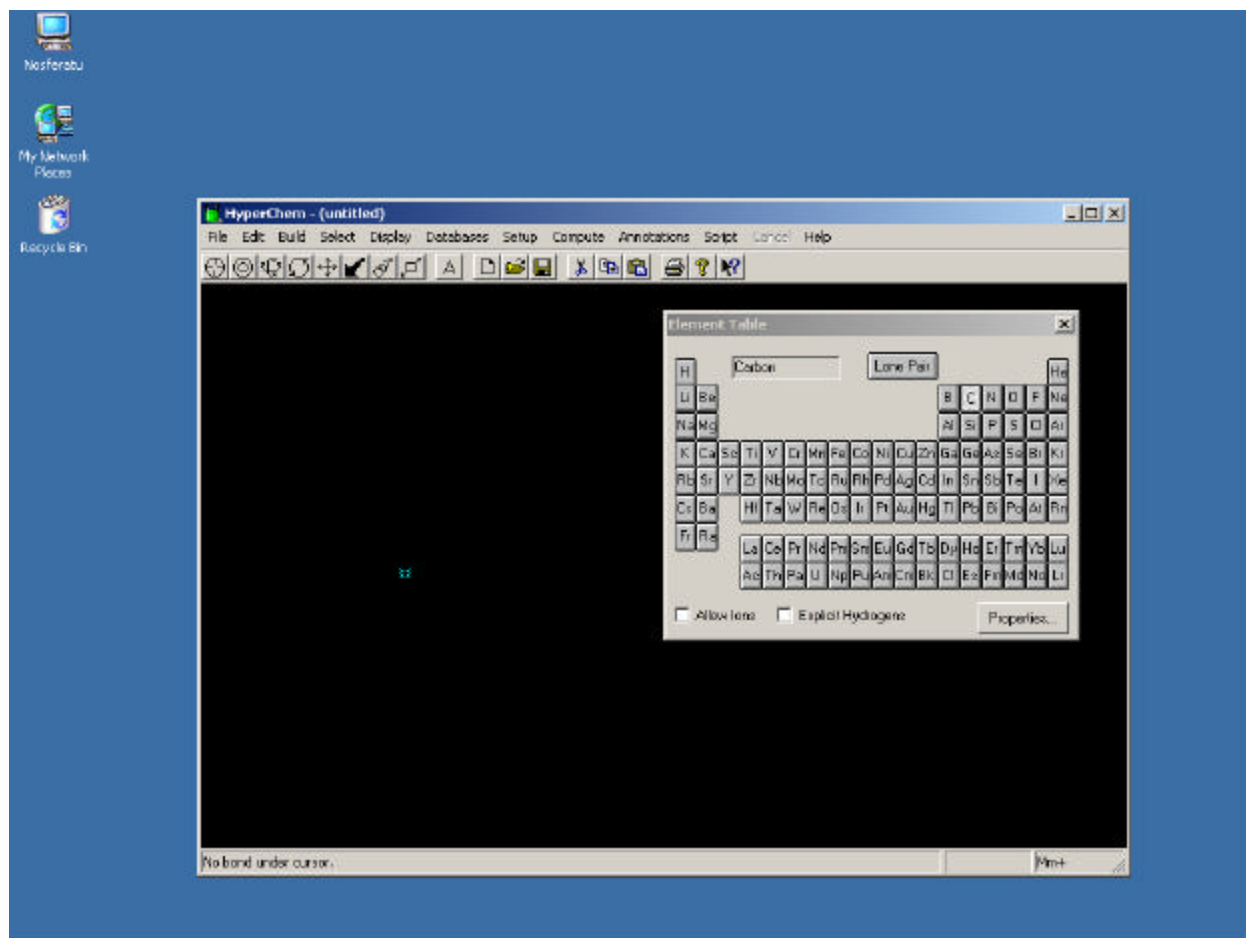
(For a color copy of this experiment, see WebCT.) Before we begin building a molecule, go to the “display” and “rendering” option.



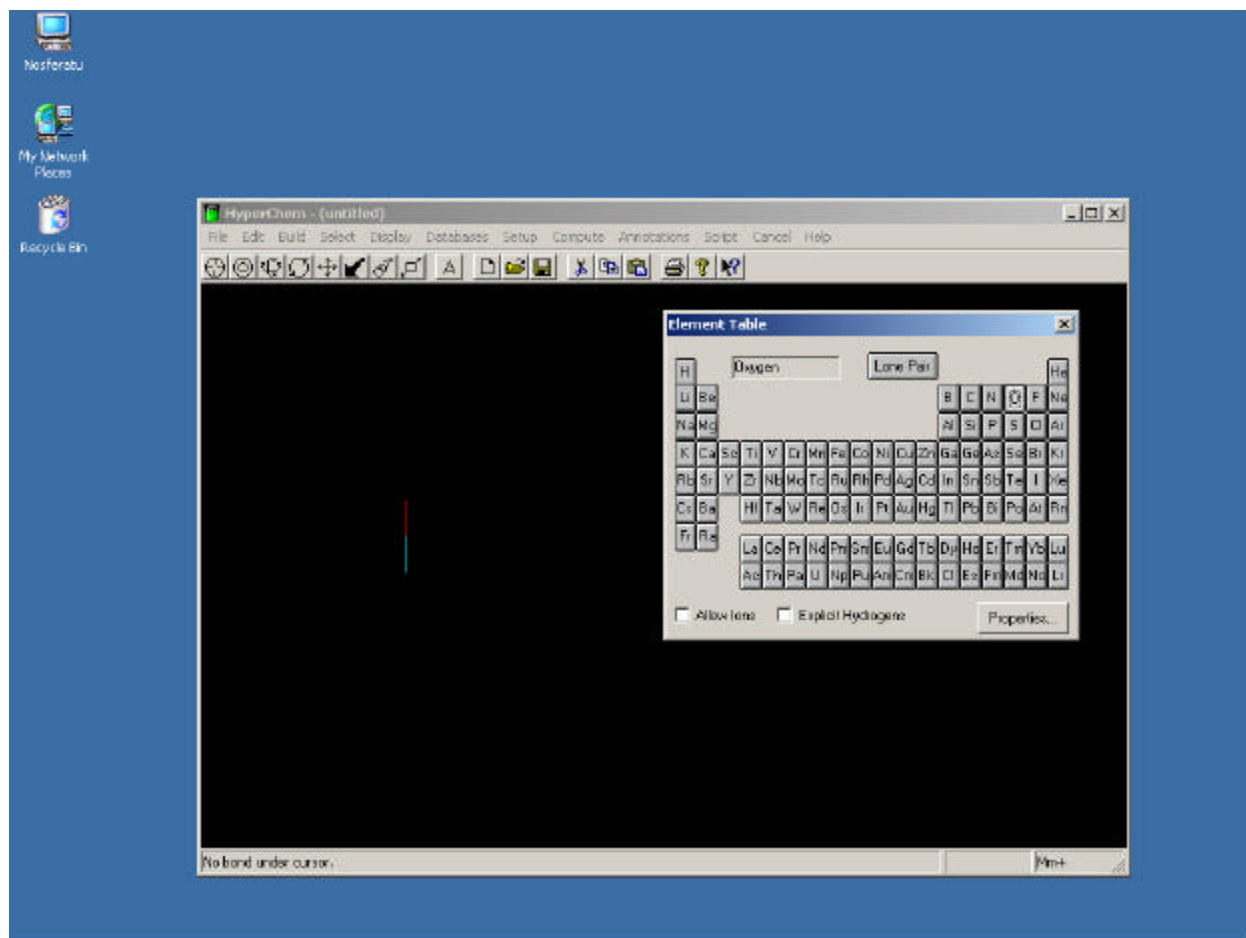
This will open up a dialog box; be sure “stick” is activated (as opposed to the other options). This will make it easier to build the molecule. Now, to build a molecule, double click the build tool, and it will bring up a periodic chart.



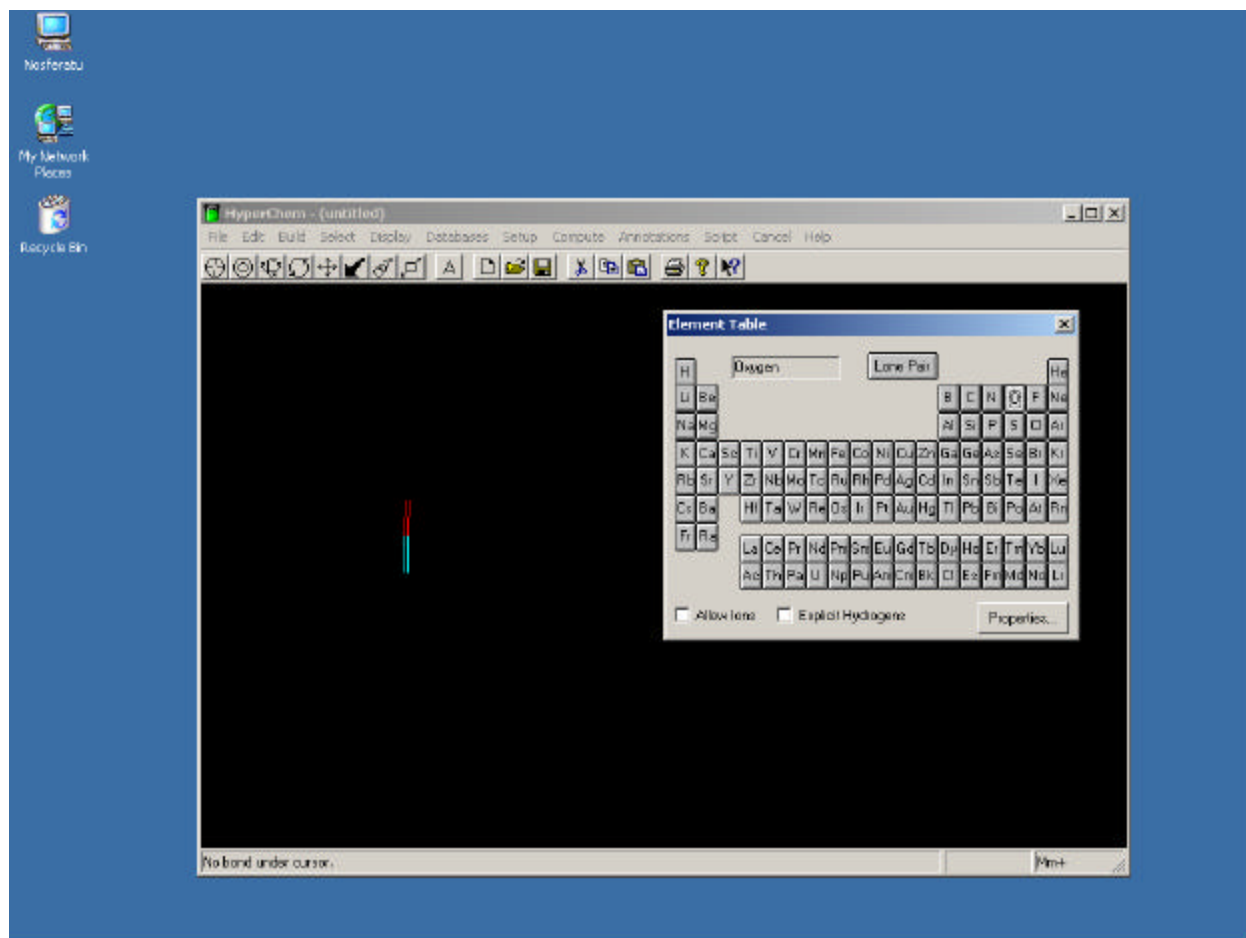
To build something like, say, formula 2, begin by clicking on C to be sure it is activated. Notice that the periodic chart is the “active window”; you will have to click once in the black space to make HyperChem the “active window”; the second time you click in the black area, a carbon should appear as a relatively small blue item (kind of a circle, but not really; you’ll see what I mean).



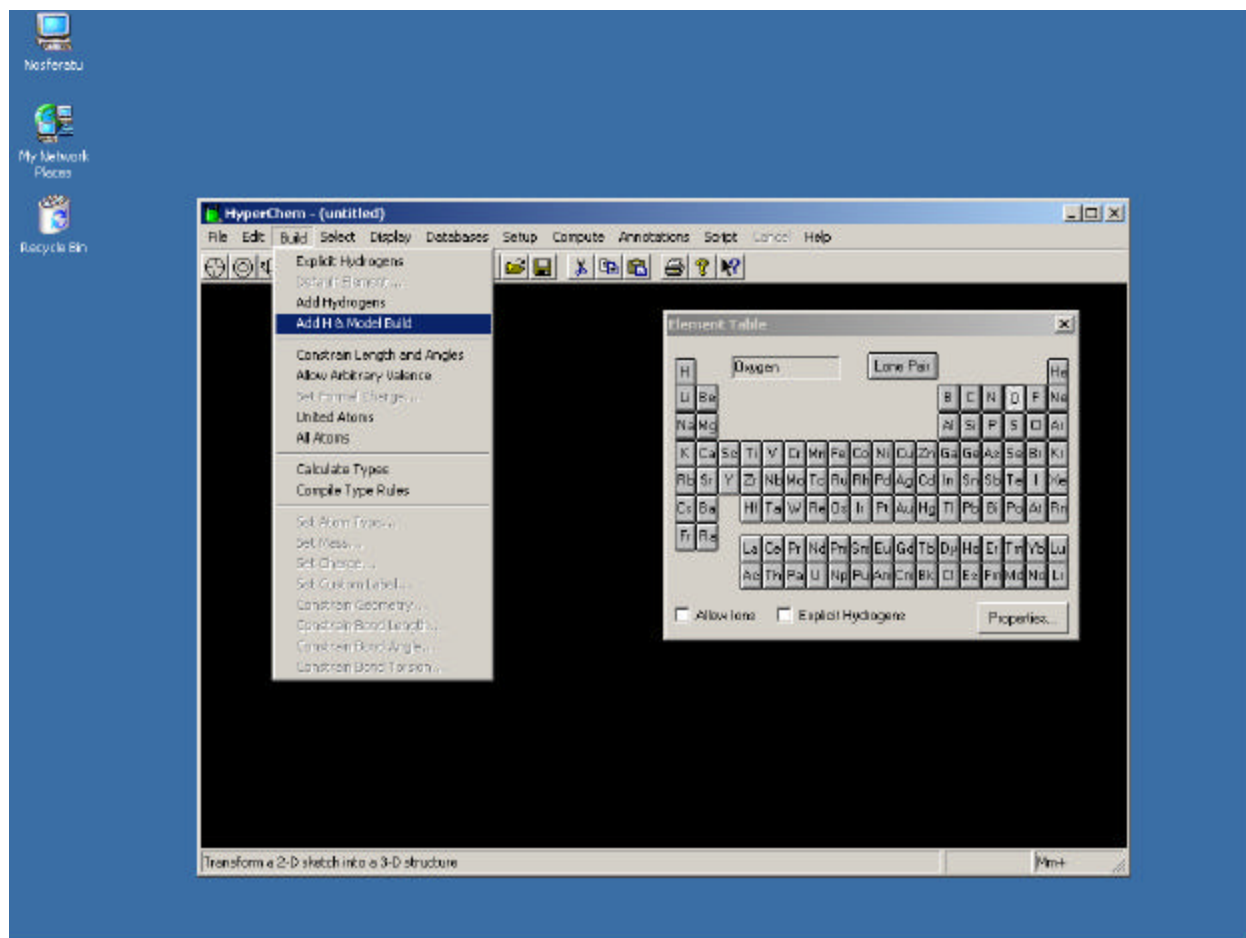
Now, we will want to attach an oxygen to this carbon. Click on oxygen in the periodic box. Place the cursor on the carbon in the black area, and draw a line up. If no line appears, the periodic box was still the “active window;” just try it again until you get a line that is blue on one side (the carbon) and red on the other.



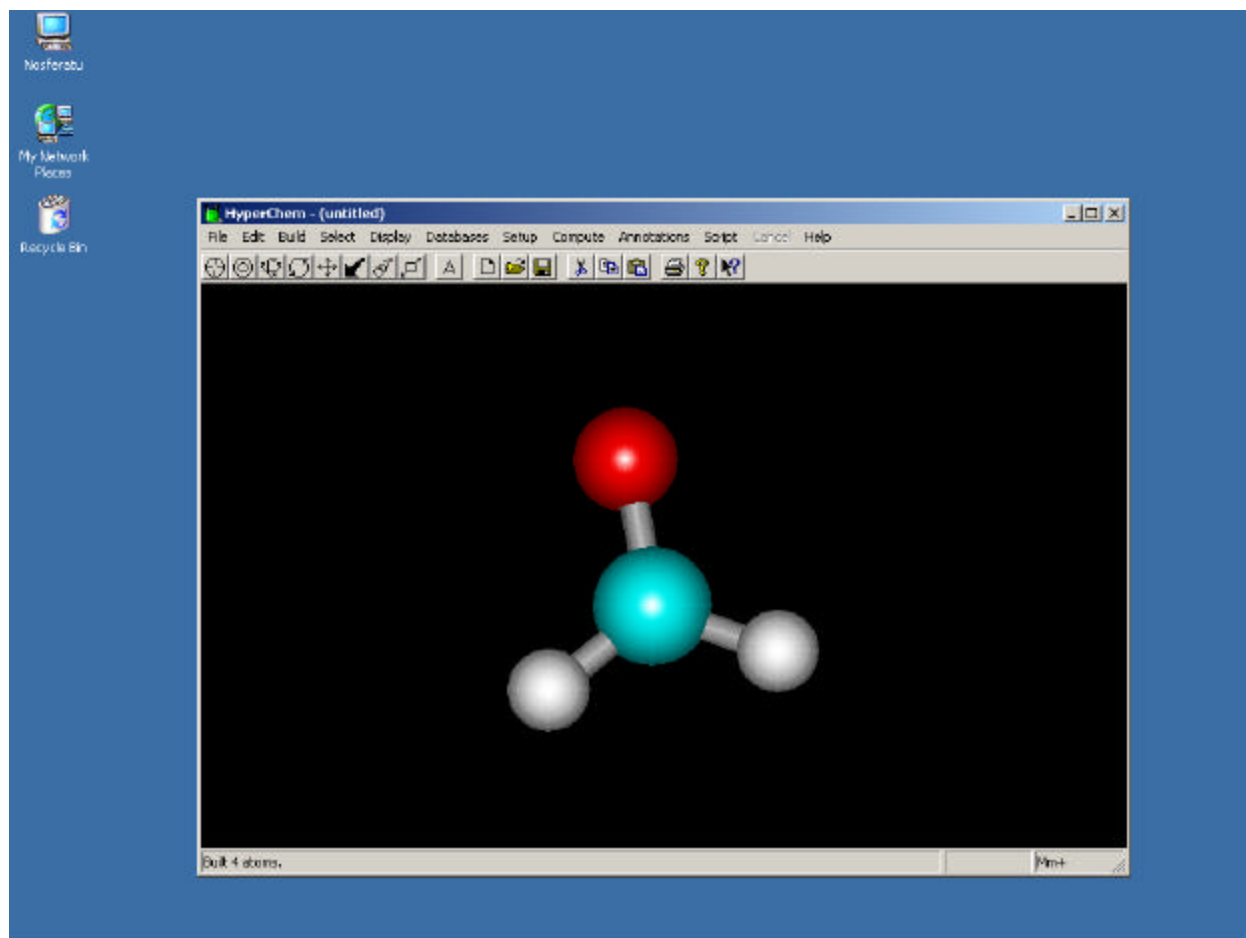
This represents a SINGLE BOND between carbon and oxygen. To make it a double bond, click on the bond until you see two parallel lines.



To undo a bond, right click on the bond instead (we need not do that now; we have what we want). Now, everything has been added except for hydrogen. Go to “build” and “add H and model build” and Hyperchem will complete the structure for you, complete with the correct bond lengths and angles.

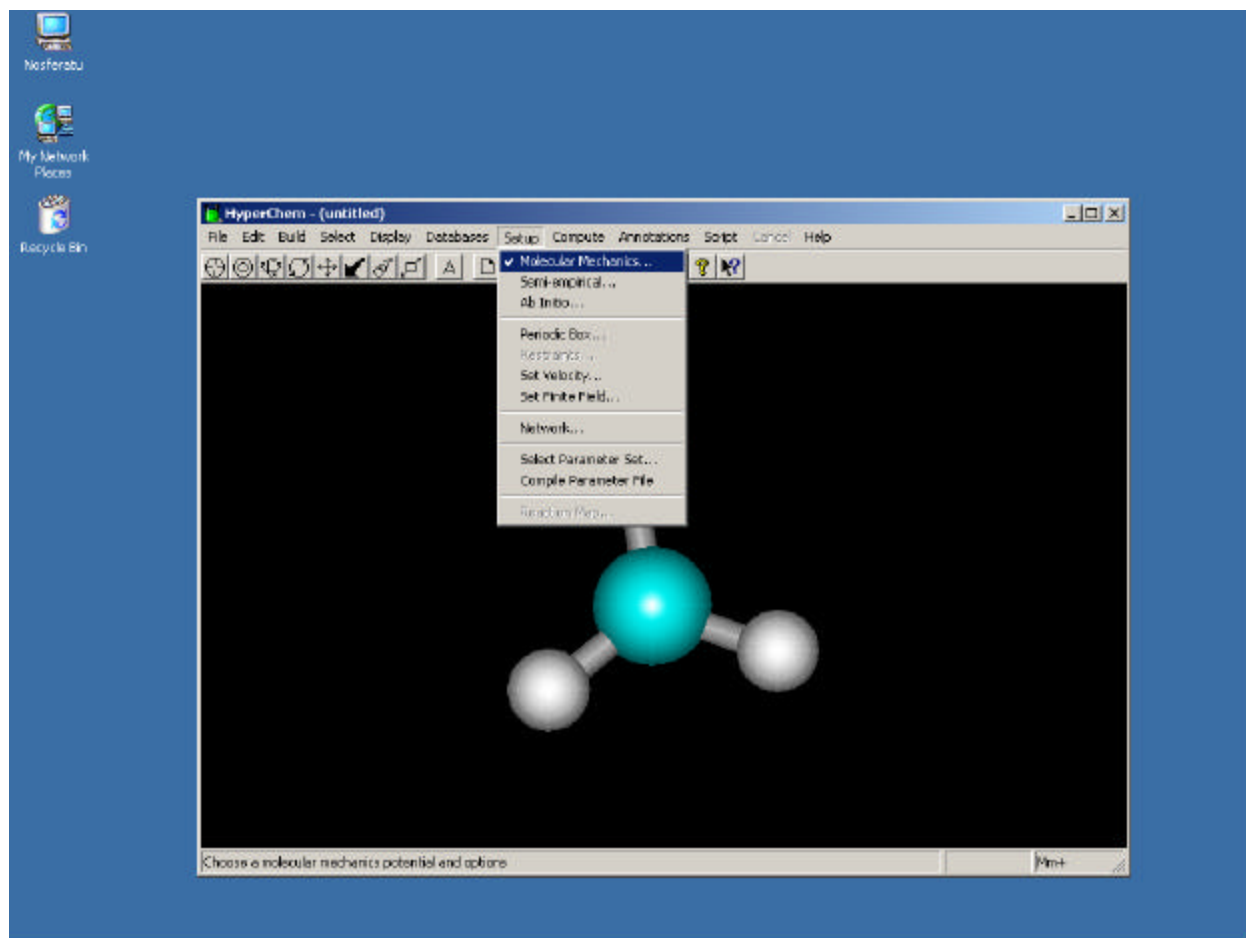


Close the periodic table window, and choose the “rotate tool.” Change the display to “balls and cylinders” option and look at the molecule carefully while rotating it. Compare the structure from HyperChem with what you predicted using VSEPR. Draw the structure as best you can.

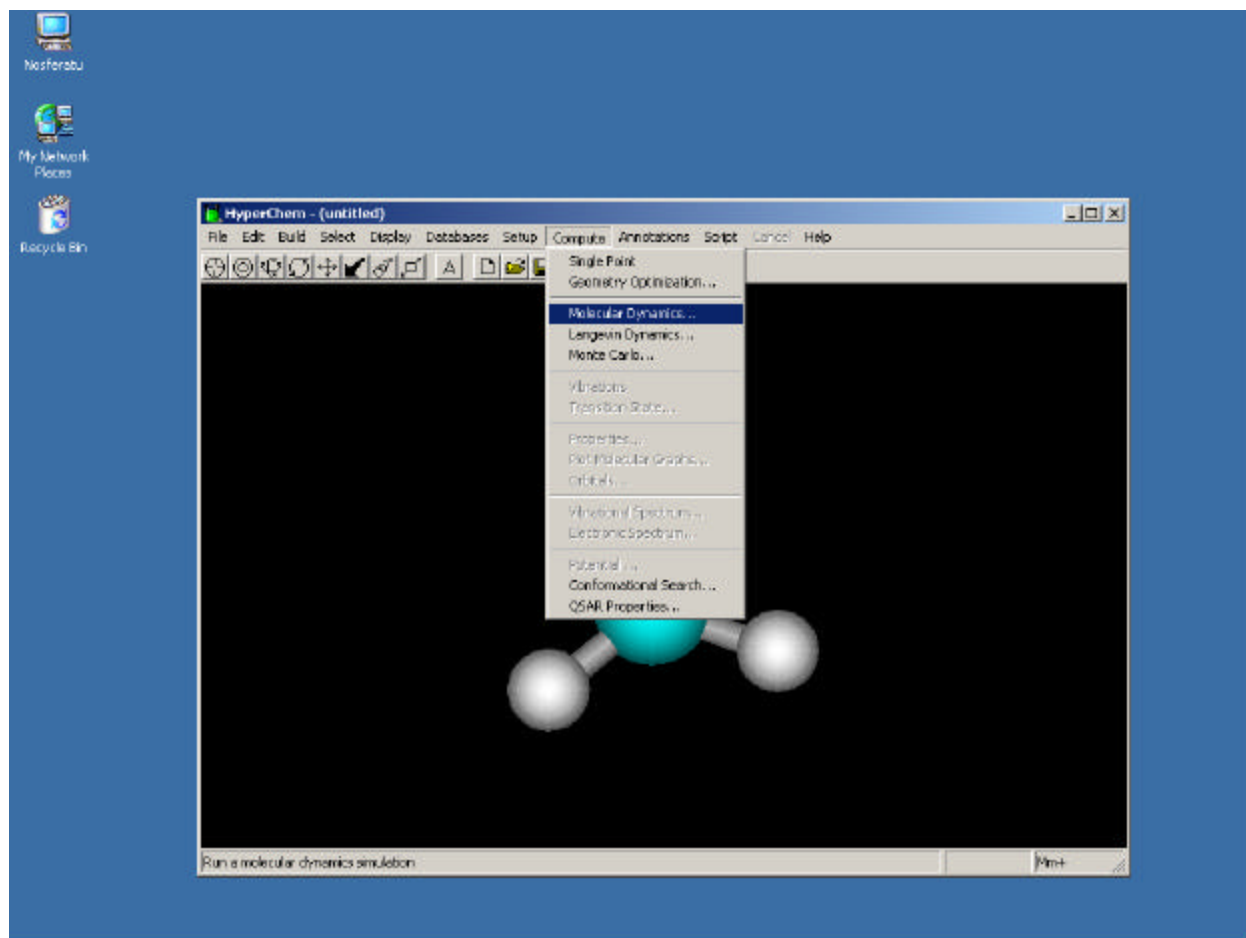


Do this for all structures above. Each time, you'll have to go to "file" "new" to open a new page (do not save the structures). This tool is designed to help you visualize the molecules you are working with. Feel free to try the various rendering options, but before building a new molecule, change the rendering back to "stick."

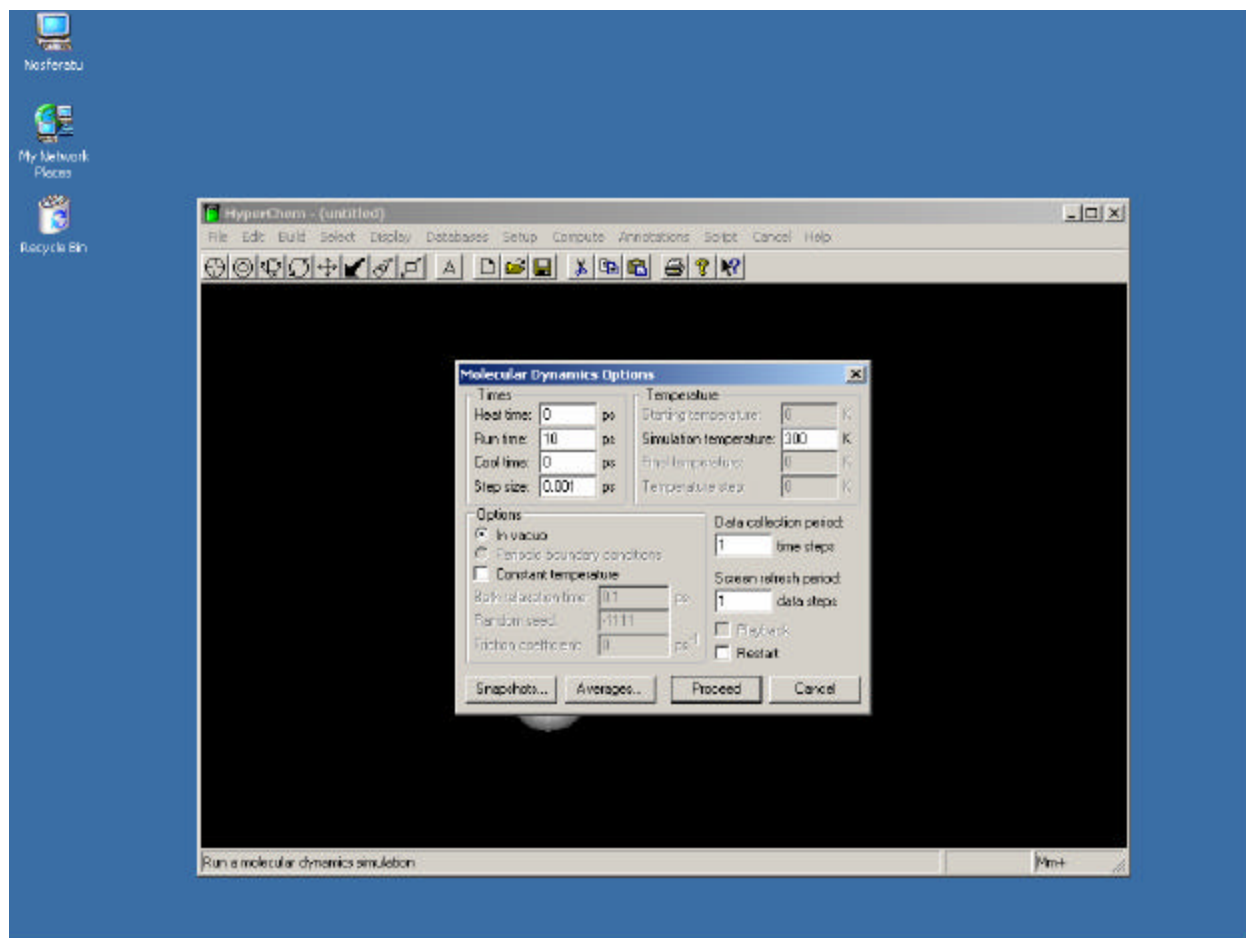
For at least a few of the molecules (you can do it for all of them if you like), let HyperChem simulate the motions the molecule can have (this is just ONE of the computational tools available in HyperChem). With the molecule on the screen, click on "setup" and "molecular mechanics."



This will open up a dialog box. Be sure that “MM+” is selected, and close the dialog box by clicking “OK.” Molecular mechanics is a technique chemists use to simulate motion of a molecule. Don’t worry about any of the options. Now, click on “compute” and “molecular dynamics...”.



Be sure the options are set as below:



Once the settings are in (heat time 0 ps, run time 10 ps, cool time 0 ps, step size 0.001 ps, simulation temperature 300K, in vacuo selected, constant temperature deselected, 1 time steps and 1 data steps), click proceed and watch the screen carefully. Take observations.

Experiment 10: Gas Laws

Purpose: To become familiar with the behavior of gases

PPpppphhhhh: Yeah, that's right, I typed it...

Background:

None

Introduction:

When one thinks of the gas laws in terms of the medical field, respiration immediately springs to mind. After all, respiration allows us to exchange oxygen for use in the cells with carbon dioxide, the bi-product from the cells, and since we are not aquatic animals, our respiration takes place in a gaseous medium. Since respiration takes place in a gaseous medium, it is subject to gas laws. Two of the most significant laws governing respiration are Boyle's Law and Poiseuille's Law, but let's not forget Charle's law, without which temperature calculations would be impossible.

Let's begin with a review of volume. Volume we all know. A one dimensional structure such as a line has length, a two dimensional structure such as our shadows have area and as three dimensional creatures, we have volume. Volume is an effect of the third dimension, which is where *most* of us exist, although there may be those who seem to be from dimensions other than our own. Anyway, let's start out with a length, say 1 cm. If we attach another length to the end of the first one, also 1 cm, we have defined a box, which has area. The area is the two sides multiplied together, or, 1 cm^2 . To add a third dimension, we can place a 1 cm line connected at the same point where the first two lines are connected and at a right angle to each of them. The only way this is possible is to place it perpendicular to the plane defined by the first two lines, thereby defining a box. The volume of the box will be the length of all three sides multiplied together, or 1 cm^3 . This volume is one cubic centimeter, or 1 cc. It is also, by definition, 1 mL. Thus, $1 \text{ cm}^3 = 1 \text{ cc} = 1 \text{ mL}$. The mL is the connection to liters, L, which is the metric unit used for volume.

Pressure, on the other hand, is a force per unit area. Force is mass (a measure of the quantity of matter) times acceleration, $F=ma$. The most familiar force in this country is the pound, lb. Weight is a force. The acceleration is acceleration due to gravity, which is smaller on the moon. The amount of matter we would have on the moon, m , would be constant and thus, our mass would be constant, but because the acceleration constant due to gravity, a , would be smaller on the moon than it is on the earth, our weight would be less on the moon than here on earth. So, if we take a weight and divide that weight by an area, we have force. For instance, suppose there is a person that weighs 180 lb with feet that have a surface area (in shoes) of about 99 in^2 . Then the pressure that person is exerting on the earth as he or she is standing still is $180 \text{ lb}/99 \text{ in}^2 = 1.82 \text{ lb/in}^2 = 1.82 \text{ psi}$. ("psi" stands for "pounds per square inch".)

Boyle's law relates gas volume to pressure. Boyle carefully measured how the volume of a gas changes as he varied the pressure on that gas. He discovered that for a system with a fixed amount of gas (n) and temperature (T), a plot of volume versus pressure gave a straight line with a negative slope. A more common way to state this would be to say that as pressure increases, volume decreases. We say that for constant n and T , volume is inversely proportional to

$$V \propto 1/P|_{n,T}$$

pressure;

$$VP = k|_{n,T}$$

The vertical line above with the subscript " n,T " is a mathematical symbolism used to remind us that this is true only if the number of moles of the gas n and the temperature T remain constant. A proportionality can be converted into an equation with the introduction of some constant, in

$$V = k/P|_{n,T}$$

our case k , even if we do not know the value of this constant. Thus,

or, since k must be constant,

$$V_1 P_1 = V_2 P_2$$

for any two states 1 and 2 (that is, for any initial state 1 to any final state 2 where we vary either pressure or volume.

Charles' decided that he was rather more interested in the relationship between the volume of a gas and the temperature of the gas at constant number of moles n and pressure p . Holding the pressure constant on the cylinder (as simple as not adding any weight to the cylinder, making its pressure equal to atmospheric pressure), he measured the volume of a gas as he heated and cooled the cylinder.

Charles discovered on plotting volume and temperature that there was a direct proportionality, that is, as temperature increased, so increased volume. He wrote the corresponding proportionality as

$$V \propto T|_{n,P}$$

The next step, as with Boyle, is to remove the proportionality by adding a constant;

$$V = rT$$

where r is some constant, or

$$V/T = r$$

Since r is constant for any two states 1 and 2, we can write

$$V_1/T_1 = V_2/T_2$$

Unfortunately, there IS a problem with Charles.

See, if you have one of the temperatures set at zero, then the equation becomes undefined. How does one circumvent this problem? Well, Charles' did so by extrapolating his data all the way to $V=0$. Such a condition, where the volume occupied by the gas is zero, is only possible for an ideal gas, and since there are no ideal gases, this becomes a hypothetical limit. It is not possible, and yet, in this impossible situation, Charles' noticed something wonderful; no matter what gas, or mixture of gases he used, no matter what pressure he kept the gas, no matter how many moles of gas he started with, these lines *all* extrapolated to *exactly the same temperature*; -273.15°C .

No matter what he did, he could NEVER reach a temperature in his extrapolations below this temperature; that means that this must be the theoretical limit of the temperature scale. Recall that temperature is directly related to kinetic energy, or motion. Doesn't it make sense, then, that there must be a point where there is no more motion, where the temperature is so cold that all motion in a molecule actually *stops*? And once this state is reached, is it possible to have a lower temperature, since temperature is related to motion of molecules? No, of course not, because we can never have less motion than absolutely no motion at all. This temperature is called "absolute zero"; it is the coldest temperature theoretically possible, and it corresponds to a state where there is no molecular motion at all; no movement, no vibration, nothing.

We can use this fact to get around the "undefined equation" problem. If we take the centigrade temperature scale, and add to it -273.15 , then we get a new temperature scale where the temperature can *never* go below zero. In fact, since this was a theoretical limit only, we can never reach absolute zero either, so we will always have a positive number for temperature. We call this temperature "Kelvin" (the corresponding absolute temperature based on the Fahrenheit scale is called "Rankine"). Whenever working with temperature in the gas laws, you must *always* convert to Kelvin.

In respiration, Boyle's Law is the one that relates to the action of the lungs. As one expands his/her diaphragm, the volume of the lungs increases resulting in a decrease in pressure inside the lungs. Because there is an open airway keeping the lungs in contact with the open air, fresh air will rush into the lungs to equilibrate the pressure. As one contracts her/his diaphragm, the volume of the lungs decreases, thereby increasing the pressure inside the lungs. Now air will

rush out of the lungs in order to equilibrate the pressure. How quickly the air will rush through the airways is a function of the radius of those passages and the pressure difference between the two ends.

This is Poiseuille's Law. Poiseuille determined empirically in 1840 that the mass of air

$$Q = \frac{p\Delta p}{8\eta l} r^4$$

that will flow through a tube per unit time can be determined by

Here, Q is the mass of air that can flow through a tube per unit time (such as, for instance, g/sec), Δp is the difference in pressure between one end of the tube and the other (without a pressure difference, there would be no flow), η is the viscosity of the air (the higher the viscosity, the greater the resistance to flow; for instance, water has a low viscosity, honey has a high viscosity), l is the length of the tube, and r is the radius of the tube. The mass of a gas is related to the number of moles of gas through the average molecular weight of the gas, $n=Q/M$, where n is the number of moles and M is the average molecular weight of the gas. The volume of the gas is directly related to the number of moles of gas, $V=nRT/P$ where V is the volume, R is the Universal Gas constant, T is temperature and P is the pressure of the gas. Thus, we find that

$$V = \frac{pRT\Delta p}{8\eta lMP} r^4$$

$V=QRT/MP$, or $Q=VMP/RT$. Thus, we can substitute for Q , and on rearrangement, we find V is the volume of air flow per unit time. Of course, π , R and 8 are all constant, and typically we can take T , η , l and M to be constant as well. Therefore, typically we adjust the pressure difference to get the volume per unit time that we want through a tube of radius r .

Consider a patient with emphysema. The condition causes a constriction of the airways, such that the radius of the airways decreases. Poiseuille tells us that all else being the same, this constriction will decrease the amount of air they can get per second (since the length and air viscosity is constant). In fact, because it is a fourth power relationship ($r^4=r*r*r*r$), the amount of air the patient can get is *greatly* diminished. Poiseuille also tells us that the patient will try to make up for this deficit by increasing the pressure difference between the atmosphere and their lungs. Boyle tells us that this can be accomplished by increasing the volume of the lungs more drastically, which will greatly decrease the pressure inside the lungs, thereby creating a greater pressure difference when compared with the atmosphere. Therefore, the patient will breath harder in order to more greatly increase the volume of their lungs in an effort to get more air. Now, of course, we have a problem, because as the patient breaths harder, they are doing more work. This requires more oxygen as well as producing more carbon dioxide which must be eliminated more quickly (also affected in an analogous manner as oxygen), which will require the patient to breath even faster through restricted airways in order to get the extra oxygen and eliminate the excess carbon dioxide. Can you imagine what that must be like?

Experimental Methods:

This experiment is broken down into three parts, one to demonstrate Boyle's Law, one to demonstrate Charles' law, and the other to demonstrate Poiseuille's Law. It does not matter which you choose to do first. **Wear your eye protection and aprons at all times. Follow all safety guidelines strictly.**

Boyle's Law: Obtain a modified "Leur Lock" portion of a syringe, its plunger and a ruler. **If the syringe has a plunger already partially in it, return it for a syringe which has no plunger in it!** The syringe has been modified in two ways; first, no needle is available. Second, the end of the syringe has been sealed off to prevent air from escaping out of or going into the barrel once the plunger is in place.

2. If necessary, place the plunger back into the barrel. Test the syringe for air leaks by pressing the plunger with your thumb as far into the syringe as possible. Carefully listen for any hissing sounds. If you hear any hissing sounds, or if the plunger does not return to the top of the syringe on release at any point in the experiment, the syringe has an air leak and will not work. Report this to your lab supervisor and get a new syringe.

3. Remove the plunger from the syringe and carefully measure the inner diameter of the syringe with a ruler. Record your results in centimeters on the report sheet. Replace the plunger.

4. Using an adjustable clamp, fasten the syringe vertically to a ring stand. The syringe must be attached with the plunger on the top such that a mass can be balanced on top of the plunger. Make sure the syringe is as vertical as possible.

5. Obtain a book with a known mass from the instructor. Record the mass of this object in grams on the report sheet.

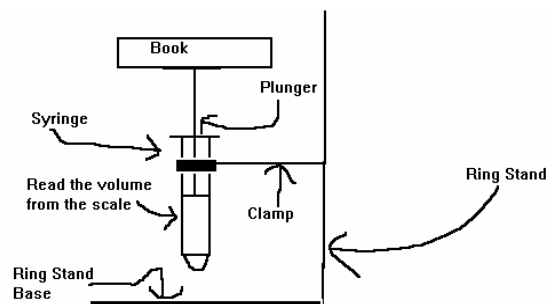
6. Carefully balance the object on the flat surface on top of the plunger. You want this object to be balanced such that it is not touching anything and can sit on the syringe without falling. This *will* require patience.

7. Once the object is balanced, push down on the object slightly to force the plunger down. Release and allow the plunger to rise back up. When the plunger has stopped moving, allow it to sit undisturbed for about 10 seconds.

8. Carefully read the volume of the syringe from the scale on the syringe barrel to the nearest mL. **Be careful not to let the book fall on you if it should fall off of the syringe!** Record this volume in mL on the report sheet.

9. Remove the object from the plunger and allow it to return to the top. If any sign of a leak ever appears throughout this experiment, report it immediately to the lab supervisor.

10. Repeat steps 4 through 8 for *at least* 5 different masses.



Charles' Law: Obtain a sealed syringe with a plunger already in place and a thermometer. **Never remove the plunger from this syringe!** If your does not have a plunger in it, return it for one that has the plunger already in it and positioned in the middle of the volume range.

1. Read the thermometer and record the temperature in degrees celcius. **Caution! You're your thermometer contains silver colored mercury, remember that mercury is toxic. Be VERY cautious when handling these thermometers to avoid breaking them. Should one break, do NOT attempt to clean the broken thermometer up yourself; report it to your lab supervisor IMMEDIATELY!** Record this temperature in the Charles' law table. Read the volume on the syringe and record it next to the room temperature.

2. Prepare an ice bath by filling an appropriate sized beaker about 2/3 full of ice and adding water almost to the top of the ice. As this is chilling, begin heating a second beaker to boiling over a Bunsen burner. **Always be cautious when using an open flame or around hot water!**

3. Place the plunger in the ice bath for several minutes, along with the thermometer. When the temperature on the thermometer stabilizes, record the temperature in the table. Read and record the corresponding volume from the syringe.

4. Once boiling, place the thermometer and syringe in the boiling water. **CAUTION! Be careful NOT to allow the syringe to get too close to the edge of the beaker, or it will melt!** Allow the thermometer and syringe to remain in the boiling water for several minutes. Once the temperature has stabilized, read the temperature and corresponding volume. Record these values in the table.

5. IF your lab supervisor has an additional temperature bath, ask him/her to put your syringe into it (**CAUTION! This will be TOO cold for you to do without DIRECT supervision; if she/he is NOT standing nearby, wait until he/she returns before proceeding!**). Ask the temperature, and record it while you wait for the syringe to cool. Read the volume of the syringe when you are instructed to do so.

Poiseuille's Law: There are several additional safety precautions to observe in this part of the experiment. Please read this section ahead of time and follow all additional precautions carefully. Although this part of the experiment involves no toxic chemicals *per se*, remember that **all laboratory equipment and surfaces are contaminated with chemicals that can be toxic. Therefore, avoid contact of any of these items or anything that comes in contact with them with your mouth.** You should have available to you at least three different types of straws with different diameters.

1. Obtain from your instructor one of each of the minimum of three straws for each member in your lab group with varying diameters (a minimum of three straws total for each person in the group). Place these on a clean paper towel until you are ready to use them. **Do not allow the straws to come into direct contact with any lab surface.**

2. If provided, write down the diameter of each straw size, or carefully measure the diameter of each straw and record these diameters in cm in the report sheet. **Keep careful track of the side of the straws that you used to measure these diameters since they have now**

come in contact with a laboratory ruler. Consider this side of the straw to be contaminated.

3. Cut the side of the straws that have been measured off of the two longest straws with scissors such that all three straws are now the same length. **Keep careful track of the side of the straws that have come in contact with the scissors. This is now the contaminated side.** Discard the pieces of the straws that have been removed.

4. Have your lab partner time you for this part; with the smallest straw first, draw in a deep breath, and time how long it takes you to exhale completely through that straw. Try to exhale as rapidly as possible. Be careful not to allow yourself to pass out or cause physical harm by trying to blow too hard. If you cannot exhale completely (you run out of breath too soon), take the time your partner measured and record it as ">time", to indicate that it takes longer than the time recorded.

5. Repeat step 4 for all straws.

6. Have your partner(s) repeat this part of the experiment beginning from step 3.

Calculations:

Boyle's Law:

Calculate the radius of the plunger from its diameter (radius=diameter/2). Convert from cm to inches (1 in = 2.54 cm). For simplicity of calculation, we assume the plunger is flat (even though we know it is not). Therefore, we can calculate the area of the plunger as $A=\pi r^2$, where r is the radius of the plunger in inches and π is about 3.141592654.

Convert the mass of the object into pounds, lb (1 kg = 2.2046 lb). To calculate the pressure exerted by the mass, simply divide the weight of the object in pounds by the area of the plunger in square inches (psi). Convert from psi into atmospheres (1 atm = 14.696 psi).

The total pressure on the gas trapped in the syringe is the pressure exerted by the mass (see above) plus the pressure exerted by the atmosphere, which we will assume to be 1 atm. Thus, for total pressure, put in the pressure exerted by the mass plus one. On a clean piece of graph paper, carefully plot the volume of the gas (y axis) versus the total pressure (x axis). Draw the best straight line that you can through the experimental points.

Take the inverse of each pressure and fill in the column 1/Pressure. On a clean piece of graph paper, carefully plot the volume of the gas (y axis) versus 1/Pressure (x axis). Draw the best straight line that you can through the experimental points. Calculate the slope of this line. This slope is the value of the constant $k(=nRT)$.

Charles' Law:

Create a graph to plot volume versus temperature, but be SURE that the temperature scale extends *at least* to $-600\text{ }^{\circ}\text{C}$, and that the volume scale extends to 0 mL. Carefully plot your data points. Extrapolate the line with a straight edge to $V=0$, and read the resulting temperature. This is your estimate for absolute zero.

Poiseuille's Law:

From the diameter of the straws, calculate their radius. Calculate the fourth power of this radius ($r^4 = r*r*r*r$).

Gas Laws Experiment: Boyle, Charles and Poiseuille Pre-Lab Questions

- 1) If we have a system where the volume of gas that can flow through a tube in 1 second is 4.2 L, what volume will be able to flow through the tube if we halve the length of the tube?

- 2) What is the function of the objects we are resting on top of the syringes in the Boyle's Law portion of this experiment?

- 4) What must we be careful of in the Charles' law portion of this experiment where we are putting the syringe in the boiling water?

- 5) What specifically are we measuring with the graduated cylinder in the Poiseuille's Law portion of this experiment?

- 6) Explain why it is better to have a sharp needle rather than a dull one when receiving a shot in terms of pressure and the amount of force necessary to break through the skin.

- 7) Why is it important to keep track of the contaminated side of the straw in the Poiseuille's Law part of this experiment?

Gas Laws Experiment: Boyle, Charles and Poiseuille Report Sheet

Boyle's Law:

Syringe Inner Diameter: _____ cm

Syringe Inner Radius: _____ cm

Syringe Inner Radius: _____ in

Syringe Inner Area: _____ in²

Object Mass (g)	Object Mass (kg)	Object Mass (lb)	Pressure Exerted by Object (psi)	Pressure Exerted by Object (atm)	Volume of Gas (mL)	Total Pressure (atm)	1/(Total Pressure) (atm ⁻¹)

Charles' Law:

Temperature in °C	Volume in mL

Poiseuille's Law:

Straw Diameter (cm)	Straw Radius R (cm)	R^4 (cm ⁴)	Time for run 1 (sec)	Volume for run 1 (mL)	Time for run 2 (sec)	Volume for run 2 (mL)	Time for run 3 (sec)	Volume for run 3 (mL)

Gas Laws Experiment: Boyle, Charles and Poiseuille

Post-Lab Questions

- 1) Does the plot of volume versus total pressure demonstrate the inverse proportionality of volume and pressure? If so, why? If not, what were you expecting to see?
- 2) From the plot of volume versus 1/pressure, calculate the slope. This is the value for the constant k .
- 3) Since we know from the ideal gas law that $k=nRT$, then $n=k/RT$. Assume that the gas inside the cylinder is at 19°C (approximately room temperature). Calculate the number of moles of gas in the cylinder, n ($R=82.06 \text{ mL}\cdot\text{atm}/\text{mol}\cdot\text{K}$). HINT! Don't forget to convert your temperature to Kelvin!
- 4) What is your estimate for absolute zero from your plot of volume versus temperature? List as many sources of error as you can.
- 5) In the Poiseuille part of the experiment, how did the volume of gas change as the radius of the straws change?
- 6) What sources of error exist for this experiment, and how would these sources of error affect your results?
- 7) You have a very non-technical patient suffering from emphysema. Explain in your own words using simple terms how Boyle's Law and Poiseuille's Law are teaming up against him or her.
- 8) Explain, in simple terms, exactly what is happening at absolute zero (assuming it can be reached). Why is this a theoretical limit, rather than a real limit?
- 9) A patient arrives in the emergency room who needed an emergency Tracheotomy. A hole was cut into this patient's Larynx at the base of the neck, and a tube, rather smaller than the Trachea, was inserted into the Trachea. Two effects are going on; the length of the breathing tube is shorter, but the radius is smaller. How will each of these effect the volume of air the patient can receive? Looking carefully at Poiseuille's Law, do you expect the net flow to be smaller or greater? Why? (HINT! Look at the *powers* involved in the radius and length of tube.)

Experiment 11: Acids and Bases

Purpose: To gain experience in the properties of acids and bases

Purvey: To sell or provide (and I'll bet you thought it meant something else, didn't you?)

Background:

See "Basic Laboratory Procedures"; Litmus paper

Introduction:

It does not take a great stretch of the imagination to recognize that, in some way, acids and bases are nothing more than a specialized class of ionic compounds, at least in the Arrhenius definition. So why, then, are acids and bases important enough to warrant their own place in a first year course? Because, when we are talking about acids and bases, we are talking about the chemistry of water-based solutions (typically), and water is of critical importance to us as human beings.

Think about it; 70% of the earth is occupied by water. YOUR BODY is 90% water. When we talk about water chemistry, we are talking about the chemistry that keeps both this planet, and us, alive, and while water occupies every day of our lives, in one way or another, it is more exotic than you might think.

Oh, sure, I know what you're saying; "what's so exotic about water? I can get it from my tap; I use it for a hundred things every day, it falls out of the sky!" That's true; on this planet, water is abundant. But did you ever stop to consider that water is the *only compound* known to science that actually expands on freezing? Were this not true, the oceans would be frozen solid, and life would be impossible on this planet. Of all the compounds known to science, the only one that expands...is water.

Today we will be playing with acids and bases, just trying to gain some experience in their properties and behavior, and keep in mind, as we are doing so, that we are playing with the compound that is most important to life on our planet!

In the second part of the experiment, we will be looking at the properties of salts in water. This might seem like an odd experiment to be throwing salt around, but, salts can be thought of as the bi-products of an acid/base neutralization, where the anion is the left over part from the acid, and the cation is the left over part of the base. For example, Ca_3PO_4 can be formed from H_3PO_4 (one hydronium added for each negative charge of the anion) and $\text{Ca}(\text{OH})_2$ (one hydroxide ion added for each positive charge of the cation). Would you expect the solution from the calcium phosphate to be acidic, neutral, or basic?

Procedure:

You will need the Pasco pH sensor for this experiment; start your Pasco system with the pH probe as usual. You will not need the graph for this experiment. Set up for manual data collection; we will be typing in concentrations for the first part.

Part I: pH of common household products

In the reagent area, you will find 0.1 M HCl and 0.1 M NaOH. Work with each of these in turn; starting with the HCl, add 1 mL to 100 mL of water to make 0.001 M HCl. Then dilute further by taking 1 mL of the 0.001 M HCl and adding it to 100 mL of water. Do this a total of three times (you will have four solutions of HCl). Repeat this in a separate set of containers for the NaOH. Measure the pH of each of these solutions and record the result. Remember to rinse off the pH probe with distilled water VERY THOROUGHLY between each solution and before the first solution.

You will find a collection of common household chemicals. For any solids, add a little bit to water. Test each chemical (or their corresponding solutions) using the pH probe. Record your results.

Part II: Salts

You will find a series of salts in the lab. Take a spatula tip full (just a little bit, maybe half the size of a pea) and dissolve it in a little water in a clean container. Before taking the pH, try to determine the identity of the acid and base that were mixed to make this salt (or that COULD be mixed to make this salt). Predict if you expect the solution to be acidic, basic or neutral. Using the Pasco system, measure the pH of the solution and record your findings.

Calculations:

None.

Observations:

Solution of HCl	pH	Solution of NaOH	pH
1×10^{-1}		1×10^{-1}	
1×10^{-3}		1×10^{-3}	
1×10^{-5}		1×10^{-5}	
1×10^{-7}		1×10^{-7}	

Observations:

Material	pH	Acidic, basic, or neutral?	strongly or weakly?

Observations:

Salt	Prediction: acidic, basic or neutral?	pH	Acidic, basic or neutral?	Strongly or weakly?

Observations:

Pre-lab Questions:

1. We are making a serial dilution of HCl and NaOH; what does this mean?
2. How do you suppose we will decide if a solution is “strongly” or “weakly” acidic or basic?

Post-lab Questions:

1. What was the biggest surprise to you in the household items?
2. Some salts you could predict the pH of in solution, some you could not. Why?

Experiment 12: Le Chatliere's principle

Purpose: To gain experience with Le Chatliere's principle

Purloin: To steal

Background:

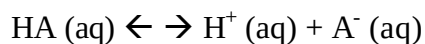
See "Using the Pasco System" and
"Basic Laboratory Procedures"; Pipettes

Introduction:

Equilibria are critical to life. I mean, if chemical reactions were not reversible, we would be one-shot machines. Our metabolism, which is nothing more than the series of chemical reactions that keep us alive, would run through their cycle one time, and we would expire. This does not happen, because most of the reactions in our metabolic pathways are reversible; what is used up in one step, is regenerated in the next.

Le Chatliere tells us that, when it comes to equilibria, we are allowed to be bullies. We can push the equilibrium to the reactants by adding products, or to the products by adding reactants. We can even push the equilibrium around by changing the conditions, such as temperature.

Today, we will be playing with what is probably one of the most important equilibria to us as humans; acid/base equilibria. Essentially, we are going to create a buffer solution, and play with it for a bit. We'll be using an (unknown) organic acid; the general equilibria will be



We are using an organic acid because they are all weak, and therefore appropriate choices for creating a buffer solution. Recall the equilibrium constant, and equilibria calculations as you've learned them in class, and we will see, I hope, some of these principles come to life.

Procedure:

You will need to start the Pasco system with the pH probe. Use the standard methods for starting the system; you will want to choose manual data collection.

Part I: Determination of the equilibrium constant K_a

You will find two bottles in the lab, one of an unknown organic acid, and one of its corresponding salt. Both of these are the same concentration which will be provided for you in class. Using the pipettes provided, thoroughly mix 20.00 mL of each of these solutions into a

clean, dry flask. Determine the pH of the solution, mix the solution more and determine the pH again. Continue doing so until the fluctuations in pH are not more than ± 0.1 . From the pH, determine the pK_a , the K_a , of the acid, and the Gibb's Free Energy, ΔG° at equilibrium.

Part II: Influence of temperature

Pour approximately 5 mL of the mixture into two test tubes. Place one into an ice bath, and the other into a hot water bath. You will want each to reach equilibrium, so go on with at least part of the rest of the experiment, but do not allow the test tube in the hot water bath to lose too much volume. Determine the pH of each solution. Determine the heat of equilibrium ΔH° .

Part III: Influence of adding reactants or products

Place 5.00 mL of the mixture into four clean, dry test tubes using a pipette. Using the pipettes provided, put 1.00 mL of the organic acid solution into test tube 1, 1.00 mL of the organic salt solution, 1.00 mL of 0.100 M HCl into test tube 3 and 1.00 mL of 0.100 M NaOH into test tube 4. Using the methods from class, predict the pH that should be found for each of these solutions. Using the Pasco system, determine the pH of each of these solutions experimentally.

Calculations:

Part I:

Recall from class that when $[HA(aq)] = [A^-(aq)]$, the resulting solution has $pH = pK_a$, and the relationship between Gibb's Free Energy and the equilibrium constant, $K = -RT \ln K$.

Part II:

Recall van't Hoff's equation from class, and the relationship between Gibb's free energy, enthalpy and entropy; $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$; recall that we need to solve this at 25°C to determine the change in entropy.

Part III:

Recall the Henderson-Hasselbach equation from class. Use the value of pK_a determined in part I.

Observations:

Part I:

pH of solution:

pK_a of solution:

K_a of solution:

ΔG° for equilibrium:

Observations:

Part II:

pH of cold solution:

K_a at 0°C:

pH of hot solution:

K_a at 100°C:

ΔH° of acid dissociation:

Observations:

Part III:

Test Tube	Additive	Predicted [HA(aq)]	Predicted [A ⁻ (aq)]	Predicted pH	Experimental pH
1					
2					
3					
4					

Observations:

Pre-lab Questions:

1. Why are we using an organic acid as the unknown acid?
2. What is special about the mixture that allows us to determine pK_a for the acid?

Post-lab Questions:

1. Now you have the K_a , pK_a , ΔH° and ΔS° for the organic acid; any predictions on what it might be? Explain.
2. How did your predicted values of pH compare with experimental? Can you account for the differences?
3. Suppose you forgot to calibrate you pH probe; all of the readings are off, but you do not know by how much. How would this influence your calculation of K_a ? Of ΔG° ? Of ΔS° ? Which of these values could you trust, and why?

Experiment 13: molecular mass of acid

Purpose: To use a titration method to determine the molar mass of an unknown acid

Purplunk: The noise my heart makes when she says “what...with you?”

Background:

See “Using the Pasco System” and “Basic Laboratory Procedures”; pipettes, burets, balance

Introduction:

Titration is the second most accurate form of quantitative analysis tool available to chemists, second only to gravimetric techniques. Between the two, however, titrations are far more common, because they are far easier than gravimetric techniques, and still afford (typically) four significant figures in accuracy.

Today you will be using titrations to determine a very fundamental property of an unknown acid; its molecular mass. Remember, molecular mass is nothing more than the mass of the material present, divided by the moles of material in the sample. Mass is easy; put it on a balance, and you get mass. Moles we can get from the titration. If you are VERY careful, you can get VERY accurate results, and you will be graded on accuracy.

Procedure:

You will want to set up the Pasco system for manual data collection using the pH probe. You will want to have it plot pH as a function of volume of base added. Set up a buret (see general procedures) for a titration.

Standardization of the Base:

Follow this procedure IF the base solution provided has not already been standardized. If the base has been standardized for you, write down the exact concentration (four significant digits) of the base and proceed to titration of the acid.

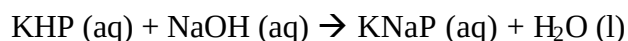
For the standardization of the base, we will be using potassium hydrogen phthalate (KHP) as our primary standard (molecular mass 204.22 g/mol). Use a CLEAN DRY spatula to measure out about 0.4 grams of KHP (you do not need exactly 0.4 g, but you do need to know exactly how much you do have to as many significant figures as possible). If your lab instructor requests you to do so, add a couple of drops of phenolphthalein. Titrate the sample either to the endpoint, or using the Pasco system to collect data as your lab instructor requests. Perform at least three titrations; if any of the runs are off by more than 0.04 mL from any other run, perform a fourth titration and discard the titration that seems to most significantly disagree with the other runs. Determine the exact concentration of the base from this step.

Titration of the Acid:

Measure out about 0.1 gram of the unknown acid (you do not need exactly 0.1 g, but you do need to know exactly how much you have and you must have at least 0.1 g to have as many significant figures as possible). If your lab instructor requests you to do so, add a couple of drops of phenolphthaleine. Titrate the sample either to the endpoint, or using the Pasco system to collect data as your lab instructor requests. Perform at least three titrations; if any of the runs are off by more than 0.04 mL from any other run, perform a fourth titration and discard the titration that seems to most significantly disagree with the other runs. Determine the molar mass of the acid from this step.

Calculations:*Standardization:*

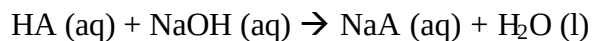
KHP is a monoprotic acid, so the reaction can be simplified as



Using standard stoichiometric techniques, calculate the molarity of NaOH for each run. Take the average and determine the standard deviation.

Molecular Mass:

The acid is monoprotic, so again, we can write



Use standard stoichiometric techniques to calculate the moles of acid for each run, then, for each run, divide the mass of the acid by the moles of the acid to determine the molecular mass. Take the average and determine the standard deviation.

Observations:*Standardization:*

Run	Mass KHP	Initial buret	Final buret	volume base	molarity
1					
2					
3					
4 (if necessary)					

Average molarity:

Observations:

Molecular Mass Determination:

Run	Mass acid	Initial buret	Final buret	volume base	moles of acid	molecular mass
1						
2						
3						
4 (if necessary)						

Average molecular mass:

Observations:

Pre-Lab Questions:

1. What is the primary standard we are using?
2. How closely should runs agree to avoid the fourth run?
3. Why do you have to be particularly careful of the tip of the buret in a titration?

Post-Lab Questions:

1. What are the major sources of error in this experiment?
2. Do you have a guess as to what the acid might be?
3. (Answer this question if you plotted the titration curve) What is the pK_a of the acid? Does this jive with your guess from question 2?

Experiment 14: Titration of Antacids

Purpose: To understand antacid neutralization power.

Purple: Favorite color of the artist originally known as Prince but then known as the artist formerly known as “Prince” but currently known as “Prince” once again.

Background:

See “Using the Pasco System” and “Basic Laboratory Procedures”; pipettes, burets, balance

Introduction:

There have been a lot of wild claims regarding antacids by the manufacturers; now you will have your opportunity to test these claims. The goal of today’s experiment is to determine how much 0.1 M HCl (roughly the molarity of HCl in stomach acid) will be neutralized per gram of antacid, per dose of antacid, and per dollar of antacid.

Procedure:Sample Preparation:

Choose three antacids; at least one liquid, at least one solid, and at least one with a mixture of active ingredients. Be sure to record the name of the antacid, the price, the number of doses per container, and the active ingredients. Weigh 1 dose of the antacid as per the instructions. If the antacid is a solid, grind it using a mortar and pestle, and place the powdered antacid into a flask. If the antacid is a liquid, measure out a dose of the antacid directly into the flask. Whether the sample is solid or liquid, add about 100 mL of water. Work with one sample at a time; you will want to perform three trials of each antacid you chose. Since you only have two flasks, one lab partner can be cleaning a flask and preparing the next sample while another lab partner is performing the titration. Each lab partner should perform approximately the same number of titrations.

Calibration of Pasco Systems:

Most probes for the Pasco system start off basically analog voltages. We must start by telling the system what voltages correspond to what pH levels. In so doing, we will use buffer solutions that are at very well known (and stable) pH.

1. Start the computer, and the Pasco software. Choose the “pH probe” sensor. Be sure that you select position “A” and that you have the temperature probe plugged into this port.

2. Choose “calibrate”; put some of the higher pH buffer to your small cup. Put the pH probe into the buffer and allow a few minutes to equilibrate. Type the correct value into the “high pH” space (for example, if it is the pH 10.00 buffer solution, type 10.00 into this space). As soon as the reading seems to have stabilized, click “read”. Return the buffer solution into the correct “used buffer” container.

3. For the low pH calibration, refill your little cup with the low pH buffer solution. Empty out your calorimeter. Put the pH probe into the buffer and allow a few minutes to equilibrate. Type the correct value into the “low pH” space (for example, if it is the pH 2.00 buffer solution, type 2.00 into this space). As soon as the reading seems to have stabilized, click “read”. Return the buffer solution into the correct “used buffer” container.

If you followed these instructions carefully, your Pasco system should be ready.

Data Collection:

In the front of the room you will find 0.1M hydrochloric acid. Choose one bottle and record the bottle number; this is the same acid you will use for the entire experiment, so use it sparingly. Fill the buret (NEVER fill a buret above your head), and set up the Pasco box to manually collect your data for you. To do so, you will want to collect data as follows;

- (1) If you have not already done so, close the calibration window by clicking OK.
- (2) Click on “sampling options”. Select “keyboard sampling”; under units put “volume of acid, mL”.
- (3) Open “table” and “graph” by dragging the appropriate icons over the pH probe icon.
- (4) Place the pH probe into the beaker with calcium carbonate; give it a couple of minutes to equilibrate.
- (5) Type “0” in for the volume added; click “read” to read the initial pH.
- (6) Add 1 to 3 drops of hydrochloric acid and give the solution a couple of minutes to equilibrate (using the pH probe to carefully stir the solution will help). Read the volume of the buret and input this for the “volume added” and click “read”.
- (7) Continue in this procedure, adding one to three drops at a time
- (8) Write the data points in your notebook. Do not count on being able to save these tables as they may be accidentally erased.
- (9) As soon as your data collection and transcription is completed, repeat the experiment with a new table and graph for the next sample.

Calculations:

On a graph of pH versus volume of acid, you will notice a rapid drop in pH at some point of the curve. At this point, all of the antacid is gone (it is called the equivalence point). Use this as the neutralization volume of the antacid. Divide the cost of the antacid by this volume to get cost per mL.

Observations:

Initial information:

Antacid Brand	Form	Dosage	Cost	Cost/Dose

Observations :

Titration

Brand	Volume run 1	Volume run 2	Volume run 3	Average volume	Cost/volume

Observations:

Pre-Lab Questions:

1. What is the approximate concentration of stomach acid?
2. What are we measuring with the Pasco system?

Post-Lab Question:

1. Look at the graphs you've generated; where is the "endpoint" (that point where you run out of antacid" indicated?
2. Measure the volume of acid neutralized for each run, and find the average volume for each antacid.
3. Calculate the volume of acid neutralized per gram of antacid, the volume of acid neutralized per dose of antacid, and \$/mL of acid neutralized.

Experiment 15: Titration of Vinegar

Purpose: To determine the concentration of acetic acid in vinegar

Purrrr-fect: Something Cat-woman used to say just before doing something villainous

Background:

See “Using the Pasco System” and “Basic Laboratory Procedures”; pipettes, burets, balance

Introduction:

Commercial vinegar is typically about 5% acetic acid (CH_3COOH). This is a classic titration in which we will determine the exact concentration of acetic acid in store-bought vinegar.

Procedure:

You will want to set up the Pasco system for manual data collection using the pH probe. You will want to have it plot pH as a function of volume of base added. Set up a buret (see general procedures) for a titration.

Standardization of the Base:

Follow this procedure IF the base solution provided has not already been standardized. If the base has been standardized for you, write down the exact concentration (four significant digits) of the base and proceed to titration of the acid.

For the standardization of the base, we will be using potassium hydrogen phthalate (KHP) as our primary standard (molecular mass 204.22 g/mol). Use a CLEAN DRY spatula to measure out about 0.4 grams of KHP (you do not need exactly 0.4 g, but you do need to know exactly how much you do have to as many significant figures as possible). If your lab instructor requests you to do so, add a couple of drops of phenolphthaliene. Titrate the sample either to the endpoint, or using the Pasco system to collect data as your lab instructor requests. Perform at least three titrations; if any of the runs are off by more than 0.04 mL from any other run, perform a fourth titration and discard the titration that seems to most significantly disagree with the other runs. Determine the exact concentration of the base from this step.

Dilution of the Acid:

The vinegar is too concentrated as it is to perform a titration, so make a dilution by putting 5.00 mL of vinegar (using a pipette) into a 50.00 mL volumetric flask and diluting as usual. This dilution means the solution you will be titrating is 10 times

weaker than the original vinegar. To take this dilution factor into account, multiply the concentration of the diluted sample by 10 to get the concentration of the original vinegar.

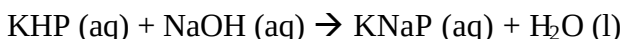
Titration of the Acid:

Measure out 10.00 mL of the diluted solution using a pipette. If your lab instructor requests you to do so, add a couple of drops of phenolphthaleine. Titrate the sample either to the endpoint, or using the Pasco system to collect data as your lab instructor requests. Perform at least three titrations; if any of the runs are off by more than 0.04 mL from any other run, perform a fourth titration and discard the titration that seems to most significantly disagree with the other runs. Determine the concentration of the acid from this step.

Calculations:

Standardization:

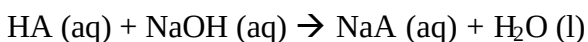
KHP is a monoprotic acid, so the reaction can be simplified as



Using standard stoichiometric techniques, calculate the molarity of NaOH for each run. Take the average and determine the standard deviation.

Concentration of the Acetic Acid:

The acid is monoprotic, so again, we can write



Use standard stoichiometric techniques to calculate the moles of acid for each run. Don't forget to take the dilution into account; once you calculate the concentration of acetic acid in the diluted solution, multiply by four to get the concentration of the original vinegar. Convert the concentration from molarity to % w/v in the standard method. Take the average and determine the standard deviation.

Observations:*Standardization:*

Run	Mass KHP	Initial buret	Final buret	volume base	molarity
1					
2					
3					
4 (if necessary)					

Average molarity:

Standard Deviation:

Observations:

Concentration of Acetic Acid in Vinegar:

Run	Initial buret	Final buret	volume base	moles of acid in dilution	molarity of acid in dilution	molarity of acid in vinegar
1						
2						
3						
4 (if necessary)						

Average molarity:

Standard Deviation:

Average % w/v:

Observations:

Pre-Lab Questions:

1. What is the dilution factor of our vinegar?
2. What volume of the diluted vinegar are we titrating?
3. What is the indicator?

Post-Lab Questions:

1. Most commercially available vinegar claims 5% acetic acid; how close did your results reflect this? What is the percent difference between what you found and what the manufacturer claims?
2. How comfortable are you with your results? If you had to testify in court, would you be willing to?
3. Site possible sources of error, and estimate if these errors would result in a final concentration that is too high, too low, or could be either.

Experiment 16: colligative properties

Purpose: To gain experience with solubility and colligative properties

Pink Floyd: A progressive rock band from the '80's. Good tunes, too!

Background:

See "Using the Pasco System"

Introduction:

Do you remember the Crystalline entity from the first season of Star Trek Light (aka Star Trek: The Next Generation)? Do you remember how the entity referred to humans? "Disgusting bags of mostly water," as memory serves. Not a bad description, since the human body is approximately 90% water. However, it probably would have been a better description to refer to humans as "disgusting walking solutions."

If our body is 90% water, that means we are 10% other "stuff". Water can be thought of as the solvent in our bodies (the solvent can be thought of as the "carrier"; that in which the solute is dissolved), while the rest of the stuff (proteins, lipids, DNA, nutrients, waste, and a plethora of other things) can be thought of as the solute (the "active ingredients" in a solution; what makes the solution of interest). Typically, the solvent is the compound present in greater amounts, but this is not always the case; it is more generally correct to think of the solute as the active ingredient, that is, the reason for us to pick up the solution in the first place, while the solvent is the carrier for the solute.

It is not technically correct to speak of "heterogeneous" or "homogeneous" solutions. All solutions, by definition, must be homogeneous. To be a true solution, you must have an even distribution of solute throughout every part of the solution. If you have an uneven distribution, then you have a mixture, which is heterogeneous. In fact, the line probably ought to have been "disgusting walking mixtures." Mixtures are cloudy in appearance caused by the diffraction of light off of the separate regions of the mixture (called the "Tyndall" effect). Even a mixture that looks homogenous (or advertised to be so as in the case of milk) is in fact not a solution at all if it is cloudy.

Solubility refers to the amount of solute that can be dissolved in a solution at a given temperature (and pressure if the solute is a gas). Notice that it does *not* speak to how long it takes to dissolve, just the maximum amount. This means that things like stirring, which makes things dissolve faster, will not influence solubility, just how long it takes for the solvent to dissolve. The proof is trying to dissolve excess solute in a solution that has already reached its solubility limit (called "a saturated solution"). Temperature will influence solubility, as will pressure but only if the solvent is a gas.

The strongest influence of solubility is the nature of the solute and the solvent. There is an old rule of thumb when discussing solubility; "like dissolves like." Although there are exceptions to this rule, generally speaking it means that polar solutes will dissolve in polar solvents, and non-polar solutes will dissolve in non-polar solvents. This provides interesting insight into substances, with a quick and convenient experiment to

test polarity. It also provides insight around the home. If you wanted to remove peanut butter from a container, for example, you know that water will not work. Well, since water will not work, it might be worth your while to try a non-polar solvent first, like cooking oil. Once the peanut butter is gone, the cooking oil can be easily removed by detergent.

The presence of a solute in a solution will influence the properties of the solution. That a solute will dissolve in a solvent to any extent means that the interaction between the solvent and solute molecules (or ions) is more energetically favorable than the interactions between the molecules (or ions) of the solute alone. In other words, the presence of the solute will attract solvent to itself and hold onto it strongly. This results in stronger intermolecular interactions in a solution than you would have in the solvent alone. As such, certain properties will change.

These changes (freezing point depression, boiling point elevation and vapor pressure depression) depend on the concentration of the solution, but not on the identity of the solute. That is, the same concentration of any solute will produce the exact same properties. We call these “Colligative properties.”

Experimental Methods:

You will eventually want to use the Pasco temperature probe for this experiment.

Diffusion:

Fill a large beaker with water. Place it somewhere on your bench where it can sit undisturbed for the entire lab period. When the water seems calm, add one drop of ink and be careful not to disturb the beaker. Record your observations. Allow the beaker to sit undisturbed as you continue on with the rest of the experiment, periodically taking observations.

Your professor may have a demonstration of a natural semi-permeable membrane set up. If so, take observations on the demonstration.

Solubility:

Solubility of Gases:

1. Ammonia (**CAUTION! CORROSIVE, TOXIC, LACRYMATOR**) is highly soluble in water. Your professor will demonstrate this for you with what is known as the “Ammonia Fountain Demonstration.”

Solubility of Compounds in Polar and Non-Polar Solvents:

Water is a polar solvent, hexane is non-polar. In six test tubes, put 1 mL of water into each. In six different test tubes, put 1 mL of hexane (**CAUTION! Toxic, highly flammable**) into each. Pair the test tubes, one with water and one with hexane, into four pairs. Add equal amounts of each of the following to each of the two test tubes within a pair;

- (1) sodium chloride (an ionic compound); the tip of a spatula full (about 0.1 g)
- (2) calcium phosphate (**CAUTION! Toxic**) (an ionic compound); the tip of a spatula full (about 0.1 g)
- (3) table sugar (sucrose, a polar covalent compound); the tip of a spatula full (about 0.1 g)
- (4) Naphthalene (**CAUTION! Toxic, flammable, toxic fumes**) (typical ingredient of urinal cakes; a non-polar covalent compound); the tip of a spatula full (about 0.1 g)
- (5) Ethylene glycol (**CAUTION! Toxic**) (typical ingredient in antifreeze; a polar covalent liquid); about 3 drops (enough to see)
- (6) Paraffin Oil (**CAUTION! Toxic, flammable**) (a non-polar covalent liquid); about 3 drops (enough to see)

Agitate each solution by flicking the test tube. Describe what happens in each.

Effect of Temperature on Solubility:

Using a clean beaker, begin heating about 50 mL of distilled H₂O. Place each of the following into three medium sized test tubes:

- (1) 2 g potassium sulfate K₂SO₄ (**CAUTION! Toxic**)
- (2) 7 g sodium sulfate Na₂SO₄ (**CAUTION! Toxic**)
- (3) 5 g potassium chloride KCl (**CAUTION! Toxic**)

When the water reaches a temperature between 30 and 40°C, add 10 mL of warm water to each of the test tubes. Agitate well, but not all of the solute will dissolve. Continue heating the water.

Once you have taken your observations, place the test tubes in the heating water. Continue heating the water, carefully agitating the solutions periodically, until the water is about 100°C (about boiling). Record your observations.

Colligative Properties:

We are going to make three solutions of more or less equal concentration. To do so, make each of the following solutions in three different test tubes:

- (1) Put 20.2 g of potassium nitrate (0.2 mol) and 20 mL of water, and agitate to dissolve.
- (2) Put 38.4 g of sugar (0.2 mol) and 20 mL of water, and agitate to dissolve.
- (3) Put 11.14 mL of ethylene glycol (0.2 mL) and 20 mL of water, and agitate to dissolve.

Heat each test tube carefully in turn until the liquid begins to boil, with the Pasco temperature probe in the solution. Record the temperature when the solution first begins to boil. Calculate the value of “i” for each substance.

Calculations:

Colligative properties:

The equation for boiling point elevation is $\Delta T_b = iK_b m$, where m is molality (moles of solute per kilogram of solvent), and for water, the boiling point elevation constant, K_b , is $0.512\text{ }^{\circ}\text{C/m}$.

Observations:

Diffusion:

Observations:

Solubility of Ammonia:

Observations:

Solubility

Solute	Hexane	Water
Sodium Chloride		
Calcium Phosphate		
Sugar		
Naphthalene		
Ethylene Glycol		
Paraffin Oil		

Observations:

Temperature and solubility:

Solute	Low Temperature	High Temperature
Potassium Nitrate		
Sugar		
Ethylene Glycol		

Observations:

Colligative Properties:

Solute	Boiling Point	Value of "i"
KNO ₃		
sugar		
ethylene glycol		

Name:

Date:

Solubility and Colligative Properties Pre-Lab Questions

- 1) List the salts we are using in part one and label them as polar or non-polar.
- 2) What similarities do you notice in the salts used in the effect of temperature on solubility? What differences?
- 3) What Colligative property are we studying?

Solubility and Colligative Properties Post-Lab Questions

- 1) Do you expect that cooking oil is polar or non-polar? Explain your reasoning.
- 2) What did you notice about the freezing point depression? Can you explain it?
- 3) Look at the compounds we used for boiling point elevation and the value of “i” that you calculated for each. What do you suppose “i” stands for?

Experiment 17: calorimetry

Purposes: To measure heat transfer

Ping: A noise made by expensive machines in operating rooms to impress the administration

Background:

See “Using the Pasco System” and “Basic Laboratory Procedures”; pipettes, balance

Introduction:

Ever wonder how it is that scientists come to draw the conclusions that have put science where it is today? Today’s experiment is split into two parts; the first is designed to get you to practice asking questions, blind, as it were, much as scientists were a hundred years ago. Nowadays, it is easy to draw on the conclusions of scientists gone by, Dalton’s atomic theory, the Bohr atom, as convenient starting points. But, what do you do if you have no convenient starting point? Then the starting point becomes the most basic; pure observation.

The next part of the experiment has several purposes; among them are to familiarize you with the data collection/analysis systems we have in the lab, that is, the Pasco systems. These are automated data collection instruments, but such instruments are meaningless without the necessary experimental set-up. For this reason, we take advantage of a basic thermodynamic principle; the conservation of energy.

A “calorimeter” is an instrument which is designed to be very well thermally insulated to prevent significant heat loss over short periods of time. Filled with water, they work on the principle that any heat lost by one object placed within the calorimeter will be exactly equal to the heat gained by the water; $q_{\text{object}} = -q_{\text{calorimeter}}$. Calorimeters today can be very sophisticated to be sure, but you can use very simple calorimeters to get good results as well. For example, today’s calorimeter amounts to little more than a Styrofoam cup.

Experimental Procedure:*Part I: Scientific Inquiry*

Ever wonder what the “scientific method” is? Depending on your source, there are a series of steps that delineate what scientists are supposed to do to answer a question, but the fact is that the scientific method is a cyclic pattern of observation and hypothesis, each feeding on the other in turn.

Imagine how chemists deduced atomic structure; have you ever seen an atom? We see the results of atoms, but not the atoms themselves. Thus, as experiment after experiment gave insight into the behavior of individual atoms, hypothesis after hypothesis were formed as to the structure of the atom. Each new hypothesis would breed a new group of experiments, and the results of the experiments would be used to modify, or just flat out abandon, earlier hypotheses.

You have before you a group of sealed canister. It is your task to determine what is within each canister ***without opening them up!*** Start by simple observation; *play* with the canister to familiarize yourself with it. Once familiar, take a guess as to what is within the canister (your first hypothesis). Discuss these ideas with your colleagues (your other group members) and determine how you can design an experiment to test the hypothesis.

Keep careful track of your thought processes and experiments. Answer the following questions, describing what you believe the canisters contain and explaining how, exactly, you came to your conclusions:

- (1) What is the size of the object(s)?
- (2) What is the shape of the object(s)?
- (3) What are the physical characteristics of the object(s)?
- (4) What do you think the canister contains?
- (5) How secure are you in your conclusion on the contents?
- (6) List the kinds of evidence you used to arrive at your conclusion about the contents of each canisters

Part II: Calorimetry – Specific heat

The calorimeters we are using today are basically just Styrofoam cups with lids. None the less, they are sufficient for today's experiment. We have to begin, though, by calibrating our instruments.

Calibration of Pasco Systems:

Most probes for the Pasco system start off basically analog voltages. We must start by telling the system what voltages correspond to what temperatures. In so doing, we will take advantage of two simple facts; water freezes at 0°C and boils at 100°C. Thus, thermometers are not necessary at these temperatures since we know them. (In fact, these temperatures are the basis of the Celsius temperature scale; this is not a coincidence.)

1. Start the computer, and the Pasco software. Choose the "temperature probe" analog sensor. Be sure that you select position "A" and that you have the temperature probe plugged into this port.
2. Choose "calibrate"; to your Styrofoam cup, add about equal quantities of ice and water. Put the temperature probe into the ice water and allow a few minutes to equilibrate. Choose the low temperature as "0.00"; make sure the units are set for °C. As soon as the reading seems to have stabilized, click "read".
3. Follow this step exactly as written; the faster you can do the last part of it, the better your results will be. For the high temperature calibration, set the temperature to "100". Empty out your calorimeter. Then, as quickly as you safely can, fill the cup with boiling water, return to your desk and quickly place the temperature probe into it and put the lid on it. ***Remember that you are handling boiling water; don't rush. Walk carefully back to your desk. If something slows you down, we can always start over again; it is not worth putting yourself or others at risk for burns by rushing or working recklessly.*** As soon as the reading has stabilized, click "read".

If you followed these instructions carefully, your Pasco system should be ready.

Data Collection:

In the front of the room are several samples of objects in boiling water, including glass and metal. These have a known starting temperature (what is it?). You will be asked to calculate the specific heat of all of them. To do so, you will want to collect data as follows;

- (1) If you have not already done so, close the calibration window.
- (2) Click on “sampling options”. Select “slow” and move the slider until “10s” is selected. This will take a data point every 10 seconds once the run is started.
- (3) Open “table” and “graph” by dragging the appropriate icons over the thermometer icon.
- (4) Fill the calorimeter with water, then empty the water out. Without drying, measure and record the mass of the calorimeter. Do not forget to “tare” the balance *before* placing it on the scale.
- (5) Fill the calorimeter about 2/3 full of water. Measure and record the mass of the calorimeter with the water in it.
- (6) Place the temperature probe into the calorimeter water and place the lid over the calorimeter. Click “record” and allow at least 2 minutes of data collection (12 data points). If, after this time, the temperature looks stable to you, *without stopping the data collection*, bring the calorimeter to the front of the room and select an object. Record the label of the object, and with the tongs, remove the object and place it in the water in the calorimeter. Replace the calorimeter lid, and as quickly as you safely can, return to your desk and put the temperature probe back into the calorimeter. Keep in mind that in this time, a few data points may have been collected that are, for all intensive purposes, meaningless. Don’t let this worry you.
- (7) Watch the graph. You want it to look as if you have a straight line for at least two minutes. As soon as you are convinced that the line is stable, click “stop”.
- (8) Remove the object and dry it off. Measure and record the mass of the object.
- (9) Write the data points in your notebook. Do not count on being able to save these tables as they may be accidentally erased.
- (10) As soon as your data collection and transcription is completed, return the object to the front of the room. Empty the water out of the calorimeter but do not dry it. Repeat steps 5-10 for the remaining objects.

Part III: Calorimetry – Heat of Reaction

We are interested in how much energy is released when we react an acid (HCl) with a base (NaOH). There is one problem, however; there will also be heat released when we dilute the HCl and the NaOH, so we will have to measure these “heat of dilutions” so we can subtract them in our experiment.

Make sure your coffee-cup calorimeter is clean and dry. Get 20.00 mL of distilled water in a beaker, and 20.00 mL of sodium hydroxide in the calorimeter. Using your Pasco probe, determine the temperature of the water. Dry off the probe, and determine the temperature of the sodium hydroxide. Start collecting temperature points with the probe in the sodium hydroxide, and quickly but carefully (without splashing) pour the water into the sodium hydroxide. Quickly put the lid on the calorimeter, and measure the

temperature change. Swirl carefully (to avoid splashing); continue measuring the temperature until it is stable, or slightly decreasing.

Pour the sodium hydroxide solution down the drain (with lots of running water). Clean and dry the coffee cup calorimeter, and dry the beaker. Get 20.00 mL of hydrochloric acid in the beaker this time, and 20.00 mL of distilled water in the coffee cup calorimeter. Just as you did before, obtain the temperature of both solutions, being sure to rinse off the temperature probe and dry it before transferring it from one solution to another. With the probe in the hydrochloric acid, begin taking data points. Pour the HCl solution carefully but rapidly (to avoid splashing) into the coffee cup calorimeter. Cap off the calorimeter. Swirl it and watch the temperature. Continue taking the temperature until the temperature has leveled off or begins to slightly decrease.

Pour the HCl solution down the drain with lots of water. Clean and dry both the beaker (remember it had HCl in it) and the coffee cup calorimeter. Get 20.00 mL of HCl in the beaker and 20.00 mL of NaOH in the coffee cup calorimeter. Once again, determine the temperature of both solutions, being careful to rinse off the temperature probe and to dry it between solutions. Begin taking temperature measurements of the NaOH, and quickly, but carefully (to avoid splashing) pour the HCl into the NaOH. Cover the calorimeter and swirl carefully. Continue taking temperature readings until the temperature levels off, or until it drops slightly. All solutions can go down the drain with running water; clean your equipment. Return the calorimeter.

Calculations:

For any heat transfer which does not involve phase changes, we have $q = mc\Delta T$, where $\Delta T = T_{\text{final}} - T_{\text{initial}}$. Here, q is heat (measured today in “calories”), m is mass (measured in grams), T is temperature (measured in degrees Celsius), and c is the specific heat. For water, we have $c_{\text{water}} = 1 \text{ cal/g}^\circ\text{C}$. Since $q_{\text{object}} = -q_{\text{calorimeter}}$, we have $m_{\text{object}} c_{\text{object}} \Delta T_{\text{object}} = -m_{\text{water}} c_{\text{water}} \Delta T_{\text{water}}$. Thus, $c_{\text{object}} = -m_{\text{water}} c_{\text{water}} \Delta T_{\text{water}} / m_{\text{object}} \Delta T_{\text{object}}$.

The mass of the object was measured directly. To determine the mass of the water, subtract the mass of the calorimeter from the mass of the water and calorimeter.

Plot a graph of temperature versus time (by convention, we always mean plot “y versus x”, so to say “plot temperature versus time” implies that temperature belongs on the y axis, while time belongs on the x axis). From the graph, determine ΔT_{water} , and the final temperature. Since heat is being absorbed by the water, ΔT_{water} should be a positive number.

The initial temperature of the object is taken to be 100°C (since, after all, it started in boiling water. The final temperature of the object should be the same as the final temperature of the calorimeter if you were patient enough. Thus, $\Delta T_{\text{object}} = T_{\text{final}} - 100$, and should be a negative number.

Calculate the specific heat of each object and report them. From a table of specific heat values, can you predict what the objects were made of?

Heat of Reactions:

You will need to calculate the heat of dilution for both the HCl and the NaOH. For NaOH, determine the initial temperature by the initial temperature of the NaOH and the water; if they are different, use the average as your initial temperature. The final

temperature is the maximum temperature reached on mixing. The heat of mixing, then, is simply

$$\Delta H_{\text{NaOH,mix}} = m\Delta T$$

where ΔT is the change in temperature and m is the mass of the total solution (take it to be 40 grams; we are assuming the density is 1.00 g/mol). In an analogous fashion, determine $\Delta H_{\text{HCl,mix}}$.

The total energy change, ΔH_{total} is determined the same way as above, but use the run where you mixed the HCl and NaOH. To find the heat of reaction, note that

$$\Delta H_{\text{total}} = \Delta H_{\text{rxn}} + \Delta H_{\text{NaOH,mix}} + \Delta H_{\text{HCl,mix}}$$

Use the values you've calculated to find ΔH_{rxn} ; this is the heat of reaction. Divide this value by the number of moles of acid (or base, whichever is less) as calculated by the concentration of the acid or base and the volumes we've used to determine the molar heat of reaction.

Observations:

Part I: Scientific Method

Observations:

Part II: Specific Heat

Object	Final T	Initial T	DT	Object

Observations:

Part III: Heat of Reaction

Run	Mixing...	with...	DT	DH

Observations:

Pre-Lab Questions:

1. Explain the principle of calorimetry that allows us to measure heat changes.
2. Would an insulator have a very high or very low specific heat?
3. We cannot just measure heat of reaction; what other two heats are necessary for this calculation?

Post-Lab Questions:

1. Did you get the correct object in part I? Do you think you should have? Knowing what you know now, do the observations you made make sense?
2. What do you suppose are the major sources of error in the specific heat and the heat of reaction experiments?

Experiment 18: kinetics

Purpose: To determine the rate law of a chemical reaction

Puree: To chop so fine the material becomes liquid-like

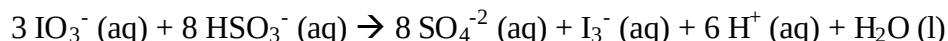
Background:

See “Using the Pasco System”

Introduction:

Kinetics is the study of reaction rates, that is, how fast reactions occur. In your first thought, you might not think this sounds terribly exciting, but consider this; if explosive reactions occurred more slowly, they would not explode. If diamonds converted to graphite more quickly, you would not want them. If the metabolic rate is not correct in your body, you could either starve to death while eating very well (if the rate is too low) or burn up while eating hardly anything at all (if the rate is too high). In fact, your body has many intricate methods for very carefully controlling the reaction rates in the metabolic pathways.

For today’s experiment, we will be using an old demonstration reaction called the “Iodine Clock Reaction.” The net ionic equation for this reaction is



Starch is added as an indicator because it will form a dark blue complex with $\text{I}_3^- (\text{aq})$. The form of the rate law equation is $\text{rate} = k[\text{IO}_3^- (\text{aq})]^x [\text{HSO}_3^- (\text{aq})]^y$. In this experiment, we will determine the values of x, y and k.

Procedure:

You will want to use the Pasco colorimeter. Set it up using the standard procedure. Choose “red” for the color (we are choosing red, because the complex is blue; this means the complex absorbs most strongly, we expect, in the red wavelength). Set the graph for “absorbance” versus “time”; set the frequency to 2 Hz for automated data collection.

There are two solutions available; the iodate solution, and the sulfite solution. You will want to get a sample of each in individual clean, dry containers. You will also want distilled water to be available. If the concentrations are given, record them; however, the concentrations may not be given (this is not a problem).

In a small test tube, put about 1 mL of the iodate solution. Quickly put 1 mL of the sulfite solution, and allow the solution to stand. Wait for a reaction to occur. Take observations; this is so you can see the reaction once before using the Pasco equipment.

Measure out two milliliters of iodate solution and put it into a cuvette. Place the cuvette into the colorimeter. One partner should be ready to press “record”, while the

other pours the solution. When you are ready, one partner should pour two milliliters of the sulfite solution into the cuvette and quickly close the colorimeter cover, while the other partner starts recording the data. You are waiting for absorbance to rapidly increase; it should level off shortly thereafter.

Clean and dry the cuvette and test tubes. In one test tube, place one milliliter of iodate solution, and one milliliter of distilled water. In the other, place two milliliters of sulfite solution. Repeat the above procedure for using the colorimeter to collect data.

For the third trial, after cleaning and drying the cuvette and test tubes, place two milliliters of iodate solution into one test tube, and one milliliter of sulfite, plus one milliliter of water in the other. Again, run the reaction to collect data.

Calculations:

From the graphs, determine the rate of reaction for each of the three runs in seconds. Notice that between run two and run one, we doubled the concentration of iodate while keeping the concentration of sulfite constant. From the data, determine the value of x . In an analogous manner, determine the value of y from runs one and three.

Choose one of the runs, and determine your experimental value of k .

Observations:

Reaction observations:

Run	[Iodate]	[Sulfite]	Rate=1/time (sec)
1			
2			
3			

Observations:

Pre-Lab Questions:

1. How are we varying the concentrations of the reactants?
2. At one point, I mentioned that the exact concentration of the reactants is not important. Is this true? Why or why not?

Post-Lab Questions:

1. What did you find for the overall order of the reaction?
2. List possible sources of error; how could this experiment be improved?
3. You were asked to perform the reaction outside of the colorimeter once before collecting data. Consider the graphs that you obtained. Do these graphs make intuitive sense based on this first reaction?

Experiment 19: qualitative analysis

Purpose: To determine the cations present in an unknown solution

Pernoid: Not a real word

Background:

See “Basic Laboratory Procedures”; Litmus paper, Bunsen burner

Introduction:

One of the oldest questions asked by chemists is “what is it?” As I understand it, the Native Americans would chew a particular type of leaf to relieve headaches; chemists wanted to know what was in it that did so. The result is aspirin.

The concept of extracting organic materials and analyzing them to determine their chemical composition is beyond the scope of this course (in fact, these topics are covered in Organic Chemistry). However, in this experiment, we will be performing the General Chemistry equivalent; qualitative analysis.

In qualitative analysis, we ask simply “what is in it.” We are not worried about how much there is (this is “quantitative analysis”), just, yeah or nay, is this metal present? It will take you several weeks to complete this procedure; this experiment replaces a “lab practical,” because in it, you will have to be particularly wary of your technique, observation skills, note taking skills, and labeling skills.

Your technique will be tested because cross contamination can cause “false positives” (that is, you will think ions are present that really are not). What’s more, if you fail to clean your equipment carefully, you can cause contamination that will result in the same problem. Also, be careful of the equipment you choose; we are looking for metal ions, so choosing things such as metal scoopulas can cause problems. At the same time, you will have to keep in mind what ions are (or might be present) in each container at all times. You will not be able to keep all solutions you make, but you must be very careful not to discard a solution you will need later on. Remember that we are beginning with a mixture of all possible ions; do not throw away solutions if you are not sure that it is no longer needed.

Your observation skills will be critical. Often it is challenging to tell if a positive is sufficient to call it positive, or determine if it is simply a contaminant. We will have ways to help you differentiate, but you will have to always be wary of what you add, and what happens.

Note taking is of critical importance. The smallest observation, which does not seem to be significant when you first write them down, could be the determining factor when deciding what is present and what is not. All too often, a student will ask me a question, such as, “what does this mean?” I usually cannot help, not because I am not willing, but because I was not present to witness the entire procedure, so I often do not know where the student is or what might have happened to get them there. The only witness is the analyst; any lost notes, then, are simply lost forever.

Finally, labeling will be very important, and is keyed in with your note-taking. This week, you will be storing solutions that you might not need for three more weeks. If you do not label the solutions carefully, you will be faced with the uncomfortable situation in which you are trying to remember “what this was” and “which one do we need for...”

Remember that you will be graded on accuracy. If you lose track of where you are, or if something goes wrong, you will have to go back to the first step to separate out ions from earlier groups. This will go faster than the first time since you will not have to perform the verification tests for ions you already know are present, but it will be time consuming none the less.

Procedure:

Initial Concepts:

To keep track of what is happening, there are a few initial concepts that you should keep in mind. These steps are, primarily, all we are doing.

Separation:

The order of the tests are critical, because one ion can often lead to a false positive later in the analysis for another ion. Separation steps are designed to separate these earlier ions from later ones. Typically, separation is based on solubility; add a reagent, to which some ions will precipitate but others will not, and separate the solid from the liquid. In a separation, you will want to keep both the precipitate (typically containing the ions for the immediate tests) and the decantate (the liquid, containing ions for later analyses).

Washing:

Sometimes, tests later on can be sensitive, so we have to make sure the separation was fairly efficient. In this case, we will wash the precipitate (typically) to make sure there are no other ions of appreciable concentration to cause problems. Washing is usually based on solubility. Once the solid and liquid are separated, you will add a wash solution to the solid, agitate the solution, re-centrifuge and decant. Most of the time, you want to keep the solid, but (unless otherwise denoted) discard the wash solution.

Verification:

The first indication that the ion you are seeking is present is the precipitation in the separation, but how do you know that the precipitate contains a specific ion (especially since many separation techniques will separate out more than one ion at a time)? The verification step is your way to ensure that the precipitate formed is indeed the ion you are seeking. A negative on the verification step means the ion was not present, and the precipitate was something other than what you thought it was.

Preparation of “Support Solutions”:

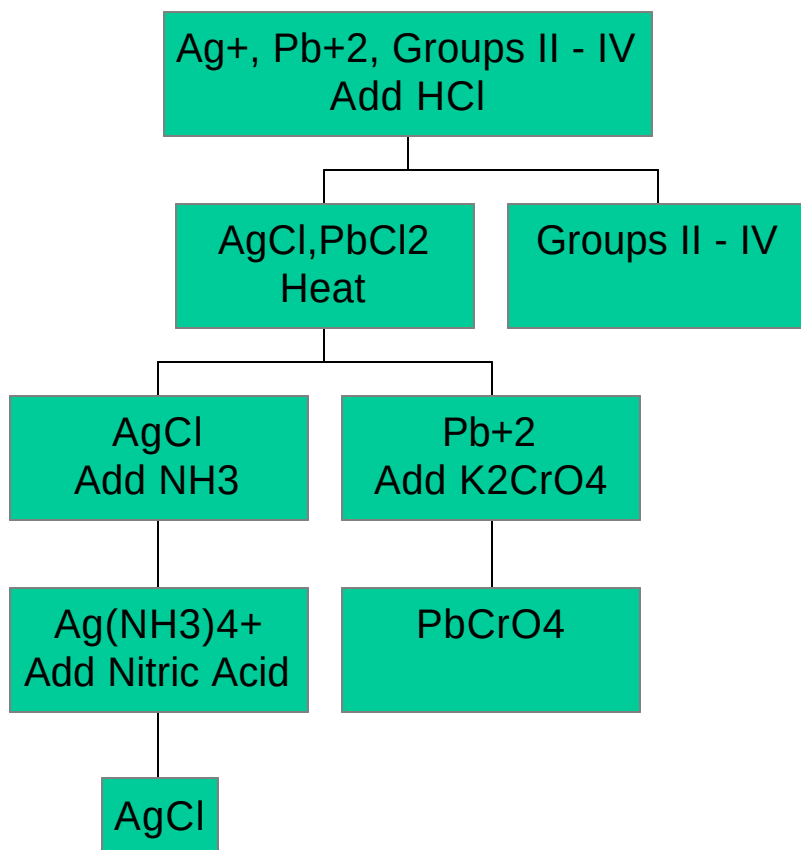
There are two principle issues that you will have to resolve when running these analyses. The first is “what does a positive test look like?” Some are obvious, some are more subtle; it is always very helpful to be able to see a positive to compare the unknown with. The second question is “how do I know it is not a contaminant?” To answer these questions, you will want two solutions in addition to the unknown; the standard, and the blank. Run all three samples (the standard, the blank and the unknown) simultaneously. Anything you do to one solution, do them to all solutions, and always remember to keep them separate. Many qualitative analysis schemes recommend running the standard, then repeating the procedure for the unknown. It is more efficient, and therefore faster, if you can keep the solutions separate so you can run all three at the same time.

Standard:

The standard contains all possible ions for the entire procedure. To make it, add 5 drops of each ion's nitrate into the same clean dry test tube (AgNO_3 , $\text{Pb}(\text{NO}_3)_2$, $\text{Bi}(\text{NO}_3)_3$, $\text{Cu}(\text{NO}_3)_2$, $\text{Sn}(\text{NO}_3)_4$, $\text{Ni}(\text{NO}_3)_2$, $\text{Co}(\text{NO}_3)_2$, $\text{Mn}(\text{NO}_3)_2$, $\text{Al}(\text{NO}_3)_3$, $\text{Cr}(\text{NO}_3)_3$, $\text{Ba}(\text{NO}_3)_2$, $\text{Ca}(\text{NO}_3)_2$, and KNO_3 .

Blank:

The blank is the opposite of the standard; it contains NO ions. Use the appropriate volume of distilled water. If a positive ever appears in the blank, this is an indication that somewhere, the solutions picked up a contaminant, and you will want to re-run the tests to ensure that you have no false positives.



Group I: Ag^+ and Pb^{+2}

Group I ions (which traditionally include Hg^{+2} as well) are put together because they all have insoluble (or relatively insoluble) salts of chloride. This fact allows for easy separation of these ions (aside from lead, as its chloride is still partially

soluble) from other ions in the solution.

Separation of Group I from Groups II through V

To one milliliter of your solution, add dilute HCl dropwise until no more precipitate seems to form. Do not add the HCl too quickly, since excess chloride ions can cause the silver to re-dissolve. Centrifuge and decant; keep both the solid (which contains group I ions) and the decantate (which contains groups II through IV ions). Add one drop of HCl to the decantate to verify that no more precipitate forms; if precipitate does form, repeat this separation step.

Separation of Ag^+ from Pb^{+2}

The solubility of lead (II) chloride increases with temperature, while silver chloride remains insoluble. Add one milliliter of distilled water; heat the solution in a hot water bath, but do not allow the solution to come to a boil. Decant the solution hot (you will not be able to centrifuge). Repeat this step one more time.

Verification of Ag^+

To the solid silver chloride, add concentrated ammonium hydroxide dropwise until the solid dissolves (if it is not silver, it will not dissolve). Once all of the silver has dissolved, acidify the solution with dilute nitric acid. If the solid was silver chloride, the precipitate should re-form.

Verification of Pb^{+2}

To the warm liquid containing lead ions, add potassium chromate, K_2CrO_4 . If the decantate contained lead ions, yellow lead (II) chromate precipitate will form.

Group II:

Group II ions are characterized as having insoluble sulfide salts. For our analysis, we will be looking for Pb^{+2} (remaining from group I), Bi^{+3} , Cu^{+2} , and Sn^{+4} (many qualitative analysis schemes also include mercury, arsenic, cadmium and antimony in this group). Because hydrogen sulfide gas is toxic, we will be using thioacetamide, which decomposes readily to produce hydrogen sulfide at lower, and therefore safer, concentrations.

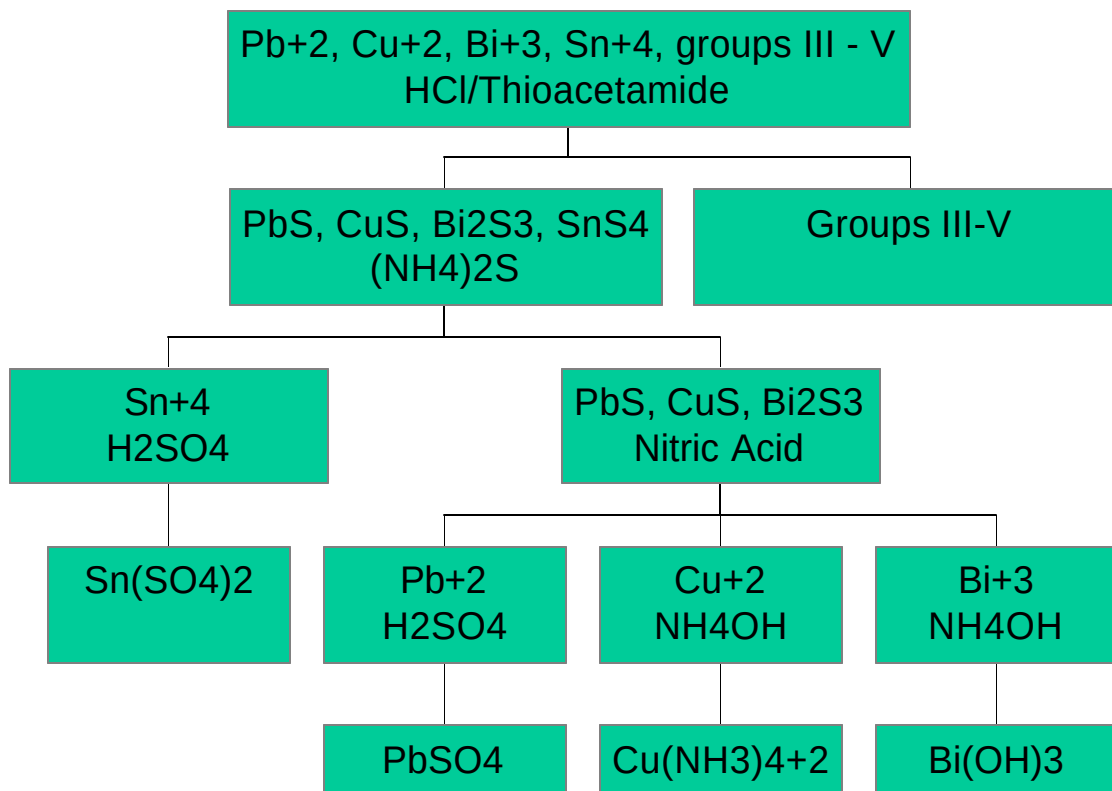
Separation of group II from groups III – V:

Heat the decantate from group I to reduce the volume. When the volume is about $\frac{1}{4}$ to $\frac{1}{2}$ the original volume, remove from the heat. Add concentrated HCl dropwise until the solution has a pH of less than 1 (use pHdrion paper to test). When acidified, add 5 drops of thioacetamide. Heat the solution to cause the decomposition of thioacetamide. Centrifuge the mixture. The solid contains group II ions.

Check the pH of the decantate and be sure it is still acidic. Add two more drops of thioacetamide, and heat. If a precipitate forms, centrifuge and add the solid to the precipitate to that from the previous step. Repeat this step until addition of thioacetamide produces no more precipitate. Keep the liquid as it contains groups III through V ions.

Separation of Sn^{+4} from Pb^{+2} , Bi^{+3} , Cu^{+2} :

To the precipitate, add 1 mL of ammonium sulfide and agitate the solution. Tin should dissolve into the liquid, while other ions remain solid. Centrifuge and decant.



Verification of Sn^{+4} :

Add dilute HCl to the decantate from the previous step until it registers acidic to litmus paper. Formation of a precipitate indicates the presence of tin.

Separation of Pb^{+2} , Bi^{+3} , and Cu^{+2} from other impurities:

To the precipitate from the separation of tin step, add 1 mL of dilute nitric acid. All ions should dissolve, leaving behind only solid impurities. Centrifuge and decant. Discard any remaining precipitate; separate the decantate into three portions.

Verification of Pb^{+2} :

Add dilute sulfuric acid to one portion the above decantate; the formation of a precipitate verifies the presence of lead. Notice that you should regard lead as present if it shows positive in EITHER group I or group II (or, naturally, in both).

Verification of Bi^{+3} :

Add ammonium hydroxide to the second portion of the above decantate until the solution is alkaline according to litmus paper. Formation of a precipitate is verification of the presence of bismuth.

Verification of Cu^{+2} :

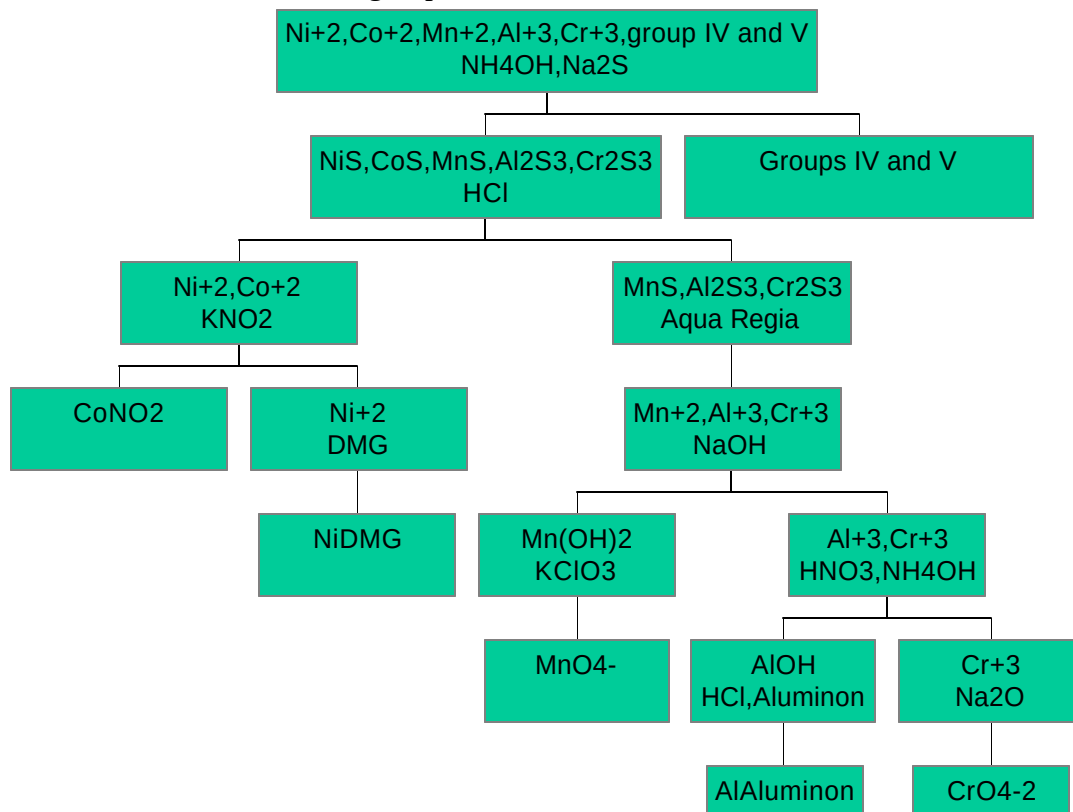
Add concentrated ammonium hydroxide dropwise to the final portion. If a precipitate forms, continue adding ammonium hydroxide until it dissolves. A blue color of the solution is a verification of copper.

Group III:

Group III ions precipitate as sulfide ions, just as group II, except in an alkaline solution rather than acidic. These ions (Ni^{+2} , Co^{+2} , Mn^{+2} , Al^{+3} , and Cr^{+3} ; many qualitative analysis schemes include Fe^{+3} and Zn^{+2} in group III) precipitate on the addition of ammonium sulfide, leaving behind groups IV and V.

Separation of group III ions from groups IV and V:

Heat the decantate from group II in a hot water bath to reduce the volume to



approximately $1/4^{\text{th}}$ the original volume. To this, add $1/2$ mL of concentrated ammonium hydroxide. Check the alkalinity with litmus paper; if it is not alkaline, continue to add concentrated ammonium hydroxide until it is. Once alkaline, add $1/2$ mL of sodium sulfide. The group III ions should precipitate out as sulfides. Centrifuge and decant; the solid is to be tested for group III ions; hold onto the decantate to test for groups IV and V ions.

Separation of Ni^{+2} and Co^{+2} from Mn^{+2} , Al^{+3} and Cr^{+3} :

Add 1 mL of dilute HCl to the precipitate; this should dissolve the nickel and cobalt salts, while leaving the rest as solids. Centrifuge and decant.

Verification of Co^{+2} :

To the decantate from the above separation step, add potassium nitrite, KNO_2 , dropwise. If cobalt is present, a yellow precipitate should form. Continue adding potassium nitrite until no further precipitate seems to form. Centrifuge and decant.

Verification of Ni^{+2} :

To the decantate from the above step, and $1/2$ mL of dimethylglyoxime (DMG). A red precipitate (not just a red color to the solution; it should be cloudy) indicates the presence of nickel.

Separation of Mn^{+2} from Al^{+3} and Cr^{+3} :

Dissolve the solid from the above separation step using as little aqua regia as possible. Once the precipitate has dissolved, neutralize it with 10 M NaOH dropwise until the solution shows alkaline to litmus paper. If a precipitate forms, it is manganese hydroxide. Centrifuge and decant.

Verification of Mn^{+2} :

Add just a little bit (about $1/4^{\text{th}}$ the size of a pea) of potassium chlorate, KClO_3 , to the decantate from the above step. A pinkish or purple color is the confirmation of manganese.

Separation of Al^{+3} from Cr^{+3} :

Neutralize the decantate from the above separation step with concentrated nitric acid. Add concentrated ammonium hydroxide dropwise until the solution is alkaline. If a precipitate forms, it is aluminum. Centrifuge and decant.

Verification of Al^{+3} :

Dissolve the precipitate from the above separation step using dilute HCl. Add several drops of aluminon reagent. If a precipitate forms, it is aluminum.

Verification of Cr^{+3} :

Add a LITTLE BIT of sodium bismuthate, Na_2BiO_3 , to the decantate from above. A yellow color is verification of chromate.

Group IV:

These are the last of the ions that can be separated by their insoluble salts. Group IV includes Ba^{+2} and Ca^{+2} (and many qualitative analysis schemes include Sr^{+2} in this group as well). They are precipitated out as carbonates.

Separation of Group IV from group V:

Verify that the decantate from the separation step of group III is still alkaline; if it is not, add concentrated ammonium hydroxide until it is. Once alkaline, add 1 mL of ammonium carbonate; barium and calcium will precipitate out as carbonates. Centrifuge and decant. Once separated, dissolve the solid in hot dilute acetic acid. Split the solution into two parts.

Verification for Ba^{+2} :

To one portion from the above step, add a few drops of potassium chromate. A precipitate, barium chromate, confirms the presence of barium.

Verification of Calcium:

To the other solution, add a few drops of sodium oxalate. The presence of a precipitate verifies calcium.

Group V:

Group V ions can be thought of as “everything left over.” We will assume this “group” contains only K^+ , although many qualitative analysis schemes include Na^+ , Mg^{+2} and NH_4^+ as well.

Flame test for K^+ :

Take an iron loop and “clean” it by putting it into a hot flame until the loop glows red hot (be careful not to burn yourself). Allow the loop to cool, and dip it in the remaining decantate from the group IV separation step. Be sure a drop is on the wire. Put the drop into the flame; a red flame verifies potassium.

Calculations:

None.

Observations:

Design your own observation page.

Pre-Lab Questions:

Make an outline of each procedure before you perform the steps.

Post-Lab Questions:

Discuss your results, which you trust, which you do not and why. Discuss possible sources of error.

Experiment 20: Beer's Law

Purpose: To gain an understanding of basic spectroscopy

Pfffft: Cartoon sound made when a character passes wind

Background:

See “Using the Pasco System” and “using HyperChem”

Introduction:

It is far too easy to fall into the trap of expecting information to be given in a classroom setting. When working to resolve a problem, scientists often do not have anybody to simply give them the answer, and are usually lucky if they can find some body of work that is at least close enough to the type of problem they have to give them an idea of where to begin with their work. A big part of what we are trying to accomplish in this course is to get our students to stop expecting answers to be given, and start thinking about how to approach problems to solve them by themselves without the help of an “expert.” This is not an easy thing to do, since it does involve a new way of thinking, but we are confident in your ability to make this transition, and will help you out in this endeavor. Once you realize that you *can* do this, you may well find that it is the single most important thing you can learn in college; *how* do I solve a problem that is *beyond* what we’ve done in a textbook?

Towards this end, we will be providing you with a series of problems, and would like you to work with your lab partners to devise an approach to resolve each problem. We’ll start off with very straightforward problems, where the path is relatively clear, but as time goes on, you can expect less and less direction to foster creative independent thought. Unfortunately, regardless of how clever one is in experimental procedure, there is no meaning unless it is clear how to package and present the results. For this reason, we’ll have to begin with experiments that are easy to perform and see, but with a primary focus of searching a topics background and reporting results.

This experiment represents a simple straightforward example of just such a project. It is not terribly challenging in and of itself, so it makes a convenient platform on which to build several new skills, such as:

- Performing literature searches
- Writing scientific papers
- Analyzing data in a statistically relevant manner
- Preparing scientific presentations
- Giving scientific presentations
- And using the PASCO system for data acquisition and analysis.

As you perform this experiment, we will be giving you guidance on each of these topics, and although we expect good results from you, we will be giving primary focus to each of the aforementioned goals as the project proceeds, and would like you to focus on these points as the primary purpose of this experiment.

Experimental background:

Believe it or not, you already have an inherent understanding of Beer's law. If I were to give you two samples of water and ask you which has more iron contamination, you'd choose the darker one as the most likely suspect. Also, if I were to put the iron contaminated water in a large fish tank and ask you which direction would look darker, you'd say it would be darker looking from the side because the path is longer through the water.

This is all Beer's law says; the solution is darker if the concentration (amount of substance per unit volume of the solution) is higher and/or if the path length is longer. So, you might ask, why, then, isn't the law named after me? Well, that's a great question, and there are two answers. First, it's a lot of fun watching a Board of Regents member turn blue with anger whenever they think we are teaching something, *anything*, that might possibly involve an alcoholic beverage. Secondly, and most importantly, because Beer was the first one to do a careful quantitative study of the relationship between absorption and concentration (whoever Beer was; I've checked several references, most of which referenced Beer's law, but say nothing of Beer himself, or, herself, as the case may be). When we say quantitative, what we mean is the first study where great care was taken to accurately measure absorptivity and concentration for many substances and many concentrations in an effort to derive a general mathematical expression relating absorption and concentration.

Absorption is just that; it's the amount of light absorbed by the solution. The more light that is absorbed, the darker the solution is, and absorptivity is a measure of absorption. Absorption should not be confused with transmittance, which is the amount of light that gets through the solution (denoted by the symbol I for intensity), and is always measured relative to the amount of light entering the solution (denoted by the symbol I_0 for the initial intensity of light entering the solution). Thus, we have transmittance,

$$T = \frac{I}{I_0}$$

or, since transmittance is more commonly written as a percentage, we have percent transmittance,

$$\%T = \left(\frac{I}{I_0} \right) * 100$$

Absorption can be shown to be related to transmittance logarithmically:

$$A = \log(T) = \log(\%T) - 2$$

We find that the relationship between absorptivity and concentration is *linear*. Beer's law is written as

$$A = \epsilon lc$$

where A is absorptivity, ϵ is the absorptivity coefficient which is a constant for any given chemical (it's why one chemical is darker at the same concentration as another), l is path length of light through the solution (which we work very hard to keep constant during these studies so $\epsilon l = \text{constant}$), and c is the concentration of the chemical. Because concentration is linear with absorptivity, and absorptivity is logarithmic with respect to percent transmittance, chemists, who are just as lazy a bunch as anybody else, *much* prefer using absorption for spectroscopic measurements so we don't have to worry about working with logarithms in our calculations, despite the fact that it might be a little bit more intuitively simple to think of amount of light getting through the solution rather than the amount of light absorbed by the solution.

Experiment:

You will be given a series of solutions with known concentrations (at the time of the writing of this experiment, the exact substance we will use and the concentrations have not yet been determined), along with at least one solution with an unknown concentration. Because this is the first time you will have used the Pasco system, the instructions to follow are relatively complete (although you should also take this opportunity to explore the software); however, keep in mind that in future experiments, considerably less detail will be given. Be sure to learn the system as you use it.

Starting and initiating the Pasco data collection system*Starting the Pasco system:*

Obtain a colorimeter, and plug the jack into the "Analog channel A" plug. Turn on the Pasco "black box", and start the computer. When the computer has booted, start the science workshop software package, found in the science workshop folder. Once the software has started, you will see a picture of the Pasco "black box" on the screen. Click and drag the picture of the analog jack on the screen to the picture of the "Analog channel A" plug; you will see a box highlighting the plug when you are on it. Release the mouse button and a dialog box with a series of choices will appear. Scroll down the list and choose "colorimeter" (NOT "light sensor"), and a picture representing the colorimeter detector will appear under the "Analog channel A" plug picture.

Calibrating the Pasco system:

We need to calibrate the Pasco system for it to work properly. What this means, essentially, is telling it what "0" is and what "maximum" is. Double click the picture of the light sensor that is now below the picture of the "Analog channel A" plug. Physically set the colorimeter to 0%T on the box. When the current value seems to stabilize (it will oscillate, but the oscillations should be around the same value), click on the "Read Low Value" box. This will set the low reading, or 0% Transmittance.

Now, place a cuvette with distilled water in the colorimeter, being sure that the SMOOTH sides to the cuvette are facing the light path. (HINT! NEVER handle a cuvette on the smooth sides; you want this to be as clean as possible.) Set the colorimeter to the correct color. (HINT! Discuss the options with your partners and choose the color that is most *absorbed* by the substance. Remember; the color that is absorbed is the color you do *not* see in the solution.) When the current value again stabilizes, click on the "Read High Value" box. This has set the 100% Transmittance reading. Your Pasco system now knows what the low and high values are, and will automatically calibrate itself. Close the dialog box.

Setting up the sampling options:

The Pasco system can either operate with automated sampling or manual sampling. Automated sampling is superb for taking data over time; however, our solutions will not change over time (hopefully). Therefore, we want to set the Pasco system for manual sampling.

To do so, click on the "sampling option" button. In the dialog box, choose the "keyboard" option. Type in "concentration" for parameter, and the correct concentration units for units (probably % w/v or % weight/volume, because it is so common and simple; 3% w/v means 3 g of solute for every 100 mL of solvent). Close the dialog box.

Now, click and drag the picture of the table to the picture of the "analog channel A" box, just as you did when choosing the colorimeter. Choose the "Absorbance" option. If you would like a second box to use percent transmittance for a comparison, repeat the procedure. In the table(s), make sure "concentration" is showing by clicking on the appropriate button on the table (the "add column" button, which has a funny little picture that's difficult to describe). Also, click and drag the "digital meter" button, and whatever additional output you'd like to see.

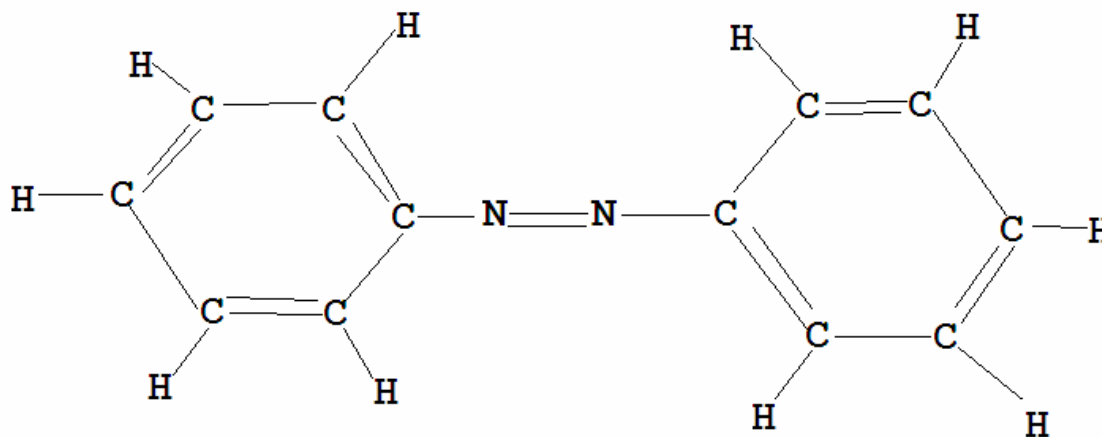
Collecting the data:

Now you are ready to run the experiment. Click the "record" button to start the experiment. A dialog box will appear that prompts you to input your first concentration. Make sure that you rinse the cuvette carefully with distilled water between each run. Put your first standard solution in the cuvette, and place the cuvette into the colorimeter. Type the concentration in, and when you are sure you are ready, click the "read" button. As soon as you have, the system will read the absorbance, and record both the concentration and absorbance which should appear in your table. You are then prompted for the second standard; repeat this procedure, starting with rinsing the cuvette, for each standard. When all standards are finished, have the Pasco system read the absorbance for each unknown you are given.

Be sure to print out the data when you are finished. We will discuss certain procedures in lab (such as considerations in rinsing cuvettes), and how to manipulate your data as well. Be sure to keep an accurate lab notebook as you perform this procedure. A list of questions you will want to discuss as part of the report will be covered when we discuss how to write a lab report.

HyperChem component:

Azobenzene (an acaricide: kills parasites such as ticks and mites) has the following structure:



Using the build tool, build this molecule in HyperChem. Be careful to get the correct double bonds. Remember that with the “add H and model build” tool, you need not put hydrogens in explicitly.

Choose “Setup...Semiempirical.” Choose “AM1” and click “OK.” Go to “Compute...Geometry Optimization” and click OK to keep the default values. Allow the system to find the optimal geometry for the AM1 basis set.

Next choose “Computer...Vibrations” and allow HyperChem to complete the calculations. Finally, go to “Compute...Vibrational Spectrum.” This gives you the calculated values where peaks should be found. Clicking on any one of these lines will show you the wavenumber of that line, its intensity and frequency. Clicking “OK” will show the particular vibrational mode of the molecule that gives rise to this particular color of light being absorbed if “animate vibrations” is selected. This shows how light energy is absorbed by molecules; by exciting vibrational modes of the molecule.

How does this spectrum compare to the true spectrum of azobenzene? What color do you suppose it is?

Calculations:

Generate a calibration curve for the standards. From the light absorbed and the calibration curve, determine the concentration of the unknown solution.

Observations:

Sample	Concentration	Absorbance

Observations:

Pre-Lab Questions:

1. Chemists much prefer to measure absorbance over percent transmittance. Why?
2. Explain Beer's law in your own words.

Post-Lab Questions:

1. The story comes down of a graduate student defending his Ph.D. dissertation, in which a member of his dissertation committee asks, "explain Beer's Law." Being unfamiliar with the law, he responded "Darker beer is better." He was credited with giving a correct response. Why?
2. Come up with as many every-day examples as you can of Beer's law.

Chemistry Laboratory Notebooks

Introduction: For an experimental scientist, there is no tool more important than the laboratory notebook. In the real world, the laboratory notebook is a legal document; they are often subpoenaed and used in court cases. A poorly kept notebook ultimately could result in millions of dollars lost to a company.

For this course, you will be required to keep and maintain a laboratory notebook, and graded on how well you keep it. Although there can be variances in style, there are several headings that should be common to all laboratory notebooks. Following is a guide of what I expect in the notebooks for this class and some helpful hints on keeping a good one. Unless otherwise noted, all headings are required in the order presented.

Experiment title and date: Start each experiment by putting the title of the experiment at the top of the page and the date.

Purpose: One or two sentences on exactly what we hope to achieve in the experiment. Often it is too easy to perform the steps in an experiment while losing sight of what it is we are trying to accomplish. The purpose is the “big picture”, the brass ring to keep your eye on. Keep it very brief; for example, “To gain experience with a variety of chemical reactions.”

Introduction (optional): The purpose of the introduction is to show the relevance of the experiment. Two or three paragraphs should be devoted here, which can include, for example, how the experiment relates to class, or how it has implications on some other aspect of life that you might be interested in. Ultimately, a few years down the road, this should remind you of why we were doing the experiment in the first place. Keep this brief if you choose to include it.

Data and Observations: Write down all raw data and observations. Data should be in tables whenever possible; use the data sheet in the lab manual as a guide but do not write the date in the lab manual as this is not your lab notebook, and you want all of your raw data in your notebook. Observations should be plentiful. Well kept observations could be important in tracking down problems if the experiment does not come out with the results you are expecting. You will be graded on the number and quality of observations taken.

Calculations: Write down ALL calculations involved in the experiment, including separate ones if you were asked to repeat an experiment several times. If there were no calculations for a given experiment, simply write “none” for this section.

Results: SUMMARIZE your findings. This should correspond with your purpose and be very brief. If we are asked, for example, to determine the percent acetic acid in vinegar, then here just write the individual calculated values for each run and the final average concentration.

Discussion: In two or three paragraphs, discuss your findings, conclusions, sources of error and thoughts on the experiment. How can it be improved? Do you trust the results? Why or why not? Think of any interesting applications of how this might be used elsewhere?

Questions: Answer all post-lab questions. As with the pre-lab questions, you need not write the questions here if you don't want to, but be sure that it is easy to ascertain which answer corresponds with which question.

Do's and Don'ts of Laboratory Notebooks:

DO write the page number on top of EACH page;

DO include an index on the first page or two of the notebook and keep it current;

DO write in non-erasable medium;

DO write only on the right side of the page; this saves the left side for scratch paper in case you need to perform a calculation during the experiment;

DO cross out any errors with a single line, keeping it legible, then date and initialize near the cross-out;

DO get my initials before you leave lab (this is proof that you performed the experiment in lab; lab reports without my initials will receive a “0” grade);

DON’T tear ANY pages out;

DON’T write in pencil or erasable ink

DON’T write ANYTHING outside of the notebook; write everything directly into the notebook;

DON’T use “White-Out” or blot out errors completely

Lab Preview Questions: This is MY proof that YOU are ready for the experiment. You must turn in your pre-lab questions on a separate sheet of paper *before* the experiment begins. You need not re-write the questions if you don’t want to; however, this section should be written such that it is clear which answer corresponds with each question (typically simply by using the same numbering scheme as in the book).

Plotting Experimental Data

Prepared By: Richard E. Bleil, Ph.D.

Introduction:

Often in lab experiments it becomes necessary to make a graph, but unfortunately, many of us have not been given procedures for making a proper graph. As a result, several common mistakes tend to occur; improper scales are chosen resulting in graphs that do not utilize the entire piece of graph paper; straight lines are drawn through only two points; axes and titles are not correctly labeled. If you have never been taught how to make a graph, it is nothing to be embarrassed about. In fact, perhaps it is the educational institutions that you have attended that ought to be embarrassed for having not taught you. In an attempt to ensure that no students go through chemistry without having had this instruction, I've written this little guideline for you broken up into individual sections for (hopefully) ease of understanding. If you have any questions, comments, or suggestions, please do not hesitate to call (298-3399 x-5658) or email (rbleil@hotmail.com) or stop by (G-12A) to see me.

Determining Data Ranges:

One of the most common mistakes made is choosing a scale that utilizes only a very small corner of the graph paper rather than the entire piece. This results in graphs that can be difficult to see and larger than necessary experimental error should it be necessary to find the slope, intercept and/or use in predictions. Therefore, it is important to be able to choose a scale for the graph that is appropriate.

To begin, we have to decide what kind of graph we are making. There are three general categories; (1) graphs to cover only the experimental points; (2) graphs that must extend to some smaller number well beyond the smallest experimental point (often zero); and (3) graphs that must extend to some greater number well beyond the largest experimental point. Regardless of the type of graph that one must make, begin by examining your data. You'll have a set of data that includes both "x" and "y" data. Generally, "x" is taken to be exact, while "y" is the measured quantity, that is, the quantity that has experimental error. For instance, suppose I were to make a graph of boiling points versus molecular weight. You'll find a table of such data below in the appendix. The boiling points I measure experimentally. This implies that there could (and will) be some experimental error associated with this measurement. The molecular weight, however, can be easily calculated with very little error (much much smaller error than the boiling point). Therefore, I make my measurement with the greatest error (boiling point) the "y" axis, and the measurement with the smallest error (molecular weight) the "x" axis.

For both the x and y axis, determine your largest and smallest value. If your graph must extend to some range other than that covered by the experimental data, use these as your limits instead. For instance, perhaps my graph of boiling points versus molecular weight must cover very small molecular weights, say down to the molecular weight of methane CH₄, which has a molecular weight of 16 g/mol. However, my

experimental data has molecular weights ranging from 86 g/mol to 142 g/mol. Then I will make a plot that runs from at most 16 g/mol through at least 142 g/mol. For ease of calculation, I may choose numbers close to our limits, such as, say, 10 g/mol to 150 g/mol as my limits. Doing this tends to make it easier to figure out the scale.

As for the y axis, it is a little more difficult. We want to choose a minimum value that we expect will be low enough include the boiling point of a chemical with molecular weight of 10 g/mol. It would be much easier if we knew we wanted the graph to go to some finite amount. Looking at the experimental data, it looks to me as if the boiling point for 10 g/mol should not be less than, say, 50°C, so this is the value I will choose for the minimum value for the boiling point. If it turns out that this guess is far too small or far too large, I'll choose a new minimum value and replot.

Choosing a Graph Orientation:

Once I have examined my experimental point ranges (both x and y axis), I'll choose an orientation for my graph. Since graph paper is *usually* longer than it is wide (this may not be true if a space for graphing is provided), then I am free to choose if I want the x axis to be along the width or length of the paper. The axis with the widest range of values I will make the length of the paper. In our case, the boiling points, that is, the "y" axis, has the greater differences between minimum and maximum values. Thus, I'll make this the length of the paper, and the molecular weights the width.

Setting Up the Axes:

Now look at the graph paper carefully, examining the number of divisions along both the length and the width of the paper. The particular piece of graph paper that I have chosen has 39 divisions along the width and 52 divisions along the length. I'll need a little space to draw in the axes, so I will leave 4 divisions along the bottom and 2 divisions from the right, making my effective graph paper 50x35. I'll draw in the graph axes now and label them. Notice that my labels include *both* the label title *and* the units in parentheses.

We'll set the minimum values for the lower left-hand corner of the axes I've just drawn. Now I have to choose the appropriate value for each division. This is as much an art as anything else, but to get some idea, with 50 divisions spanning 50 to 200 °C, I expect each division along the y axis to be about $(200-50)/50$, or about 3 °C for each division. Before drawing anything, I'll check this by hand. I notice that this will take my scale up to 200 °C, so I will draw them in. First, I'll draw a mark every fifth line. Typically this can be every fifth or tenth line for ease. Then, I'll write in the temperature at each of these new divisions, and add "1 div = 3 °C" to the axis title.

Similarly, along the x axis, I expect each division to have a value $(150-10)/35$, or 4 g/mol per division. Checking this by hand, I see that it is exactly right, so, as with the y axis, I'll draw in the value every fifth division and add the line "1 div = 4 g/mol" to the x axis title. Now my axes are set, and the graph will take the entire sheet of graph paper.

Plotting the Data and Drawing Experimental Lines:

Now I plot the points. Each point I'll plot as carefully as possible, estimating the distance between divisions when necessary. Each experimental point I'll circle once they're plotted. You may notice that in our example, the points do not line up as nicely as perhaps we would like. This makes the task of drawing a straight line all the more challenging.

If we simply draw a line from the first point to the last point, then we have drawn a two-point curve and there was no reason to take more experimental measurements than just these two points. Instead, we want a line that best fits as much of the experimental data as possible. To do so, place a straight edge amongst the experimental data. If the curve *must* intercept at zero (or some other point), fix this side of the straight edge. Typically, however, there are no requirements for the intercept. Now move the straight edge such that when you draw the line, it looks as if the error above the line is about equal to the error below the line.

At this point, you're probably wondering what I mean. Well, if we assume that the error is uniformly distributed amongst all of the data. That means that I should have as much error that is too high as error that is too low. Recall that we assumed the "x" values are exact, and the only error is in the "y" values. So, if we were to measure the distance from each point to the line drawn for those points above the line, this sum should be equal to the sum of the distances from the points below the line to the line. Now, we needn't be so exact as to actually take these measurements and do the summations, but try to draw a line so that if we were to do this, the two errors would be as close as possible.

In some graphs, this is easier to do than in others. Typically (but not always), we'll have about as many points above the line as below (in our example, it is two above and three below). However, really we are drawing a line so the points look as randomly distributed about that line as possible. There are mathematically exact ways to determine the line that best fits data, but we shall not go into this here. If you would like to learn how to do this (it is called "Linear Regression"), please consult a statistics textbook or stop by my office and ask. I'll be happy to teach this technique to any who would like to learn it on an individual by-request basis.

Discarding Data:

Most of the time, we cannot discard data. In our present example, the points are so greatly spread out that I cannot assume that any data can be discarded. Typically, the only time it is acceptable to discard *one* experimental point is if it is clearly far off from the line that could be drawn without it. That is, if all of the points lie very very close to a line and one point is extremely far off, then most likely that one point is an exception and can be discarded. However, typically too few points are taken to be able to do this. If you feel that you have a point that *should be* discarded for any reason, talk with your professor and ask his/her opinion before discarding.

Reading a Point:

So now we want to find a point. We want to use our results to estimate the boiling point of methane (16 g/mol). Now we draw a line from the "x" axis at 16 g/mol)

up to the line and read the corresponding boiling point. This turns out to be about 55 °C. This is actually a *terrible* result, but it is what we found. Always be true to your data. If a result is far from correct, write a little paragraph on what you think may be the reason behind the poor result.

Title:

Always include a title with your graph, as well as your name and the date. The title is always something along the lines of “Graph of y versus x:”. When we use “versus”, by convention, it is always y versus x. You may include a sub-title if you like, which is often helpful for similar plots from different sets of data.

REMEMBER:

- (1) ALWAYS use the ENTIRE piece of graph paper.
- (2) ALWAYS label your axes including units and division sizes.
- (3) ALWAYS draw the line through as many experimental points as possible.
- (4) ALWAYS include a title, name and date.

These few rules will help to ensure that you make the best possible graphs.

Appendix: Example of graphing; boiling points versus molecular weight.

Experimental data:

Molecular Formula/Name	Molecular Weight in g/mol	Boiling Point in °C ¹
C ₆ H ₁₄ / Hexane	86.18	123
C ₇ H ₁₆ / Heptane	100.21	104
C ₈ H ₁₈ / Octane	114.23	124
C ₉ H ₂₀ / Nonane	128.26	145
C ₁₀ H ₂₂ / Decane	142.29	198

¹ In order to simulate experimental data, this column was obtained by taking the published boiling points for these liquids and randomly adding +/- 15% error.

Graph Ranges:

x axis range (molecular weight): 10 - 150 g/mol (this will be the width of the graph paper)

y axis range (boiling point): 50 - 200 °C (this will be the length of the graph paper)

Graph Paper Divisions:

x (width): 39 before axis; 35 after axis; $(150-10)/35 = 4$ g/mol per division

y (length): 52 before axis; 50 after axis; $(200-50)/50 = 3$ °C per division

Boiling point of methane (16 g/mol) from the line: approx. 55 °C.

Factor Label Method

What is it? The factor label method is an algebra intensive method of problem solving utilizing units.

Where would I use it? The factor label method is generally applicable to most problem solving situations.

What is the benefit? The factor label method has two main benefits making it a powerful technique well worth the time and effort to learn it. First, utilizing the factor label method can often predict when an answer is wrong. Second, utilization of the factor label method often removes the necessity of memorization of many formulas and equations.

What are the drawbacks? Being a problem solving system, it is not for everybody. It seems cumbersome at first, difficult to get a handle on, and other problem solving techniques exist that work well also. However, many students have never been taught *any* problem solving techniques, and would find this beneficial. However, it will take some work before the student is comfortable with the method.

The Factor Label Method Foundation:

The factor label method is based on a very simple idea; the idea of unity. Yep, unity, the number one.

One is the loneliest number that you'll ever do. Two can be as bad as one, it's the loneliest number since the number one. At least this is what "Three Dog Night" would have us believe, but in fact, one is the most powerful number that you'll ever do. It has some wonderfully mystical and powerful properties, although these properties are so well known and so common that most have never even really considered them. If you multiply a number by one, the number is not changed. If you divide a number by one, the number is not changed. And the way that we write the number one can be varied, twisted and bizarre.

Sometimes one is assumed. Take the number 12. The number 12 implies 12×1 , 1×12 , or even $12/1$. It's this last representation that we'll dwell on. We like the representation $12/1$, because it allows us to write division by 12 as a multiplication. For instance, $3/12 = 3 \times (1/12)$. So dividing by a number is identical to multiplying by the fraction 1 divided by that number. You'll see how this idea becomes significant a little later, but first I would like to continue with another way in which we can represent the number one.

As we know, twelve is a number. It is a number, in fact, equal to 12. But what is 12 divided by one dozen? Well, since $1 \text{ dozen} = 12$, then

$$\frac{12}{1\text{dozen}} = \frac{12}{12} = 1$$

Then, isn't writing 12/dozen a form of writing 1? And since there are 2.54 cm in each inch, then isn't 2.54 cm / 1 in another form of writing 1? Or 5,280 ft / mile, or 1 yd / 3 ft all ways of writing the number one? And aren't these constants all equal to dozen/12, 1 in/2.54 cm, 1 mile/5280 ft, or even 3 ft/ yd, respectively, since, after all, 1/1=1? We call these quantities "conversion factors."

A conversion factor is a fractional representation of the number 1. Since they are representations of 1, we are free to change the numerator and denominator *at will*, which you will eventually see is a very important characteristic in the factor label method. Another concept that is important is that of *units* being *algebraic quantities*.

If we consider units to be algebraic quantities, then we can "cancel out" equivalent units in numerators and denominators. For instance, everybody knows that if we have 3 feet, and we want to find out how many inches that is, then we simply multiply 3 by 12. However, in the factor label, we multiply 3 feet by 12 inches/foot. Recognizing the equivalents of feet and foot (feet is just plural foot!), then we can write

$$3\text{foot} * \frac{12\text{inch}}{1\text{foot}} = 36\text{inch}$$

Notice how "foot" cancels, since it is in both the numerator and denominator, just as an algebraic quantity "x" would cancel if it were both in the numerator and denominator, as in

$$x * \frac{y}{x} = y$$

The factor label method in action:

So now we've got it. Conversion factors are equivalent regardless of what is in the numerator and what is in the denominator, and units cancel algebraically. In the final analysis, all we really need to do is make sure that the units all cancel such that the units of the final answer match the units we seek.

All we really need to do is make sure that the units all cancel such that the units of the final answer match the units we seek!

I felt that that needed reiteration. It's the heart of the factor label method, however, so it warrants this reiteration.

To be sure that the units cancel, there are a few simple points to keep in mind.

(1) ALWAYS use consistent units! (For instance, feet and cm are length units, but feet don't cancel out cm!!)

- (2) ALWAYS write the units down AVOIDING superfluous units (made up units that don't belong)!!
- (3) ALWAYS be sure that the units cancel out by having one in the denominator and the other in the numerator or vice versa.
- (4) ALWAYS be sure the units of the final answer match the units you are seeking.

Beware, however, of one point. The factor label method will tell you if an answer is wrong if the units are not the units you want. However, if a unitless number is needed (such as π), then the units will match, but the answer will still be wrong.

A few examples:

Example 1: Recently, I have been putting together an electric race car track (remember the electric slot cars? No? Am I showing my age?). The track occupies what would be my bedroom were I married, but since I'm a bachelor, it is more important to me to have an electric race car track than a bed. Anyway, I have 120 track pieces, an average of 9 inches each, on a scale of 1:87. What is the scale length of my track in miles?

GIVEN: 120 track pieces, 9 inches/piece, scale 1:87

FIND: miles of scale track

CONVERSION FACTORS: We will find we need 12 in/ft, and 5,280 ft/mile. In a problem, you typically will not know what conversion factors you will need ahead of time, but as you perform the problem, you will see how they are required. For instance, if you know the conversion factor 5,280 ft/mile, and you are at "inches" in the problem, then you will know you need to convert from inches to feet, or 12 in/ft.

Let's set it up and see what we get:

$$120 \text{ piece} * \frac{9 \text{ inch}}{\text{piece}} * \frac{87 \text{ inch}}{\text{inch}} * \frac{\text{foot}}{12 \text{ inch}} * \frac{\text{mile}}{5,280 \text{ foot}} = 1.48 \text{ mile}$$

Notice how all units cancel, piece with piece, inch by inch, foot cancels foot. The term "87 inch / 1 inch" is our scale factor. By convention, scale factors have really no units. I just included "inch" terms to make it easier to see how I got that term, but even so, the inches in this conversion factor would cancel with themselves, making the scaling factor unitless. You may also notice that the "grammar" is not quite correct, since foot is singular but there are 5,280 of them. However, using "feet" rather than "foot" can lead to confusion as to whether or not a unit can cancel, so I always just use singular units and try to avoid plurals.

Example 2: Some calculations are made much easier by not requiring the memorization of some formula, which sometimes means you can even perform a calculation without requiring that you have covered that subject in class! For instance, suppose you have a gold bar that measures about 9 inches by 5 inches by 2 inches. Given the density of gold is 19.3 g/mL, how many pounds will this bar weigh?

GIVEN: gold bar 9"x5"x2", 19.3 g/mL

FIND: pounds

CONVERSION FACTORS: We will need $1 \text{ mL} = 1 \text{ cm}^3$, 2.54 cm/in. , 2.2046 lb/kg , $1,000 \text{ g/kg}$. In a problem, you would have a table of conversion factors that you can choose from.

First, let's find the volume independently of the factor label method. We know we need volume, because mL, milliliter, is a volume. We have a height, width and length, and the volume of the gold bar is height times width times length, or

$$9\text{inch} * 5\text{inch} * 2\text{inch} = 90\text{inch}^3$$

Notice that even here I used labels. Now that we have our volume, let's solve this problem using the factor label method.

$$90\text{inch}^3 * \left(\frac{2.54\text{cm}}{\text{inch}}\right)^3 * \frac{\text{mL}}{\text{cm}^3} * \frac{19.3\text{g}}{\text{mL}} * \frac{\text{kg}}{1000\text{g}} * \frac{2.2046\text{pound}}{\text{kg}} = 62.8\text{pound}$$

A gold bar weighs over 60 pounds! Consider THAT the next time you see a movie where they are handling gold bars single handedly!!

Notice above that we needed to convert from in^3 to cm^3 , but all we had was the conversion factor 2.54 cm/in. If we had multiplied $90 \text{ in}^3 * (2.54 \text{ cm/in.})$, we would have seen that only ONE of the "inch" units would have canceled, leaving us with $228.6 \text{ in}^2\text{cm}$, a meaningless number. To cancel all three inches in the in^3 term, we had to cube our conversion factor 2.54 cm/in. as shown above. Another approach would have been to cube this number before hand and use the conversion factor $16.387 \text{ cm}^3/\text{in}^3$. *Carry out this calculation and be sure that your answer matches that shown in order to be sure that you are "cubing" correctly!*

Example 3: The nearest star is about 4.5 light years away (one light year is the distance light will travel in a vacuum in one year). The Apollo spacecraft travelled about as fast as a bullet, which we can approximate as $1,200 \text{ ft/second}$. Suppose that we stock a craft which travels at $1,200 \text{ ft/sec}$ with a crew of men and women who are to create a new generation to take control of the ship every 20 years. How many new generations will be required before reaching Alpha-Centauri?

GIVEN: 4.5 light years, $1,200 \text{ ft/sec}$, 1 generation/20 years

FIND: generations

CONVERSION FACTORS: we will need $3.0 \times 10^{10} \text{ cm/s}$ (speed of light), 365.25 days/year , 24 hr/day , 60 min/hr , 60 sec/min , 2.54 cm/in. , 12 in/ft

OK, first, let's see what distance we are talking, thereby converting this into a standard distance/speed problem.

$$4.5\text{year} * \frac{365.25\text{day}}{\text{year}} * \frac{24\text{hour}}{\text{day}} * \frac{60\text{min}}{\text{hour}} * \frac{60\text{sec}}{\text{min}} * \frac{3.0 \times 10^{10}\text{cm}}{\text{sec}} = 4.3 \times 10^{18}\text{cm}$$

Now, let's finish up the problem.

$$4.3 \times 10^{18}\text{cm} * \frac{\text{inch}}{2.54\text{cm}} * \frac{\text{ft}}{12\text{inch}} * \frac{\text{sec}}{1,200\text{ft}} * \frac{\text{min}}{60\text{sec}} * \frac{\text{hr}}{60\text{min}} * \frac{\text{day}}{24\text{hour}} * \frac{\text{year}}{365.25\text{day}} * \frac{\text{generation}}{20\text{year}} =$$

186,000 generations

This problem was inspired by an old episode of Star Trek.

Example 4: In the last few problems, the starting point has always been the number with one single unit (as opposed to this *per* that), so here's a problem that is slightly different. The velocity of light (as we've seen) is 3.0×10^{10} cm / sec. What is this speed in miles / hour?

GIVEN: 3.0×10^{10} cm/sec

FIND: velocity in miles / hour

CONVERSION FACTORS: We'll need 2.54 cm / in, 12 in / ft, 5280 ft / mile, 60 sec / min and 60 min / hour

OK, in this case the starting point is pretty obvious, but this won't always be the case. If you're not sure where to start, start anywhere and just work it out. This *can* lead to the problem of having the correct units but in the inverse arrangement. This problem can be solved by simply inverting the answer, as will be seen in the following example. As for the present example, let's work it out.

$$\frac{3.0 \times 10^{10}\text{cm}}{\text{sec}} * \frac{\text{inch}}{2.54\text{cm}} * \frac{\text{foot}}{12\text{inch}} * \frac{\text{mile}}{5,280\text{foot}} * \frac{60\text{sec}}{\text{min}} * \frac{60\text{min}}{\text{hour}} = 6.7 \times 10^8 \text{ mile/hour}$$

Notice in this one that we let the unit "sec" slide along until we first converted from cm to mile.

Example 5: Finally, not all problems will go along seamlessly. One problem that shows up periodically is when you get the correct units, but in the inverse order (that is the numerator and denominator are switched). There are two ways to solve this difficulty, one is to go back and start over again with the starting point put in the inverse way that you started before (for instance, if you started with feet / sec, then we would use sec / feet). The other is to notice that taking 1. / the answer will put the units in the right way, which is simply one divided by the answer. You may want to initially try both ways, one to verify the other, until you are comfortable with this little "factor label trick". Here's an example problem. If we add 150,000 cal to 100. g of water with a specific heat of 1.0 cal / g °C, what will be the temperature change of the water in degrees Celsius?

GIVEN: 150,000 cal, 100. g, 1.0 cal / g °C

FIND: Temperature change in °C

CONVERSION FACTORS: no additional conversion factors needed

Notice that we have not one, but *two* single unit numbers (150,000 cal and 100. g). With which do we start? Well, without any further information, it's not clear, so we'll start with one of them, say 100 g, and we'll just keep going. So, let's set it up.

$$100.g * \frac{10cal}{g^{\circ}C} * \frac{1}{150,000cal} = 6.67 \times 10^{-4} {}^{\circ}C^{-1}$$

There are a couple of things to note about this problem. The first is that the last term is written as 1/150,000 cal. I knew I needed calories in the denominator to cancel calories in the numerator, but 150,000 calories is a single unit number. Thus, I recognize that 150,000 cal is equivalent to 150,000 cal / 1, so to get calories in the numerator, I just "flip" this number around. I did NOT add any superfluous or artificial units to the "1" in the numerator. When we write 150,000 cal in this fashion, this means that we *divide* by 150,000 cal.

Finally, we note that our final units here are ${}^{\circ}C^{-1}$. This is NOT the unit we wanted, which was simply ${}^{\circ}C$, so to get our final answer, we take 1. / our answer above, or $1./6.67 \times 10^{-4} {}^{\circ}C^{-1} = 1,500 {}^{\circ}C$. The unit ${}^{\circ}C^{-1}$ becomes simply ${}^{\circ}C$ on inversion of the answer, and we get the answer we seek. The very first time I worked out this problem, I chose mass to start with and my answer was the inverse of that which I sought. Now I know that I *should have* started with calories, which would have given the correct units. However, I chose to start with mass again here to demonstrate that when problems arise, it is not (typically) the end of reality as we know it.

Practice Problems:

Ready to try a few problems on your own? If so, let's go; following you will find a series of problems for you to practice on your own. Answers are provided, but YOU get to get there on your own!

Problem 1: Here's a classic style of problem used to practice Factor-Label. Suppose we have a farmer who has four cows ready to go to market. He trades the cow for chicken at the rate of 20 chicken per cow. He trades the chickens for pig at a rate of 15 chicken per 2 pig. On average, each pig weighs 140 lb. The pig he trades for corn, 3 bushels of corn per pound of pig. The corn is then traded for wheat at 1 bushel of wheat per 2 bushels of corn. Wheat is premium, so he trades it for rye, 5 bushels of rye per bushel of wheat. The rye he then gives to a baker, who makes 500 loaves of bread for each bushel. (I'm making these numbers up, so I have no idea how realistic they are.) The loaves of bread are given to a pet store who can feed 3 birds per loaf of bread. The birds are used to feed snakes, and each snake eats an average of 9 birds before being sold for \$32.00 each. How much money did the pet store make from the four cows?'

Answer: Wow, what a horse trader, as it were. Turns out the pet store made \$59,733,333.33. I'm in the *wrong* business!

Problem 2: OK...how about a simpler one, but one that you will need to be able to do eventually in chemistry? First a one-step calculation: The density of ethyl xanthogenic ester is 1.085 g/mL. What volume in mL would you need to get 100 g?

Answer: Even if you don't know what ethyl xanthogenic ester is, or even what density is, you can solve this problem using the factor label method (which turns out to be 92.2 mL).

Problem 3: Another one-step problem: If the molecular weight of the rather long-named compound 5 β -Pregnan-20 α -ol-3-one is 318.50 g/mol, then how many moles (abbreviated mol) would you have if you had 15.00 g?

Answer: Just what *is* this stuff? Dunno...don't care, at least not for this problem. Factor label tells us the answer is 0.04710 mol.

Problem 4: OK, now let's combine the two types of problems, and see how we do. If we need 0.5000 moles of 3-hexyne, which has a molecular weight of 82.15 g/mol and density of 0.7231 g/mL, what volume in mL do we need to get?

Answer: Two step problem here, and one of the types of problems that tends to give students much grief. The answer is 56.80 mL...how'd you do?

Basic Laboratory Statistics

In chemistry, we work very hard to control the environment to ensure that our results are well understood. We make sure that our equipment is clean to avoid extraneous reactions, we control temperature, heck, sometimes, we even control the amount of light in the room. It is not uncommon to find organic laboratories that happen to have windows to have aluminum foil over those windows to block any extraneous light, since this is typically an un-controllable and un-accounted for factor. Most importantly, we do many repetitions of the same experiment, to ensure that the results we get are not one-time flukes. To be a valuable piece of information, all experiments must be reproducible.

As you get further ahead in your studies of chemistry, you will undoubtedly learn more about probability and statistics, but this is a good time to start with some basic concepts. Although many of the formulas I will show you here are often standard in software packages such as Excel, it is important to understand the principle behind them to be able to interpret them correctly. Before we begin, though, we need to understand the two major categories of errors that can arise, and the results these will have on our findings.

Types of Errors

There are two broad categories of errors that arise, referred to as “random” or “systematic.” Any specific error you can name should fit into one of these two categories, and each has a unique impact on the results.

Random error is just that, random. These are often referred to as “human error,” although in truth these errors are far more common than that. For example, if you fail to get all of the reagent off of a piece of weighing paper, or if you accidentally get an unnoticed piece of dust in your reagent, these errors are random. They are unpredictable, and can have a variety of consequences, including either overestimating the value that you seek, or underestimating it.

Systematic errors are typically associated with the instruments that we use. For example, if your balance is not calibrated correctly, all of it's readings may be off by some fixed amount, say, for example, 0.11 g. These errors will always result in errors in one direction only; for example, all of our readings may be 0.11 g too high, or 0.11 g too low.

These errors impact “accuracy” and/or “precision.” These terms are often used synonymously, but in fact, they each have very special and unique definitions.

“Precision” is a measure of how close all of the measurements are with one another. That is, in our misaligned balance, we may get a series of readings that are all very close to one another, and yet they are not very close to what the true value ought to be. For example, five masses of 100.00 mL of water might read 100.09, 100.11, 100.10, 100.12 and 100.10. These readings are all close to one another, but they ought to be reading closer to 100.00; they are precise, but not very accurate. Sometimes, these errors can be corrected for by using the same equipment. For example, if we are using mass difference, and get the mass of a beaker first, it's mass will also be off by 0.11 g. Thus, when we subtract the mass of the beaker from the mass of the liquid and the beaker to get

just the mass of the liquid, this systematic error will automatically cancel itself. However, it is important to note that *this error cancellation will only occur if we are consistent with the instruments that we use* since it is highly unlikely that a second instrument, in this case a balance, will be off by the same amount as the other. Thus, if we switch balances in the middle of the experiment, these systematic errors will not cancel out.

“Accuracy” is how close the mean is to the true value (which may or may not be known). For example, because of the density of water, 100.00 mL of water should have a mass of 100.00 g. You can have high accuracy with low precision. For example, suppose our balance is outside on a gusty day; we might get five readings of 95.02, 107.80, 96.42, 101.33 and 99.46. The mean of this data (see below) is 100.01 g, which is very close to what it ought to be, but the values themselves are all over the place.

Naturally, we want results that are both accurate and precise. If we cannot have both, however, which would you rather have yourself, high accuracy, or high precision?

Mean, Median and Mode

The term “average” is thrown around a lot, but has very low precision in its meaning (see how I did that?). People usually mean “mean” when speaking of average, but it can also refer to median or mode. From here on out, get in the habit of using the terms “mean,” “median” or “mode” instead of “average,” as this will tell people precisely what you mean.

There are three basic ways to discuss the “middle of the value” or “average” for a set of data. The “mean” is the most common, this is simply determined by adding up the individual data points, and dividing by the total number of data points:

$$\text{mean} = \bar{x} = \frac{\sum_{i=1}^N x_i}{N}$$

Thus, for our set of data 95.02, 107.80, 96.42, 101.33 and 99.46, we have

$$\bar{x} = \frac{95.02 + 107.80 + 96.42 + 101.33 + 99.46}{5} = 100.01$$

as our mean (see “significant figures” below).

The “median” is the middle value of the data. If we re-arrange this data in increasing (or decreasing) order, we have 95.02, 96.42, 99.46, 101.33, and 107.80. Since we have five data points, the middle data point will be the third data point, or, in this case, 99.46. This is our median. If we have an even number of data points, then the median is the mean of the two center data points. Notice that this is close to, but distinctly different from our mean.

Finally, the “mode” refers to the number that is most frequently seen. We have no mode in this data set, since none of the data values repeat. Suppose, however, that we actually have six data values, where two of them were both 101.33; then the mode would

be 101.33, and the median would be $(99.46+101.33)/2=100.40$. Notice that this would also change the mean to 100.27; why?

The closer the mean, the median and the mode are to one another, the more “normalized” the data is (that is, the closer the data would fit to a curve created with no systematic errors at all).

Variance and Standard Deviation

We would like to have a measure of how close our data points are to one another, or, better still, how close they are to the mean. For any given data point, we can simply do a subtraction, $\bar{x} - x_i$, but the problem is, if we try to take the average distance from

the mean, or $\frac{\sum_{i=1}^N (\bar{x} - x_i)}{N}$, we will find that this gives us a value of zero, because of the way that the mean is defined. In other words, we will have the same error above the mean value as we have below the mean value. How can we get around this problem? Well, instead of taking the absolute difference between each data point and the mean, let's square this value instead. That is, let's find $(\bar{x} - x_i)^2$. Since the square value of any negative number is positive, NOW we can add these points together, and get a legitimate non-zero value. We call this the “Variance”:

$$\text{Variance} = V = \frac{\sum_{i=1}^N (\bar{x} - x_i)^2}{N}$$

Notice that this does not give us the mean value of each data point from the mean, but rather the square of this value. We are really not interested in the square distance from the mean, so we define the standard deviation as

$$\text{Stddev} = s = \sqrt{V}$$

The smaller our standard deviation is, the less spread out our data is.

Significant Figures

Possibly one of the most confusing subject for students is the concept of “significant figures.” One reason that this is so might be that students tend to not understand why significant figures are, well, significant. The reason is simply this; it is the quickest and easiest way for a scientist to show to the reader how reliable their data is. An older method used to be using the “ $\pm$ ” symbol. For example, suppose I want to tell you the temperature is $106.234456756316846541684 \pm 0.01^\circ\text{C}$. Two things immediately come to mind; first, why did I bother to write out all of those digits if I can only trust the result to ± 0.01 ? The second is, why bother writing all of this out, including “ ± 0.01 ”, when 106.23 means *exactly* the same thing.

THIS is what significant figures are. The assumption is an error of ± 1 in the last significant decimal point. This is why your instructor will, much to your dismay, insist that you write down “0” from time to time to show the significant figures; if we write 101.2, we are implying an accuracy of only ± 0.1 ; however, 101.20 implies an accuracy of ± 0.01 , ten times the accuracy of the previous number. The “0” will not figure into any of the calculations, but it is telling the reader that you were very careful in your measurements (only 10 times more careful than if you write 101.2).

So, what are the rules for significant figures? What is significant, and what is not? Let’s separate this into two different categories:

Numbers Less than One:

- (1) Any non-zero number is significant;
- (2) Any zero between non-zero numbers is significant;
- (3) Any zero after the last non-zero number is significant;
- (4) Any zero before or after the decimal point, but preceding the first non-zero number, is NOT significant.

I know, that last one has you confused. We say “it is not significant because it merely places the decimal point,” to which students typically scream “knowing where the decimal point is pretty %*\$(@ significant!!!” No, actually, it’s pretty %*\$(@ important, but it is not significant. Let me explain a bit further; the number of zeros before the first non-zero number can change by simply changing the units. For example, there are 4 significant figures in the number 0.2330 mm, but if we convert from millimeters to meters, this number becomes 0.0002330 m. We should not be able to change the number of significant figures simply by changing the units, so “0.000” are not significant zeros, although they are very important.

Numbers Greater than One:

- (1) Any non-zero number is significant
- (2) Any zero between non-zero numbers is significant;
- (4) Any zero after a decimal point is significant;
- (5) *if there is no decimal point shown*, then any zeros after the last non-zero number is NOT significant, however;

(6) *if there is a decimal point shown*, then any zeros after the last non-zero number IS significant

I know your head hurts; have you tried aspirin? This seems like an odd thing, so why should it matter if I wrote down a simple little “.” or not? I think of it this way, the author did not HAVE to write down that decimal point, since the zeros place it; thus, if (s)he does go through the effort of writing it, then the author is trying to tell us something. Thus, for example, if you ask how much money I have, I might write \$100 as my answer. This means it’s somewhere around \$100, but I’m not giving you an exact number. Maybe it’s as low as \$90, or as high as \$110; all you have is a ballpark figure. However, if I write \$100. as my answer, notice that I did not *have* to write that decimal point. I am telling you that, $\pm \$1$, I have 100. Maybe it’s only \$99, or as high as \$101, but you know much more precisely how much I have, and the say I tell you that I am giving you a more exact figure is with the decimal point. Notice that I can get even more precise by telling you I have \$99.32; now you know, to the penny, exactly how much I have.

Mathematical Functions and Significant Figures

Often, math is required using numbers of various significant figures. We’ll cover only two basic classes of mathematical functions:

(1) *Addition and Subtraction*: When adding or subtracting two or more numbers, do not worry so much about the number of significant figures. Instead, keep only the smallest number of digits after the decimal point that are significant in any of the numbers. For instance, when adding 101.2336 and 207.44, the answer will have only two places past the decimal point (308.67).

(2) *Multiplication and Division*: When multiplying or dividing, keep only the maximum number of significant figures as is in any of the digits involved. For example, 101.2336 has 7 significant figures, but 207.44 has only 5 significant figures, so our product can have only 5 significant figures (21,000.).

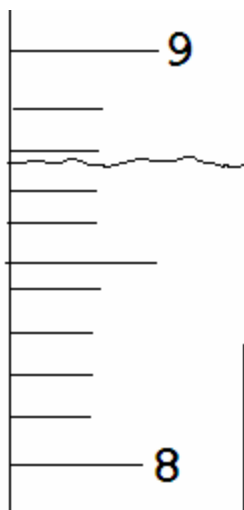
Be careful not to confuse exact numbers for inexact. For example, we know there are 12 inches in 1 foot, so if we are using this conversion factor to convert 3.0252 ft to inches, we might think we can only have two significant figures because of the factor “12.” However, this is an exact definition, and as such, it can have as many significant figures as we want. In other words, even though we don’t bother writing the zeros, there are really “12.000000000000000...” inches in 1 foot because this is an exact definition. Thus, our answer will be 36.302 inches (5 significant figures).

Reading Instruments

There are two types of instruments; analog and digital. Digital instruments are easy, just write down every number they give you, including zeros. This automatically

gives you the number of significant figures. However, analog devices are a little more tricky.

Analogue devices have some form of scale, with an indicator. In an old-fashioned thermometer, the scale is on the side, with the indicator being the level of the liquid. Other instruments, like voltmeters, for example, had a scale (usually with a portion mirrored so you always look at it from the same angle by lining up the pointer so you cannot see the image) with a pointer. Whenever you have an instrument like this, you can always estimate one significant figure more than the scale on the instrument.



0.1.

Take the following example; suppose you are measuring the liquid in a graduated cylinder, with markings every 0.1 mL, as shown in the figure to the left. We know that the liquid level is above 8.7, but less than 8.8; so what is it? (Forgive the squiggly line; it was drawn by hand.) Well, how far up does it look to you? Maybe 70% of the way? OK, so you record 8.77 in your records. Don't worry that the last number is a guess, this is what a significant figure means! The reader knows that last 7 is not significant, but if you don't record it, then the reader will assume that the first 7 was a guess. This is also why you need to record zeros; if the line were exactly on the 8.7 line, record it as 8.70, so the reader knows that your estimate is to 0.01, not just

Rounding

In the significant figures section, I intentionally left out how to round numbers. The rules for rounding are usually a little more involved than most people realize (not much, just a little). Remember that the purpose of rounding is to try to minimize errors in the final results. If we always round down, we know our answer will always be too low; if we always round up, the answer will always be too high. Through proper rounding techniques, we will get a mix of answers that are either too low or too high, and the errors have at least a fighting chance of canceling each other out so our final answer is close to the actual answer.

(1) Numbers greater than 5 always round up. Thus, to 4 significant figures, 22.347 becomes 22.35. Keep in mind that any number greater than five will round up; thus 654.235000000004 rounds up to 654.24 to 5 significant figures, because 0.005000000004 is greater than 0.005, even though it is not by much.

(2) Numbers less than 5 always round down. Thus, to 3 significant figures, 22.94432 rounds to 22.9. Again, though, remember that it is any number less than 5, so to six significant figures, 345.312499999999993 rounds to 345.312, since 0.0004999999999993 is less than 0.0005, again, not by much, but it is.

(3) Numbers exactly equal to zero depend on a convention. Some people will tell you if it is exactly 5, you always round up, or always round down, but this introduces a systematic error. Instead, use this: *by convention*, if it is exactly equal to 5, always round up IF the number proceeding the 5 is odd, and always round down if the number proceeding the five is even. Thus, to 3 significant figures, 23.45 rounds to 23.4, but 63.75 rounds to 63.8. This may sound very arbitrary to you, but that is only because it is; however, it is less arbitrary than always rounding the same way. The underlying assumption is that half the time you will round up, and half the time you will round down; thus, the errors, over a long period of time, will cancel out.